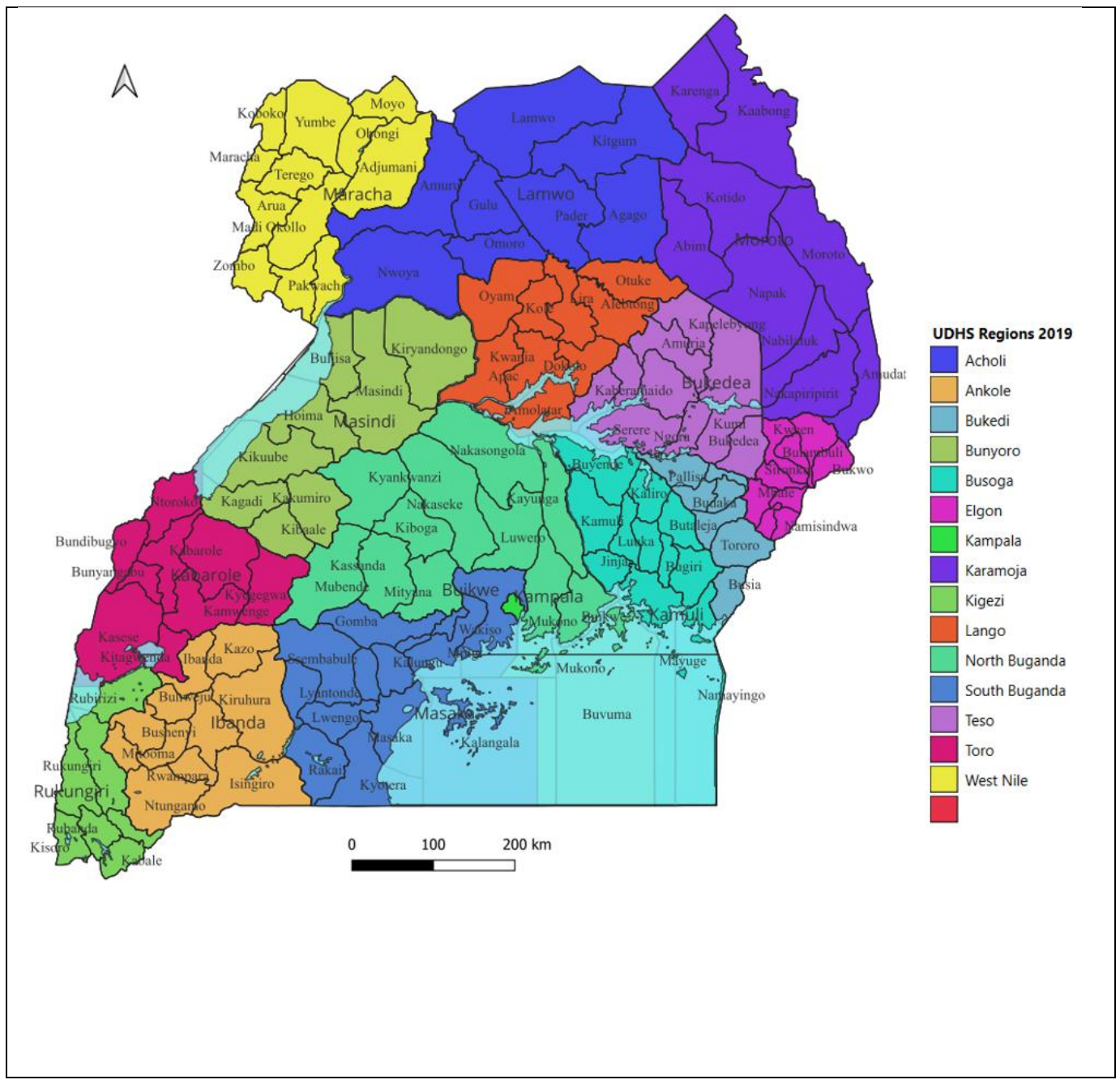
Supplementary file

Supplementary Figure S1. Map of Uganda showing the 146 districts and 15 sub regions



Supplementary Figure S2. Health-facility catchment areas derived using an adapted 60-minute two-step floating catchment area (2SFCA) method for 280 higher-level public health facilities in Uganda

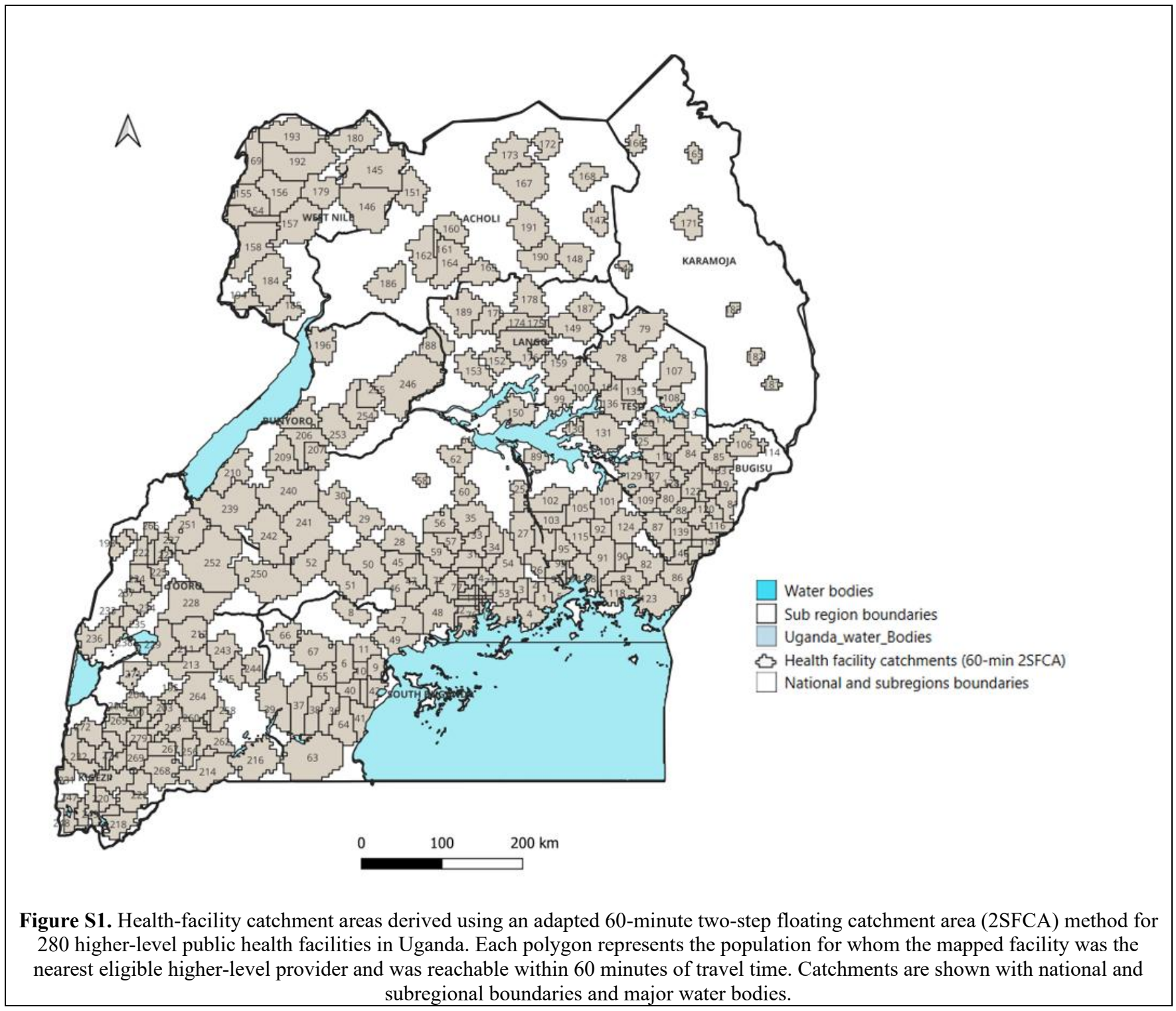


Figure S1. Health-facility catchment areas derived using an adapted 60-minute two-step floating catchment area (2SFCA) method for 280 higher-level public health facilities in Uganda. Each polygon represents the population for whom the mapped facility was the nearest eligible higher-level provider and was reachable within 60 minutes of travel time. Catchments are shown with national and subregional boundaries and major water bodies.

Supplementary Table S1. Health-facility catchment-area dataset for 280 higher-level public health facilities in Uganda. The dataset is structured with one row per facility-linked 60-minute catchment and includes facility ID, facility name, type, ownership, managing authority, longitude, latitude, subregion, district, catchment population, and linked DHIS2 counts of malaria cases, malaria deaths, SCD cases, and SCD deaths.

No.	ID	name	type	ownership	Sub region	Catchment population	Malaria cases	Scd cases	Crude malaria rate per 10,000	Crude scd rate per 10,000	Crude joint burden
1	HF0002	Buikwe St. Charles Lwanga HOSPITAL	General Hospital	Private Not For Profit	NORTH BUGANDA	117227	2943	720	251.051	61.4192	312.4702
2	HF0007	Kawolo HOSPITAL	General Hospital	Government	NORTH BUGANDA	41558	2628	711	632.3724	171.087	803.4594
3	HF0008	Lugazi Scoul HOSPITAL	General Hospital	Private For Profit	NORTH BUGANDA	77886	8130	26	1043.829	3.3382	1047.168
4	HF0019	Nkokonjeru HOSPITAL	General Hospital	Private Not For Profit	NORTH BUGANDA	120831	623	15	51.55958	1.241402	52.80099
5	HF0022	St. Francis Nyenga HOSPITAL	General Hospital	Private Not For Profit	NORTH BUGANDA	126012	3772	199	299.3376	15.7922	315.1298
6	HF0039	Butenga HC IV	HC IV	Government	SOUTH BUGANDA	126212	0	0	0	0	0
7	HF0052	Gombe HOSPITAL	General Hospital	Government	SOUTH BUGANDA	143579	2092	100	145.7038	6.964808	152.6686
8	HF0083	Maddu HC IV	HC IV	Government	SOUTH BUGANDA	44006	0	0	0	0	0
9	HF0092	Bukulula HC IV	HC IV	Government	SOUTH BUGANDA	62110	0	0	0	0	0
10	HF0093	Villa Maria HOSPITAL	General Hospital	Private Not For Profit	SOUTH BUGANDA	54373	2290	369	421.1682	67.8651	489.0333
11	HF0103	St. Joseph of the Good Shephard Kyamulibwa Ngo HC IV	HC IV	Private Not For Profit	SOUTH BUGANDA	82203	0	0	0	0	0
12	HF0118	Kololo HOSPITAL	General Hospital	Private For Profit	KAMPALA	243967	2628	711	107.7197	29.14334	136.863
13	HF0202	International Hospital Kampala (IHK)	General Hospital	Private For Profit	KAMPALA	183488	23	9	1.253491	0.490496	1.743987
14	HF0203	Kibuli HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	71046	1382	17	194.5229	2.392828	196.9157
15	HF0204	St. Francis Nsambya HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	181515	11550	601	636.3113	33.11022	669.4215
16	HF0216	Kiruddu	NR Hospital	Government	KAMPALA	238924	3525	2191	147.5364	91.70275	239.2391
17	HF0252	Kadic HOSPITAL	General Hospital	Private For Profit	KAMPALA	239469	3771	273	157.4737	11.40025	168.874
18	HF0253	Murchison Bay HOSPITAL	General Hospital	Government	KAMPALA	40079	74	0	18.46351	0	18.46351
19	HF0262	Benedict Medical centre HC IV	HC IV	Private Not For Profit	KAMPALA	99318	151	2	15.20373	0.201374	15.4051
20	HF0263	Butabika NATIONAL REFERAL HOSPITAL	NR Hospital	Government	KAMPALA	66129	299	373	45.21462	56.40486	101.6195
21	HF0264	Naguru Hospital - China Uganda Friendship	RR Hospital	Government	KAMPALA	68614	0	0	0	0	0
22	HF0281	Lubaga HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	594380	1030	383	17.32897	6.443686	23.77266
23	HF0282	Mengo HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	291689	1861	724	63.80077	24.82094	88.62171
24	HF0283	Namungoona Orthodox Hospital HC III	General Hospital	Private Not For Profit	KAMPALA	503661	0	0	0	0	0
25	HF0290	Bbaale HC IV	HC IV	Government	NORTH BUGANDA	29346	0	0	0	0	0
26	HF0298	Kangulumira HC IV	HC IV	Government	NORTH BUGANDA	109538	4109	71	375.1196	6.481746	381.6014
27	HF0304	Kayunga Hospital	General Hospital	Government	NORTH BUGANDA	213826	11399	1041	533.0979	48.68452	581.7824
28	HF0316	Bukomero HC IV	HC IV	Government	NORTH BUGANDA	86711	0	0	0	0	0
29	HF0327	Kiboga HOSPITAL	General Hospital	Government	NORTH BUGANDA	123578	3771	65	305.1522	5.259849	310.412
30	HF0354	Ntwetwe HC IV	HC IV	Government	NORTH BUGANDA	120014	0	0	0	0	0
31	HF0362	Bombo General Military HOSPITAL	General Hospital	Government	NORTH BUGANDA	170628	6308	148	369.6928	8.673833	378.3667
32	HF0376	Kalagala HC IV	HC IV	Government	NORTH BUGANDA	68178	0	0	0	0	0
33	HF0401	Luwero HC IV	HC IV	Government	NORTH BUGANDA	138002	0	0	0	0	0
34	HF0402	Bishop Asili Ceaser HC IV	HC IV	Private Not For Profit	NORTH BUGANDA	69531	0	0	0	0	0
35	HF0411	Nyimbwa HC IV	HC IV	Government	NORTH BUGANDA	166414	0	0	0	0	0
36	HF0429	Kiwangala HC IV	HC IV	Government	SOUTH BUGANDA	118581	0	0	0	0	0
37	HF0438	Kyazanga HC IV	HC IV	Government	SOUTH BUGANDA	175942	0	0	0	0	0
38	HF0446	Lwengo HC IV	HC IV	Government	SOUTH BUGANDA	142999	0	0	0	0	0
39	HF0465	Lyantonde HOSPITAL	General Hospital	Government	SOUTH BUGANDA	130642	7898	65	604.555	4.975446	609.5304
40	HF0494	Masaka REGIONAL REF HOSPITAL	RR Hospital	Government	SOUTH BUGANDA	159156	9391	2603	590.0512	163.5506	753.6018
41	HF0499	Kyannamukaaka HC IV	HC IV	Government	SOUTH BUGANDA	73642	2649	0	359.7121	0	359.7121

42	HF0505	Kiyumba HC IV	HC IV	Government	SOUTH BUGANDA	53833	0	0	0	0	0
43	HF0506	St. Joseph Kitovu HOSPITAL	General Hospital	Private Not For Profit	SOUTH BUGANDA	86572	4672	434	539.6661	50.13165	589.7977
44	HF0537	Mwera HC IV	HC IV	Government	NORTH BUGANDA	61345	0	0	0	0	0
45	HF0540	Kyantungo HC IV	HC IV	Government	NORTH BUGANDA	79995	0	0	0	0	0
46	HF0550	Mityana HOSPITAL	General Hospital	Government	NORTH BUGANDA	129409	2697	226	208.4083	17.46395	225.8722
47	HF0560	Ssekanyonyi HC IV	HC IV	Government	NORTH BUGANDA	53235	2444	0	459.0952	0	459.0952
48	HF0583	Mpigi HC IV	HC IV	Government	SOUTH BUGANDA	267404	0	0	0	0	0
49	HF0588	Nkozi HOSPITAL	General Hospital	Private Not For Profit	SOUTH BUGANDA	123986	1071	157	86.38091	12.66275	99.04365
50	HF0608	Kassanda HC IV	HC IV	Government	NORTH BUGANDA	193413	0	0	0	0	0
51	HF0615	Kiganda HC IV	HC IV	Government	NORTH BUGANDA	183110	0	0	0	0	0
52	HF0650	Mubende RR HOSPITAL	RR Hospital	Government	NORTH BUGANDA	250907	285	0	11.35879	0	11.35879
53	HF0678	Mukono T.C. HC IV	HC IV	Government	KAMPALA	261527	0	0	0	0	0
54	HF0680	St. Francis HOSPITAL Naggalama	General Hospital	Private Not For Profit	KAMPALA	249722	2040	2021	81.69068	80.92984	162.6205
55	HF0698	Kojja HC IV	HC IV	Government	KAMPALA	79842	0	0	0	0	0
56	HF0715	Kiwoko HOSPITAL	General Hospital	Private Not For Profit	NORTH BUGANDA	51687	2628	711	508.4422	137.558	646.0002
57	HF0718	Nakaseke HOSPITAL	General Hospital	Government	NORTH BUGANDA	70751	9051	173	1279.273	24.4519	1303.725
58	HF0720	Ngoma HC IV	HC IV	Government	NORTH BUGANDA	4217	0	0	0	0	0
59	HF0724	Semuto HC IV	HC IV	Government	NORTH BUGANDA	115960	0	0	0	0	0
60	HF0732	St. Franciscan HC IV	HC IV	Private Not For Profit	NORTH BUGANDA	23060	0	0	0	0	0
61	HF0741	Nakasongola Military HOSPITAL	General Hospital	Government	NORTH BUGANDA	7978	1539	13	1928.978	16.29416	1945.273
62	HF0752	Nakasongola HC IV	HC IV	Government	NORTH BUGANDA	39471	5466	304	1384.804	77.01801	1461.822
63	HF0773	Kakuuto HC IV	HC IV	Government	SOUTH BUGANDA	257871	0	0	0	0	0
64	HF0777	Kalisizo HOSPITAL	General Hospital	Government	SOUTH BUGANDA	176754	444	149	25.11964	8.429788	33.54942
65	HF0854	Bamu HOSPITAL	General Hospital	Private For Profit	SOUTH BUGANDA	96347	1587	86	164.7177	8.926099	173.6438
66	HF0859	Ntuusi HC IV	HC IV	Government	SOUTH BUGANDA	30912	0	0	0	0	0
67	HF0860	Ssembabule HC IV	HC IV	Government	SOUTH BUGANDA	113368	0	0	0	0	0
68	HF0873	Buwambo HC IV	HC IV	Government	KAMPALA	171754	0	0	0	0	0
69	HF0883	Luwunga Barracks HC IV	HC IV		KAMPALA	242339	2279	9	94.04168	0.37138	94.41306
70	HF0887	Kisubi HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	538746	444	149	8.241355	2.76568	11.00703
71	HF0889	Uganda Martyrs HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	426954	0	2	0	0.046843	0.046843
72	HF0919	Namayumba HC IV	HC IV	Government	KAMPALA	173431	0	0	0	0	0
73	HF0921	Saidina Abubakar Islamic HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	271032	511	14	18.85388	0.516545	19.37042
74	HF0926	Kasangati HC IV	HC IV	Government	KAMPALA	386003	0	0	0	0	0
75	HF0940	Mildmay Uganda HOSPITAL	General Hospital	Private Not For Profit	KAMPALA	492592	11	31	0.223308	0.629323	0.852632
76	HF0948	Ndejje HC IV	HC IV	Government	KAMPALA	65353	0	0	0	0	0
77	HF0957	Wakiso HC IV	HC IV	Government	KAMPALA	166302	0	0	0	0	0
78	HF0966	Amuria HC IV	HC IV	Government	TESO	240177	0	0	0	0	0
79	HF0975	Kapelebyong HC IV	HC IV	Government	TESO	62411	7821	192	1253.154	30.76405	1283.918
80	HF0998	Budaka HC IV	HC IV	Government	BUKEDI	230903	0	0	0	0	0
81	HF1015	Bududa HOSPITAL	General Hospital	Government	BUGISU	388002	1665	206	42.91216	5.309253	48.22142
82	HF1032	Bugiri HOSPITAL	General Hospital	Government	BUSOGA	292996	11617	551	396.4896	18.8057	415.2953
83	HF1063	Nankoma HC IV	HC IV	Government	BUSOGA	232165	0	0	0	0	0
84	HF1065	Bukedea HC IV	HC IV	Government	TESO	180776	0	0	0	0	0
85	HF1092	Muyembe HC IV	HC IV	Government	BUGISU	278504	0	0	0	0	0
86	HF1113	Dabani HOSPITAL	General Hospital	Private Not For Profit	BUKEDI	297299	3938	556	132.4592	18.70171	151.1609
87	HF1125	Busolwe HOSPITAL	General Hospital	Government	BUKEDI	195701	22309	679	1139.955	34.69583	1174.651
88	HF1130	Kabasa Memorial HOSPITAL	General Hospital	Private Not For Profit	BUKEDI	182641	1852	11	101.401	0.602274	102.0033
89	HF1147	Kidera HC IV	HC IV	Government	BUSOGA	111894	0	0	0	0	0
90	HF1165	Busesa HC IV	HC IV	Government	BUSOGA	136566	0	0	0	0	0
91	HF1166	Iganga HOSPITAL	General Hospital	Government	BUSOGA	449866	11399	1041	253.3866	23.14023	276.5268
92	HF1180	Bugono HC IV	HC IV	Government	BUSOGA	129867	0	0	0	0	0
93	HF1207	Budondo HC IV	HC IV	Government	BUSOGA	128509	0	0	0	0	0
94	HF1209	Bugembe HC IV	HC IV	Government	BUSOGA	88179	0	0	0	0	0
95	HF1234	Buwenge NGO Hospital	General Hospital	Private Not For Profit	BUSOGA	195011	2139	78	109.6861	3.999775	113.6859
96	HF1246	Walukuba HC IV	HC IV	Government	BUSOGA	31392	0	0	0	0	0
97	HF1249	Mpumudde HC IV	HC IV	Government	BUSOGA	145831	0	0	0	0	0
98	HF1250	Kakira Worker's HOSPITAL	General Hospital	Private For Profit	BUSOGA	206568	3224	54	156.0747	2.614154	158.6888

99	HF1269	Kaberamaido HC IV	HC IV	Government	TESO	91502	0	0	0	0	0
100	HF1276	Lwala HOSPITAL	General Hospital	Private Not For Profit	TESO	119681	6090	864	508.8528	72.19192	581.0447
101	HF1281	Bumanya HC IV	HC IV	Government	BUSOGA	245575	0	0	0	0	0
102	HF1312	Kamuli HOSPITAL	General Hospital	Government	BUSOGA	196516	1382	17	70.32493	0.865068	71.19
103	HF1313	Kamuli Mission Hospital	General Hospital	Private Not For Profit	BUSOGA	160864	2093	190	130.1101	11.81124	141.9214
104	HF1322	Nankandulo HC IV	HC IV	Government	BUSOGA	176006	0	0	0	0	0
105	HF1340	Namwendwa HC IV	HC IV	Government	BUSOGA	177898	0	0	0	0	0
106	HF1354	Kapchorwa HOSPITAL	General Hospital	Government	BUGISU	182969	3324	39	181.6699	2.131506	183.8014
107	HF1375	Katakwi General Hospital	General Hospital	Government	TESO	111728	22896	957	2049.262	85.65443	2134.916
108	HF1390	Toroma HC IV	HC IV	Government	TESO	45096	0	0	0	0	0
109	HF1405	Kibuku HC IV	HC IV	Government	BUKEDI	173371	0	0	0	0	0
110	HF1408	Atutur HOSPITAL	General Hospital	Government	TESO	79058	4879	4387	617.1447	554.9116	1172.056
111	HF1419	Kumi HC IV	HC IV	Government	TESO	139796	0	0	0	0	0
112	HF1421	Kadami HOSPITAL	General Hospital	Private Not For Profit	TESO	94576	3771	155	398.7263	16.38891	415.1152
113	HF1431	Kumi NGO HOSPITAL	General Hospital	Private Not For Profit	TESO	21091	1817	92	861.4956	43.62003	905.1156
114	HF1438	Kaproron HC IV	HC IV	Government	BUGISU	13961	0	0	0	0	0
115	HF1475	Kiyunga HC IV	HC IV	Government	BUSOGA	220722	0	0	0	0	0
116	HF1483	Bugobero HC IV	HC IV	Government	BUGISU	218236	0	0	0	0	0
117	HF1492	Bubulo HC IV	HC IV	Government	BUGISU	168844	0	0	0	0	0
118	HF1515	Kigandalo HC IV	HC IV	Government	BUSOGA	215196	0	0	0	0	0
119	HF1524	Bufumbo HC IV	HC IV	Government	BUGISU	132774	0	0	0	0	0
120	HF1530	Busiu HC IV	HC IV	Government	BUGISU	241986	0	0	0	0	0
121	HF1532	Cure Children's Hospital	General Hospital	Private Not For Profit	BUGISU	126686	760	5	59.99094	0.394677	60.38561
122	HF1533	Namatala HC IV	HC IV	Government	BUGISU	297370	0	0	0	0	0
123	HF1547	Buyinja HC IV	HC IV	Government	BUSOGA	390611	0	0	0	0	0
124	HF1583	Nsinze HC IV	HC IV	Government	BUSOGA	275880	0	0	0	0	0
125	HF1591	Ngora Ngo HOSPITAL	General Hospital	Private Not For Profit	TESO	100774	2139	307	212.2577	30.46429	242.722
126	HF1595	Ngora Gvt HC IV	HC IV	Government	TESO	51233	0	0	0	0	0
127	HF1600	Butebo HC IV	HC IV	Government	BUKEDI	256457	0	0	0	0	0
128	HF1609	Kanginima HOSPITAL	General Hospital	Government	BUKEDI	69718	11399	1041	1635.024	149.3166	1784.341
129	HF1617	Pallisa HOSPITAL	General Hospital	Government	BUKEDI	319861	810	579	25.3235	18.10161	43.42511
130	HF1629	Apapai HC IV	HC IV	Government	TESO	51802	0	0	0	0	0
131	HF1643	Serere HC IV	HC IV	Government	TESO	269660	0	0	0	0	0
132	HF1647	Budadiri HC IV	HC IV	Government	BUGISU	200652	0	0	0	0	0
133	HF1667	Buwasa HC IV	HC IV	Government	BUGISU	144065	0	0	0	0	0
134	HF1688	ATIRIR HC IV	HC IV	Government	TESO	52329	0	0	0	0	0
135	HF1697	Princes Diana HC IV	HC IV	Government	TESO	89809	2897	67	322.5719	7.46024	330.0321
136	HF1698	Soroti REGIONAL REF HOSPITAL	RR Hospital	Government	TESO	245592	7489	1167	304.9365	47.51781	352.4543
137	HF1722	Mukuju HC IV	HC IV	Government	BUKEDI	135857	0	0	0	0	0
138	HF1726	Mulanda HC IV	HC IV	Government	BUKEDI	42992	0	0	0	0	0
139	HF1734	Nagongera HC IV	HC IV	Government	BUKEDI	133014	0	0	0	0	0
140	HF1747	Rubongi Military HOSPITAL	General Hospital	Government	BUKEDI	93459	4349	57	465.3354	6.0989	471.4343
141	HF1751	Benedictine Eye HOSPITAL	General Hospital	Government	BUKEDI	80273	0	0	0	0	0
142	HF1752	Tororo General HOSPITAL	General Hospital	Government	BUKEDI	42651	16907	2472	3963.988	579.5812	4543.569
143	HF1754	St. Anthony'S Tororo HOSPITAL	General Hospital	Private Not For Profit	BUKEDI	73373	1374	261	187.2618	35.57156	222.8333
144	HF1769	Abim HOSPITAL	General Hospital	Government	KARAMOJA	42977	3938	556	916.3023	129.3713	1045.674
145	HF1785	Adjumani HOSPITAL	General Hospital	Government	WEST NILE	184607	13988	328	757.7179	17.76748	775.4854
146	HF1805	Mungula HC IV	HC IV	Government	WEST NILE	56104	0	0	0	0	0
147	HF1831	Kalongo Ambrosoli Memorial HOSPITAL	General Hospital	Private Not For Profit	ACHOLI	33544	16436	1408	4899.767	419.7416	5319.509
148	HF1856	Patongo HC III	HC IV	Government	ACHOLI	76697	0	0	0	0	0
149	HF1868	Alebtong HC IV	HC IV	Government	LANGO	124673	0	0	0	0	0
150	HF1882	Amolatar HC IV	HC IV	Government	LANGO	141470	0	0	0	0	0
151	HF1903	Atiak HC IV	HC IV	Government	ACHOLI	40186	0	0	0	0	0
152	HF1927	Aduku HC IV	HC IV	Government	LANGO	80505	0	0	0	0	0
153	HF1934	Apac HOSPITAL	General Hospital	Government	LANGO	116013	2482	737	213.9417	63.52741	277.4691
154	HF1965	Arua REGIONAL REF HOSPITAL	RR Hospital	Government	WEST NILE	271111	7301	802	269.2995	29.582	298.8815
155	HF1970	Oriajini HOSPITAL	General Hospital	Private Not For Profit	WEST NILE	275970	1847	556	66.92761	20.14713	87.07474
156	HF1994	Omugo HC IV	HC IV	Government	WEST NILE	269889	0	0	0	0	0
157	HF1999	Rhino Camp HC IV	HC IV	Government	WEST NILE	119694	0	0	0	0	0
158	HF2009	Kuluva HOSPITAL	General Hospital	Private Not For Profit	WEST NILE	274398	1801	50	65.63451	1.822168	67.45668
159	HF2019	Dokolo HC IV	HC IV	Government	LANGO	153861	0	0	0	0	0
160	HF2027	Awach HC IV	HC IV	Government	ACHOLI	74663	0	0	0	0	0
161	HF2028	Gulu Military Hospital	General Hospital	Government	ACHOLI	47199	1821	39	385.8093	8.262803	394.0721
162	HF2029	St. Mary's Hospital Lacor	General Hospital	Private Not For Profit	ACHOLI	175898	14835	4585	843.3888	260.6631	1104.052
163	HF2057	Lalogi HC IV	HC IV	Government	ACHOLI	60887	0	0	0	0	0

164	HF2067	Gulu Regional Referal Hospital	RR Hospital	Government	ACHOLI	214139	17764	2371	829.5535	110.7223	940.2759
165	HF2099	Kaabong HOSPITAL	General Hospital	Government	KARAMOJA	45080	5434	387	1205.424	85.84821	1291.272
166	HF2108	Karenga HC IV	HC IV	Government	KARAMOJA	17189	0	0	0	0	0
167	HF2124	Kitgum HOSPITAL	General Hospital	Government	ACHOLI	150715	15823	629	1049.864	41.73447	1091.599
168	HF2140	Naam Okora HC IV	HC IV	Government	ACHOLI	28508	0	0	0	0	0
169	HF2145	Koboko Hospital	General Hospital	Government	WEST NILE	251072	2628	711	104.6711	28.31855	132.9896
170	HF2152	Aboke HC IV	HC IV	Government	LANGO	96525	0	0	0	0	0
171	HF2167	Kotido HC IV	HC IV	Government	KARAMOJA	128765	0	0	0	0	0
172	HF2186	Madi-Opei HC IV	HC IV	Government	ACHOLI	10993	0	0	0	0	0
173	HF2190	Padibe HC IV	HC IV	Government	ACHOLI	46874	0	0	0	0	0
174	HF2205	Ayira Health Services	HC IV	Private For Profit	LANGO	156218	180	0	11.52233	0	11.52233
175	HF2206	Lira REGIONAL REF HOSPITAL	RR Hospital	Government	LANGO	90097	13408	3698	1488.17	410.4454	1898.615
176	HF2211	Amach HC IV	HC IV	Government	LANGO	51559	0	0	0	0	0
177	HF2222	PAG Mission HC IV	HC IV	Private Not For Profit	LANGO	274235	7492	1818	273.1964	66.29351	339.4899
178	HF2228	Ogur HC IV	HC IV	Government	LANGO	160407	0	0	0	0	0
179	HF2249	Obongi HC IV	HC IV	Government	WEST NILE	63956	0	0	0	0	0
180	HF2278	Moyo HOSPITAL	General Hospital	Government	WEST NILE	69781	1861	724	266.69	103.7526	370.4426
181	HF2280	Tokora HC IV	HC IV	Government	KARAMOJA	29400	0	0	0	0	0
182	HF2292	Nabilatuk HC IV	HC IV	Government	KARAMOJA	44804	0	0	0	0	0
183	HF2305	Matany HOSPITAL	General Hospital	Private Not For Profit	KARAMOJA	13632	19599	197	14377.49	144.5159	14522.01
184	HF2329	Nebbi Hospital	General Hospital	Government	WEST NILE	181821	14033	580	771.8024	31.89948	803.7019
185	HF2330	Angal St. Luke HOSPITAL	General Hospital	Private Not For Profit	WEST NILE	114766	11399	1041	993.2371	90.70619	1083.943
186	HF2353	Anaka HOSPITAL	General Hospital	Government	ACHOLI	151427	11399	1041	752.7715	68.74595	821.5175
187	HF2375	Orum HC IV	HC IV	Government	LANGO	50739	0	0	0	0	0
188	HF2376	Aber Ngo HOSPITAL	General Hospital	Private Not For Profit	LANGO	128850	1861	724	144.4311	56.1892	200.6203
189	HF2400	Anyeke HC IV	HC IV	Government	LANGO	190427	0	0	0	0	0
190	HF2410	5Th Military Division HOSPITAL	General Hospital	Government	ACHOLI	58045	0	0	0	0	0
191	HF2428	Pajule HC IV	HC IV	Government	ACHOLI	56279	0	0	0	0	0
192	HF2445	Yumbe Hospital	General Hospital	Government	WEST NILE	581762	334	724	5.741176	12.44495	18.18612
193	HF2448	Midigo HC IV	HC IV	Government	WEST NILE	179360	0	0	0	0	0
194	HF2457	Nyapea HOSPITAL	General Hospital	Private Not For Profit	WEST NILE	95962	2482	272	258.6453	28.34469	286.99
195	HF2475	Nsiika HC IV	HC IV	Government	ANKOLE	47326	0	0	0	0	0
196	HF2481	Buliisa Hospital	General Hospital	Government	BUNYORO	120553	1665	206	138.1131	17.08787	155.201
197	HF2490	Bundibugyo Hospital	General Hospital	Government	TORO	100365	13471	1315	1342.201	131.0218	1473.223
198	HF2494	Busaru HC IV	HC IV	Private Not For Profit	TORO	119966	0	0	0	0	0
199	HF2501	Kikyo HC IV	HC IV	Government	TORO	36800	0	0	0	0	0
200	HF2508	Bushenyi HC IV	HC IV	Government	ANKOLE	50128	0	0	0	0	0
201	HF2512	Kiu Teaching HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	67364	745	43	110.593	6.383223	116.9763
202	HF2513	Ishaka Adventist HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	31340	1281	16	408.7422	5.105289	413.8475
203	HF2521	Kyabugimbi HC IV	HC IV	Government	ANKOLE	51812	0	0	0	0	0
204	HF2522	Comboni HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	51408	2092	100	406.9439	19.45238	426.3962
205	HF2528	Hunter Foundation HC	General Hospital	Private For Profit	ANKOLE	30088	0	0	0	0	0
206	HF2565	Azur HC III	HC IV	Private Not For Profit	BUNYORO	206859	0	0	0	0	0
207	HF2566	Hoima REGIONAL REF HOSPITAL	RR Hospital	Government	BUNYORO	125437	14128	1777	1126.301	141.6646	1267.966
208	HF2570	Kigorobya HC IV	HC IV	Government	BUNYORO	163324	0	0	0	0	0
209	HF2578	Kikuube HC IV	HC IV	Government	BUNYORO	166217	0	0	0	0	0
210	HF2586	Kyangwali HC IV	HC IV	Government	BUNYORO	134630	0	0	0	0	0
211	HF2594	Ibanda HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	42035	4306	137	1024.374	32.59156	1056.966
212	HF2601	Ishongororo HC IV	HC IV	Government	ANKOLE	168709	4936	1	292.5743	0.059274	292.6335
213	HF2622	Ruhoko HC IV	HC IV	Government	ANKOLE	180360	0	0	0	0	0
214	HF2652	Kabuyanda HC IV	HC IV	Government	ANKOLE	287835	0	0	0	0	0
215	HF2673	Nyamuyanja HC IV	HC IV	Government	ANKOLE	25375	0	0	0	0	0
216	HF2677	Rugaaga HC IV	HC IV	Government	ANKOLE	211556	0	0	0	0	0
217	HF2695	Hamurwa HC IV	HC IV	Government	KIGEZI	71374	0	0	0	0	0
218	HF2705	Kabale REGIONAL REF HOSPITAL	RR Hospital	Government	KIGEZI	186946	1268	34	67.82706	1.818707	69.64577
219	HF2708	Rugarama HOSPITAL	General Hospital	Private Not For Profit	KIGEZI	53292	0	19	0	3.565254	3.565254
220	HF2734	Muko HC IV	HC IV	Government	KIGEZI	88238	0	0	0	0	0
221	HF2749	Mparo HC IV	HC IV	Government	KIGEZI	91127	0	0	0	0	0
222	HF2753	Bukuku HC IV	HC IV	Government	TORO	113254	0	0	0	0	0
223	HF2761	Kataraka HC IV	HC IV	Government	TORO	11443	0	0	0	0	0
224	HF2778	Kibiito HC IV	HC IV	Government	TORO	181871	0	0	0	0	0
225	HF2791	Kida HOSPITAL	General Hospital	Private Not For Profit	TORO	64154	3771	31	587.8056	4.832133	592.6377
226	HF2801	Virika HOSPITAL	General Hospital	Private Not For Profit	TORO	71989	16513	179	2293.834	24.86503	2318.699
227	HF2806	Fort Portal Regional Referral Hospital	RR Hospital	Government	TORO	99738	9429	1200	945.3815	120.3158	1065.697
228	HF2819	Rukunyu HC IV	HC IV	Government	TORO	360089	0	0	0	0	0

229	HF2830	Ntara HC IV	HC IV	Government	TORO	155691	0	0	0	0	0
230	HF2836	Kambuga HOSPITAL	General Hospital	Government	KIGEZI	72430	6342	142	875.602	19.60509	895.207
231	HF2837	Bwindi Community HOSPITAL	General Hospital	Private Not For Profit	KIGEZI	35204	1828	30	519.266	8.521871	527.7879
232	HF2843	Kanungu HC IV	HC IV	Government	KIGEZI	162171	0	0	0	0	0
233	HF2874	Kilembe HOSPITAL	General Hospital	Private Not For Profit	TORO	91641	2844	100	310.3413	10.91214	321.2534
234	HF2879	Hiima Iaa (Uci) HC IV	HC IV	Private For Profit	TORO	134312	0	0	0	0	0
235	HF2886	St. Paul HC IV	HC IV	Private Not For Profit	TORO	114729	0	0	0	0	0
236	HF2891	Kagando HOSPITAL	General Hospital	Private Not For Profit	TORO	328735	5434	387	165.3003	11.7724	177.0727
237	HF2900	Rwesande HC IV	HC IV	Private Not For Profit	TORO	91010	0	0	0	0	0
238	HF2916	Nyamirami HC IV	HC IV	Government	TORO	46975	0	0	0	0	0
239	HF2927	Kagadi HOSPITAL	General Hospital	Government	BUNYORO	518195	5434	387	104.864	7.468232	112.3322
240	HF2931	Kakindo HC IV	HC IV	Government	BUNYORO	546903	0	0	0	0	0
241	HF2932	Kakumiro HC IV	HC IV	Government	BUNYORO	391570	0	0	0	0	0
242	HF2937	Kibaale HC IV (Kibaale)	HC IV	Government	BUNYORO	262532	0	0	0	0	0
243	HF2983	Kazo HC IV	HC IV	Government	ANKOLE	86199	0	0	0	0	0
244	HF2984	Rushere Community HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	29029	1527	4	526.0299	1.377943	527.4079
245	HF2991	Kiruhura HC IV	HC IV	Government	ANKOLE	27071	0	0	0	0	0
246	HF3019	Kiryandongo HOSPITAL	General Hospital	Government	BUNYORO	284372	11157	727	392.3381	25.5651	417.9032
247	HF3033	Rubuguri HC IV	HC IV	Government	KIGEZI	59258	0	0	0	0	0
248	HF3034	Kisoro HOSPITAL	General Hospital	Government	KIGEZI	107975	2628	711	243.39	65.84867	309.2387
249	HF3036	Mutolere (St. Francis) HOSPITAL	General Hospital	NGO	KIGEZI	100020	0	0	0	0	0
250	HF3047	Kyegegwa HC IV	HC IV	Government	TORO	533214	0	0	0	0	0
251	HF3083	Kyarusozi HC IV	HC IV	Government	TORO	174158	0	0	0	0	0
252	HF3085	Kyenjojo Hospital	General Hospital	Government	TORO	486694	10855	217	223.0354	4.458653	227.4941
253	HF3103	Bwijanga HC IV	HC IV	Government	BUNYORO	127805	0	0	0	0	0
254	HF3109	Masindi HOSPITAL	General Hospital	Government	BUNYORO	133145	4401	661	330.5413	49.64503	380.1863
255	HF3118	Masindi Military/Army Barracks HC IV	HC IV	Government	BUNYORO	73661	0	0	0	0	0
256	HF3137	Bugamba HC IV	HC IV	Government	ANKOLE	70444	0	0	0	0	0
257	HF3141	Mbarara Community HOSPITAL	General Hospital	Private For Profit	ANKOLE	2661	1447	424	5437.093	1593.177	7030.27
258	HF3145	UPDF 2nd Div.Hospital	General Hospital	Government	ANKOLE	82785	0	0	0	0	0
259	HF3146	Mayanja Memorial HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	15199	78	8	51.31932	5.26352	56.58284
260	HF3149	Ruharo Mission HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	30847	189	295	61.27038	95.63366	156.904
261	HF3153	Mbarara Municipal Council HC IV	HC IV	Government	ANKOLE	45420	0	0	0	0	0
262	HF3171	Holy Innocents Children's HOSPITAL	General Hospital	Private Not For Profit	ANKOLE	181255	3398	101	187.4707	5.572259	193.0429
263	HF3186	Kinoni HC IV	HC IV	Government	ANKOLE	85125	0	0	0	0	0
264	HF3188	Bwizibwera HC IV	HC IV	Government	ANKOLE	260709	0	0	0	0	0
265	HF3207	Mitooma HC IV	HC IV	Government	ANKOLE	117562	0	0	0	0	0
266	HF3214	Karugutu HC IV	HC IV	Government	TORO	31592	0	0	0	0	0
267	HF3222	Itojo HOSPITAL	General Hospital	Government	ANKOLE	61365	15823	629	2578.514	102.5018	2681.016
268	HF3226	Ntungamo HC IV	HC IV	Government	ANKOLE	169086	0	0	0	0	0
269	HF3236	Rwashamaire HC IV	HC IV	Government	ANKOLE	118495	957	2	80.76298	0.168784	80.93176
270	HF3241	Rubaare HC IV	HC IV	Government	ANKOLE	69442	0	0	0	0	0
271	HF3255	Rugazi HC IV	HC IV	Government	ANKOLE	139662	0	0	0	0	0
272	HF3264	Bugangari HC IV	HC IV	Government	KIGEZI	80889	0	0	0	0	0
273	HF3271	Buhunga HC IV	HC IV	Government	KIGEZI	84850	0	0	0	0	0
274	HF3291	Kebisoni HC IV	HC IV	Government	KIGEZI	57926	0	0	0	0	0
275	HF3308	Kisiizi NGO HOSPITAL	General Hospital	Private Not For Profit	KIGEZI	124151	672	9	54.12745	0.724921	54.85237
276	HF3330	North Kigezi HC IV	HC IV	Private Not For Profit	KIGEZI	76159	0	0	0	0	0
277	HF3331	St. Karolii Lwanga Nyakibale HOSPITAL	General Hospital	Private Not For Profit	KIGEZI	24156	828	4	342.7683	1.655886	344.4242
278	HF3339	Kabwohe HC IV	HC IV	Government	ANKOLE	142720	0	0	0	0	0
279	HF3353	Kitagata HOSPITAL	General Hospital	Government	ANKOLE	80650	2124	5	263.3605	0.619964	263.9805
280	HF3362	Shuuku HC IV	HC IV	Government	ANKOLE	11821	0	0	0	0	0

.

Supplementary Table S2. Subregional summary of crude malaria, sickle cell disease (SCD), and joint burden across 280 health-facility catchments in Uganda

Subregion	Catchments (n)	Mean crude malaria rate	Median crude malaria rate	Mean crude SCD rate	Median crude SCD rate	Mean crude joint burden	Median crude joint burden
Karamoja	7	2357.0	0.0	51.4	0.0	2408.4	0.0
Acholi	15	584.1	0.0	60.7	0.0	644.7	0.0

Bukedi	15	510.0	25.3	56.2	0.6	566.2	43.4
Teso	18	362.7	106.1	49.4	3.7	412.1	121.4
Ankole	34	336.2	0.0	54.9	0.0	391.1	0.0
North Buganda	30	315.1	31.5	26.0	0.0	341.2	32.1
Toro	21	290.6	0.0	3.5	0.0	294.1	0.0
West Nile	15	243.9	66.9	17.8	17.8	261.7	87.1
Lango	14	185.9	0.0	9.0	0.0	194.8	0.0
Bunyoro	14	161.7	0.0	5.1	0.0	166.7	0.0
South Buganda	21	153.6	0.0	1.7	0.0	155.3	0.0
Kigezi	17	125.7	0.0	4.0	0.0	129.7	0.0
KAMPALA	26	68.1	0.0	7.0	0.4	75.0	13.2
Busoga	21	52.2	0.0	3.9	0.0	56.1	0.0
Bugisu	12	23.0	0.0	1.4	0.0	24.4	0.0

Footnote: Rates are expressed per 10,000 catchment population. Joint crude burden was defined as the sum of crude malaria and crude SCD rates. Mean values reflect sensitivity to high-burden catchments, whereas medians reflect the central tendency within each subregion.

Supplementary Table S3. Catchment-level posterior summaries from the baseline Bayesian shared-component model. Catchment-specific posterior summaries from the baseline shared-component model, including expected counts, shared effect estimates, exceedance probabilities, hotspot classification, and fitted relative risks.

Catchment id	Area index	Catchment population	Malaria cases	Scd cases	Expected malaria cases	Expected scd_cases	Shared mean	Shared sd	Exceed prob_rr_gt1	Hotspot 080	Hotspot 090	rr_malaria	rr_scd
HF0002	1	20563	2943	720	1858.297	193.0095	6.828272	0.559059	1	1	1	8.52E-04	0.019328
HF0007	2	9446	2628	711	853.6438	88.66255	7.593941	0.570109	1	1	1	0.003608	0.09044
HF0008	3	20547	8130	26	1856.852	192.8593	3.692117	0.593547	1	1	1	0.002358	7.38E-04
HF0019	4	17421	623	15	1574.352	163.5179	3.213903	0.600418	1	1	1	2.51E-04	5.92E-04
HF0022	5	26312	3772	199	2377.84	246.9711	5.346469	0.548292	1	1	1	6.67E-04	0.003271
HF0039	6	30712	0	0	2775.472	288.2706	-4.57883	3.001889	0.06359	0	0	5.85E-07	1.26E-05
HF0052	7	36720	2092	100	3318.421	344.6632	4.323834	0.550851	1	1	1	1.90E-04	8.47E-04
HF0083	8	8716	0	0	787.673	81.81058	-3.84639	3.459682	0.133117	0	0	2.48E-05	4.06E-04
HF0092	9	16039	0	0	1449.459	150.5461	-4.06775	3.117525	0.09598	0	0	2.81E-06	5.69E-05
HF0093	10	10903	2290	369	985.3143	102.3383	6.809453	0.55867	1	1	1	0.002359	0.035246
HF0103	11	17771	0	0	1605.982	166.8031	-4.11981	3.077174	0.090313	0	0	2.08E-06	4.30E-05
HF0118	12	35945	2628	711	3248.383	337.3889	6.249937	0.553381	1	1	1	2.49E-04	0.006246
HF0202	13	21561	23	9	1948.488	202.377	2.302872	0.644779	0.999823	1	1	6.31E-06	2.19E-04
HF0203	14	7544	1382	17	681.7583	70.80989	4.214161	0.594675	1	1	1	0.002973	0.003593
HF0204	15	25095	11550	601	2267.859	235.548	6.506165	0.554296	1	1	1	0.002246	0.010841
HF0216	16	37804	3525	2191	3416.383	354.8379	7.29937	0.569713	1	1	1	3.02E-04	0.017399
HF0252	17	28974	3771	273	2618.407	271.9573	5.555393	0.547781	1	1	1	5.50E-04	0.003697
HF0253	18	5788	74	0	523.0669	54.32763	1.235591	1.108442	0.867513	1	0	2.70E-04	3.80E-04
HF0262	19	13670	151	2	1235.371	128.3101	1.648259	0.801867	0.980086	1	1	9.89E-05	1.81E-04
HF0263	20	10779	299	373	974.1083	101.1744	6.75199	0.570508	1	1	1	3.17E-04	0.036388
HF0264	21	8264	0	0	746.8254	77.568	-3.49949	3.187463	0.136126	0	0	1.28E-05	2.42E-04
HF0281	22	94308	1030	383	8522.702	885.1988	4.645325	0.548164	1	1	1	1.42E-05	4.89E-04
HF0282	23	46581	1861	724	4209.568	437.2211	5.993506	0.553041	1	1	1	1.05E-04	0.003787
HF0283	24	102942	0	0	9302.964	966.2397	-5.51614	2.803956	0.024576	0	0	3.41E-08	8.38E-07
HF0290	25	7616	0	0	688.265	71.4857	-3.49535	3.255539	0.141487	0	0	1.79E-05	3.27E-04
HF0298	26	25610	4109	71	2314.4	240.3819	4.385743	0.558441	1	1	1	7.67E-04	0.001244
HF0304	27	45037	11399	1041	4070.036	422.7287	6.44837	0.55253	1	1	1	6.88E-04	0.005827
HF0316	28	15892	0	0	1436.175	149.1663	-4.12219	3.158436	0.095923	0	0	3.19E-06	6.19E-05
HF0327	29	25071	3771	65	2265.69	235.3228	4.316868	0.559308	1	1	1	7.35E-04	0.00119
HF0354	30	22487	0	0	2032.171	211.0687	-4.54338	3.227733	0.079624	0	0	1.93E-06	3.61E-05
HF0362	31	28783	6308	148	2601.147	270.1645	4.991053	0.551292	1	1	1	9.32E-04	0.002039
HF0376	32	11582	0	0	1046.676	108.7116	-3.8142	3.166314	0.114175	0	0	6.14E-06	1.18E-04
HF0401	33	29137	0	0	2633.138	273.4873	-4.57519	3.043594	0.066391	0	0	7.13E-07	1.51E-05
HF0402	34	14744	0	0	1332.429	138.3909	-4.03211	3.150809	0.100324	0	0	3.63E-06	7.11E-05
HF0411	35	33798	0	0	3054.357	317.2366	-4.6825	3.008656	0.059813	0	0	4.90E-07	1.05E-05
HF0429	36	24780	0	0	2239.392	232.5914	-4.46303	3.077398	0.073493	0	0	1.07E-06	2.20E-05
HF0438	37	38773	0	0	3503.952	363.9332	-4.8511	3.030472	0.054714	0	0	3.92E-07	8.32E-06
HF0446	38	27853	0	0	2517.102	261.4353	-4.58502	3.081399	0.06838	0	0	8.57E-07	1.76E-05
HF0465	39	25852	7898	65	2336.269	242.6534	4.319183	0.564957	1	1	1	0.001447	0.001123
HF0494	40	29288	9391	2603	2646.784	274.9046	7.760042	0.572131	1	1	1	0.001341	0.034443
HF0499	41	13401	2649	0	1211.061	125.7852	1.321376	0.93135	0.922018	1	1	0.001805	1.29E-04
HF0505	42	8760	0	0	791.6494	82.22358	-3.58381	3.213082	0.132344	0	0	1.21E-05	2.27E-04
HF0506	43	17879	4672	434	1615.742	167.8168	6.497123	0.55416	1	1	1	0.00179	0.015422
HF0537	44	14188	0	0	1282.183	133.1722	-4.0046	3.155576	0.10221	0	0	3.98E-06	7.71E-05
HF0540	45	14995	0	0	1355.112	140.7469	-4.09063	3.181529	0.099266	0	0	3.80E-06	7.28E-05
HF0550	46	29277	2697	226	2645.79	274.8013	5.347913	0.54731	1	1	1	3.85E-04	0.002998

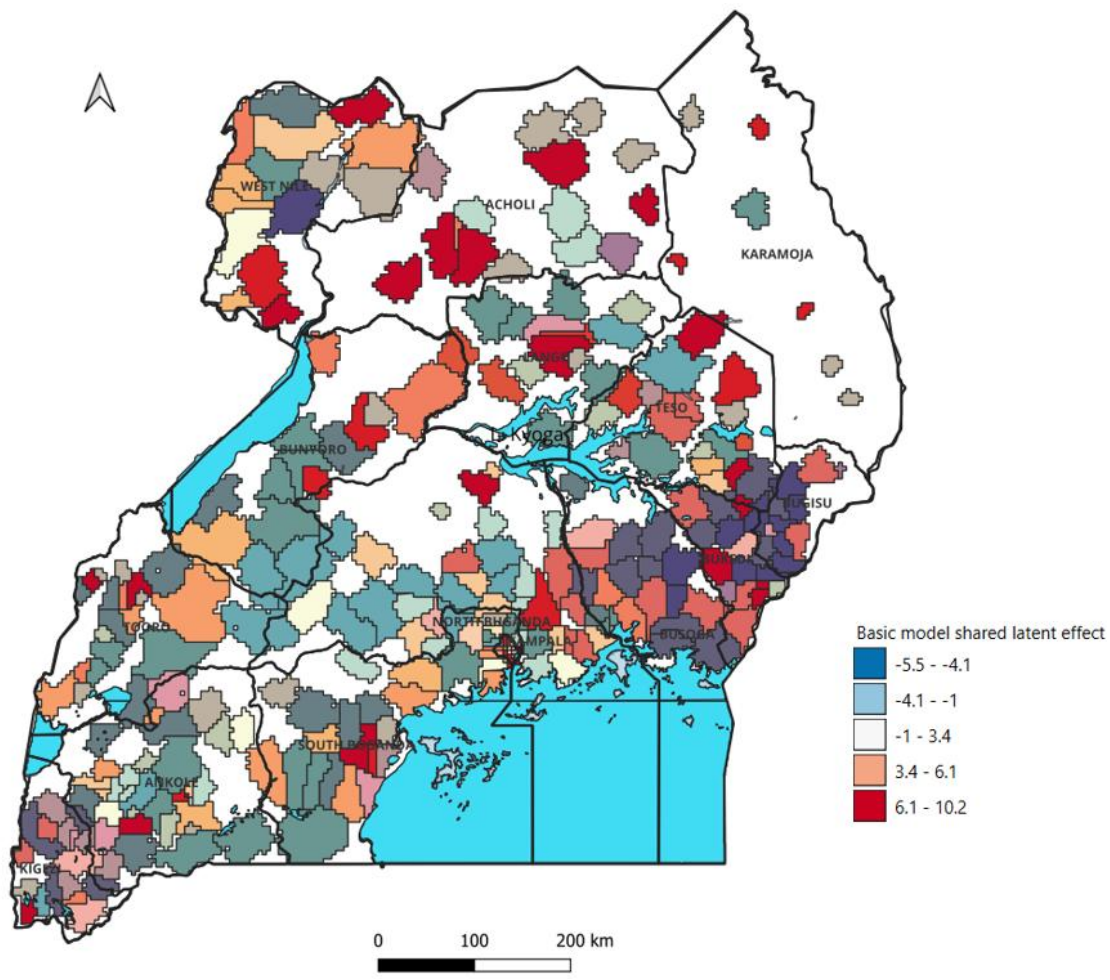
HF0560	47	10014	2444	0	904.9745	93.99394	1.588793	0.933971	0.955539	1	1	0.002983	2.27E-04
HF0583	48	44464	0	0	4018.253	417.3504	-4.8521	2.928332	0.048765	0	0	2.37E-07	5.34E-06
HF0588	49	27583	1071	157	2492.701	258.901	5.020321	0.548329	1	1	1	1.72E-04	0.002345
HF0608	50	45130	0	0	4078.44	423.6016	-4.97994	3.010629	0.049052	0	0	2.77E-07	5.93E-06
HF0615	51	28603	0	0	2584.88	268.475	-4.7096	3.162722	0.068231	0	0	1.00E-06	1.96E-05
HF0650	52	42103	285	0	3804.887	395.1894	-0.34243	1.024555	0.369107	0	0	1.97E-05	9.38E-06
HF0678	53	45197	0	0	4084.495	424.2305	-4.8371	2.906389	0.048026	0	0	2.19E-07	5.00E-06
HF0680	54	49954	2040	2021	4514.389	468.8809	6.919653	0.568631	1	1	1	1.00E-04	0.009191
HF0698	55	15464	0	0	1397.496	145.149	-3.974	3.06708	0.09754	0	0	2.70E-06	5.54E-05
HF0715	56	11149	2628	711	1007.546	104.6473	7.426064	0.567652	1	1	1	0.00259	0.06492
HF0718	57	14181	9051	173	1281.55	133.1065	5.867289	0.555944	1	1	1	0.005511	0.009811
HF0720	58	376	0	0	33.97947	3.529231	-1.52275	4.262012	0.360439	0	0	0.185514	1.851381
HF0724	59	24745	0	0	2236.229	232.2628	-4.43805	3.057274	0.073301	0	0	1.03E-06	2.13E-05
HF0732	60	5972	0	0	539.6952	56.0547	-3.3864	3.375602	0.157882	0	0	4.02E-05	6.89E-04
HF0741	61	3728	1539	13	336.9028	34.99195	4.681491	0.609397	1	1	1	0.013557	0.011519
HF0752	62	7103	5466	304	641.9047	66.67056	7.086403	0.562447	1	1	1	0.013267	0.068495
HF0773	63	46103	0	0	4166.371	432.7344	-5.03492	3.052511	0.049529	0	0	2.91E-07	6.24E-06
HF0777	64	33538	444	149	3030.86	314.7962	4.73714	0.551161	1	1	1	4.84E-05	0.001503
HF0854	65	20865	1587	86	1885.589	195.8442	4.734382	0.551991	1	1	1	4.46E-04	0.002256
HF0859	66	5596	0	0	505.7157	52.52547	-3.47441	3.541477	0.163281	0	0	7.67E-05	0.001224
HF0860	67	19756	0	0	1785.368	185.4348	-4.36954	3.19477	0.0857	0	0	2.27E-06	4.38E-05
HF0873	68	14170	0	0	1280.556	133.0032	-3.94998	3.115039	0.102392	0	0	3.61E-06	7.18E-05
HF0883	69	30893	2279	9	2791.829	289.9695	2.264354	0.645322	0.999775	1	1	2.92E-04	1.24E-04
HF0887	70	60614	444	149	5477.744	568.9383	4.142503	0.551419	1	1	1	1.48E-05	4.60E-04
HF0889	71	71727	0	2	6482.036	673.2478	-0.95532	1.019994	0.174483	0	0	3.24E-08	3.65E-06
HF0919	72	22924	0	0	2071.663	215.1705	-4.37569	3.067758	0.076884	0	0	1.23E-06	2.52E-05
HF0921	73	35279	511	14	3188.196	331.1376	2.429452	0.609188	0.999967	1	1	5.03E-05	1.35E-04
HF0926	74	51320	0	0	4637.836	481.7025	-4.95244	2.896544	0.043654	0	0	1.66E-07	3.82E-06
HF0940	75	65003	11	31	5874.382	610.1346	2.383656	0.602016	0.999962	1	1	3.64E-07	8.16E-05
HF0948	76	10195	0	0	921.3316	95.69285	-3.67021	3.157139	0.122514	0	0	7.76E-06	1.50E-04
HF0957	77	23314	0	0	2106.908	218.8311	-4.33785	3.031036	0.076194	0	0	1.09E-06	2.30E-05
HF0966	78	39980	0	0	3613.03	375.2624	-4.86145	3.014285	0.053394	0	0	3.55E-07	7.60E-06
HF0975	79	8327	7821	192	752.5187	78.15933	6.497357	0.559771	1	1	1	0.013811	0.03155
HF0998	80	41002	0	0	3705.389	384.8552	-4.84675	2.98559	0.052254	0	0	3.16E-07	6.91E-06
HF1015	81	71146	1665	206	6429.53	667.7944	4.345843	0.546213	1	1	1	4.03E-05	4.63E-04
HF1032	82	51046	11617	551	4613.075	479.1307	5.707266	0.547743	1	1	1	5.46E-04	0.002403
HF1063	83	42335	0	0	3825.853	397.367	-4.8976	2.999417	0.05125	0	0	3.06E-07	6.63E-06
HF1065	84	22892	0	0	2068.771	214.8701	-4.42564	3.112047	0.077499	0	0	1.37E-06	2.76E-05
HF1092	85	45814	0	0	4140.254	430.0218	-4.99597	3.016824	0.048857	0	0	2.72E-07	5.86E-06
HF1113	86	54227	3938	556	4900.544	508.9884	5.613341	0.546958	1	1	1	1.64E-04	0.002147
HF1125	87	37541	22309	679	3392.615	352.3693	6.243508	0.55311	1	1	1	0.001938	0.005474
HF1130	88	31671	1852	11	2862.138	297.272	2.396934	0.629462	0.99993	1	1	2.26E-04	1.39E-04
HF1147	89	16454	0	0	1486.963	154.4414	-4.15513	3.1661	0.094696	0	0	3.01E-06	5.91E-05
HF1165	90	28965	0	0	2617.594	271.8728	-4.59434	3.057802	0.066484	0	0	7.49E-07	1.56E-05
HF1166	91	82312	11399	1041	7438.612	772.6013	5.841301	0.547249	1	1	1	2.06E-04	0.001745
HF1180	92	21313	0	0	1926.076	200.0492	-4.35811	3.116744	0.081013	0	0	1.59E-06	3.21E-05
HF1207	93	22757	0	0	2056.571	213.603	-4.33834	3.046855	0.077241	0	0	1.18E-06	2.47E-05
HF1209	94	19238	0	0	1738.556	180.5727	-4.22599	3.090794	0.085768	0	0	1.84E-06	3.74E-05
HF1234	95	43021	2139	78	3887.848	403.806	3.926637	0.554702	1	1	1	1.42E-04	4.83E-04
HF1246	96	4118	0	0	372.1475	38.65259	-3.05512	3.404349	0.184749	0	0	9.23E-05	0.00153
HF1249	97	27547	0	0	2489.448	258.5631	-4.49606	3.022308	0.068425	0	0	7.61E-07	1.62E-05
HF1250	98	38980	3224	54	3522.659	365.8761	3.690954	0.5637	1	1	1	2.60E-04	4.10E-04
HF1269	99	14944	0	0	1350.503	140.2682	-4.08077	3.184179	0.099996	0	0	3.83E-06	7.42E-05
HF1276	100	20124	6090	864	1818.625	188.889	7.05393	0.560237	1	1	1	0.001842	0.024221
HF1281	101	43229	0	0	3906.645	405.7583	-4.9062	2.992752	0.050569	0	0	2.88E-07	6.30E-06
HF1312	102	38659	1382	17	3493.65	362.8632	2.572956	0.60272	0.99999	1	1	1.13E-04	1.37E-04
HF1313	103	32937	2093	190	2976.547	309.155	5.052417	0.546933	1	1	1	2.36E-04	0.001992
HF1322	104	35682	0	0	3224.616	334.9203	-4.69437	2.976214	0.057364	0	0	4.08E-07	8.96E-06
HF1340	105	39080	0	0	3531.696	366.8148	-4.79151	2.980615	0.053966	0	0	3.43E-07	7.53E-06
HF1354	106	26873	3324	39	2428.538	252.2368	3.756109	0.571306	1	1	1	5.64E-04	6.29E-04
HF1375	107	16880	22896	957	1525.461	158.44	7.383329	0.565794	1	1	1	0.009839	0.038145
HF1390	108	7587	0	0	685.6443	71.2135	-3.27252	3.104897	0.145945	0	0	1.24E-05	2.54E-04
HF1405	109	32289	0	0	2917.987	303.0727	-4.68381	3.045554	0.062035	0	0	5.84E-07	1.23E-05
HF1408	110	14010	4879	4387	1266.097	131.5014	8.982398	0.602419	1	1	1	0.003044	0.253665
HF1419	111	24695	0	0	2231.71	231.7935	-4.25384	2.936353	0.073713	0	0	7.84E-07	1.78E-05
HF1421	112	18583	3771	155	1679.363	174.4247	5.456662	0.55075	1	1	1	0.001337	0.005115
HF1431	113	3748	1817	92	338.7102	35.17968	6.534008	0.562344	1	1	1	0.015841	0.074762
HF1438	114	2355	0	0	212.8235	22.10463	-2.7946	3.716159	0.226021	0	0	7.34E-04	0.010557
HF1475	115	47838	0	0	4323.165	449.0196	-4.95687	2.96027	0.04702	0	0	2.18E-07	4.91E-06
HF1483	116	35867	0	0	3241.334	336.6567	-4.82626	3.077949	0.058439	0	0	5.13E-07	1.07E-05
HF1492	117	28252	0	0	2553.16	265.1804	-4.6347	3.110939	0.068138	0	0	8.98E-07	1.81E-05
HF1515	118	40545	0	0	3664.089	380.5656	-4.85423	2.998217	0.05272	0	0	3.33E-07	7.20E-06
HF1524	119	29019	0	0	2622.474	272.3797	-4.65763	3.107472	0.066956	0	0	8.44E-07	1.71E-05
HF1530	120	29079	0	0	2627.896	272.9429	-4.63547	3.08706	0.066603	0	0	7.98E-07	1.63E-05
HF1532	121	23806	760	5	2151.37	223.4491	1.952319	0.689372	0.997687	1	1	1.64E-04	1.24E-04
HF1533	122	56800	0	0	5133.069	533.1392	-5.17223	2.983992	0.041519	0	0	1.64E-07	3.60E-06
HF1547	123	63519	0	0	5740.271	596.2054	-5.21466	2.931992	0.037658	0	0	1.17E-07	2.65E-06

HF1583	124	52652	0	0	4758.21	494.205	-5.06672	2.961318	0.043543	0	0	1.81E-07	4.03E-06
HF1591	125	18565	2139	307	1677.736	174.2558	6.095215	0.55155	1	1	1	7.60E-04	0.010117
HF1595	126	6851	0	0	619.1312	64.30522	-3.27548	3.181337	0.1516	0	0	1.82E-05	3.57E-04
HF1600	127	44898	0	0	4057.474	421.424	-4.78022	2.878725	0.048404	0	0	2.09E-07	4.92E-06
HF1609	128	12997	11399	1041	1174.551	121.9931	7.697621	0.569232	1	1	1	0.008263	0.069968
HF1617	129	54737	810	579	4946.634	513.7753	5.583949	0.554648	1	1	1	3.32E-05	0.002192
HF1629	130	6620	0	0	598.2556	62.137	-3.40942	3.305402	0.151161	0	0	2.70E-05	4.82E-04
HF1643	131	40788	0	0	3686.05	382.8465	-4.83647	2.986569	0.05268	0	0	3.19E-07	7.03E-06
HF1647	132	35229	0	0	3183.678	330.6683	-4.80392	3.070109	0.058822	0	0	5.22E-07	1.08E-05
HF1667	133	25438	0	0	2298.856	238.7675	-4.54643	3.123654	0.072768	0	0	1.14E-06	2.28E-05
HF1688	134	8949	0	0	808.7295	83.99758	-3.62689	3.241519	0.131595	0	0	1.24E-05	2.33E-04
HF1697	135	13885	2897	67	1254.8	130.3281	4.929921	0.557954	1	1	1	0.00184	0.003991
HF1698	136	44836	7489	1167	4051.871	420.842	6.546739	0.554123	1	1	1	4.56E-04	0.00659
HF1722	137	20993	0	0	1897.157	197.0456	-4.33979	3.111055	0.081514	0	0	1.62E-06	3.26E-05
HF1726	138	10796	0	0	975.6446	101.334	-3.82019	3.229909	0.118453	0	0	8.32E-06	1.54E-04
HF1734	139	26861	0	0	2427.454	252.1241	-4.5283	3.064508	0.069749	0	0	8.86E-07	1.83E-05
HF1747	140	18869	4349	57	1705.209	177.1092	4.484511	0.562904	1	1	1	0.001496	0.001849
HF1751	141	10610	0	0	958.8356	99.58815	-3.80006	3.232219	0.119861	0	0	8.64E-06	1.61E-04
HF1752	142	8639	16907	2472	780.7145	81.08784	8.959156	0.593406	1	1	1	0.02774	0.375971
HF1754	143	13722	1374	261	1240.07	128.7982	6.222442	0.553928	1	1	1	8.94E-04	0.015739
HF1769	144	3752	3938	556	339.0717	35.21722	8.299438	0.580925	1	1	1	0.034258	0.448405
HF1785	145	39520	13988	328	3571.459	370.9447	5.469724	0.549898	1	1	1	0.001097	0.002389
HF1805	146	9776	0	0	883.4662	91.76001	-3.92381	3.430898	0.12638	0	0	1.82E-05	3.05E-04
HF1831	147	8783	16436	1408	793.7279	82.43946	8.398145	0.581669	1	1	1	0.02609	0.207216
HF1856	148	13211	0	0	1193.89	124.0018	-3.9462	3.174634	0.106926	0	0	4.79E-06	9.32E-05
HF1868	149	20616	0	0	1863.087	193.507	-4.34576	3.145014	0.083518	0	0	1.82E-06	3.67E-05
HF1882	150	22673	0	0	2048.98	212.8145	-4.40465	3.108298	0.078232	0	0	1.37E-06	2.80E-05
HF1903	151	5663	0	0	511.7706	53.15435	-3.53128	3.57479	0.161618	0	0	8.49E-05	0.001295
HF1927	152	16106	0	0	1455.514	151.175	-4.0692	3.114498	0.095686	0	0	2.77E-06	5.60E-05
HF1934	153	21637	2482	737	1955.356	203.0904	6.792724	0.559486	1	1	1	6.49E-04	0.017867
HF1965	154	65633	7301	802	5931.315	616.048	5.800647	0.547306	1	1	1	2.08E-04	0.002114
HF1970	155	48979	1847	556	4426.278	459.7293	5.687353	0.549903	1	1	1	9.43E-05	0.002631
HF1994	156	37474	0	0	3386.56	351.7405	-4.73776	2.970528	0.055365	0	0	3.66E-07	8.04E-06
HF1999	157	16857	0	0	1523.383	158.2241	-4.20468	3.185373	0.093418	0	0	3.03E-06	5.86E-05
HF2009	158	44937	1801	50	4060.999	421.7901	3.449874	0.563544	1	1	1	1.09E-04	2.85E-04
HF2019	159	28213	0	0	2549.635	264.8144	-4.55869	3.051923	0.067626	0	0	7.77E-07	1.63E-05
HF2027	160	8475	0	0	765.8936	79.5485	-3.68756	3.325089	0.133713	0	0	1.74E-05	3.05E-04
HF2028	161	8613	1821	39	778.3648	80.8438	4.872606	0.566836	1	1	1	0.003006	0.006092
HF2029	162	28907	14835	4585	2612.353	271.3284	8.338194	0.58281	1	1	1	0.002174	0.062278
HF2057	163	12103	0	0	1093.759	113.6018	-3.9188	3.225022	0.11216	0	0	6.49E-06	1.23E-04
HF2067	164	42088	17764	2371	3803.532	395.0486	7.328746	0.563411	1	1	1	0.001228	0.015194
HF2099	165	7239	5434	387	654.1952	67.94709	7.300905	0.564571	1	1	1	0.012698	0.083906
HF2108	166	3020	0	0	272.9202	28.34648	-2.88977	3.544548	0.207459	0	0	2.58E-04	0.003973
HF2124	167	29361	15823	629	2653.381	275.5898	6.4046	0.55391	1	1	1	0.002247	0.00829
HF2140	168	4459	0	0	402.964	41.8533	-3.17132	3.444868	0.178632	0	0	8.79E-05	0.001443
HF2145	169	44637	2628	711	4033.887	418.9742	6.032402	0.551513	1	1	1	1.62E-04	0.00405
HF2152	170	18498	0	0	1671.681	173.6269	-4.28184	3.176187	0.088812	0	0	2.45E-06	4.80E-05
HF2167	171	17234	0	0	1557.453	161.7627	-4.19101	3.157524	0.092203	0	0	2.69E-06	5.30E-05
HF2186	172	1735	0	0	156.7936	16.28515	-2.45146	3.653938	0.251139	0	0	0.001098	0.015755
HF2190	173	9064	0	0	819.1221	85.077	-3.6714	3.259839	0.130029	0	0	1.27E-05	2.34E-04
HF2205	174	28991	180	0	2619.944	272.1169	-0.08419	1.045802	0.467917	0	0	2.62E-05	1.83E-05
HF2206	175	20149	13408	3698	1820.884	189.1236	8.488476	0.585338	1	1	1	0.004044	0.103387
HF2211	176	7013	0	0	633.7713	65.82579	-3.53913	3.375976	0.147244	0	0	2.92E-05	5.07E-04
HF2222	177	47543	7492	1818	4296.505	446.2506	6.916701	0.559503	1	1	1	4.06E-04	0.009129
HF2228	178	29883	0	0	2700.555	280.4894	-4.62167	3.061296	0.065559	0	0	7.06E-07	1.48E-05
HF2249	179	12609	0	0	1139.487	118.3513	-3.99976	3.254229	0.109518	0	0	6.50E-06	1.20E-04
HF2278	180	12616	1861	724	1140.12	118.417	7.307036	0.567553	1	1	1	0.001432	0.05162
HF2280	181	3782	0	0	341.7829	35.49881	-3.05156	3.485928	0.19068	0	0	1.38E-04	0.002205
HF2292	182	5048	0	0	456.1925	47.38181	-3.26226	3.414909	0.169713	0	0	6.29E-05	0.001052
HF2305	183	2347	19599	197	212.1006	22.02954	7.832276	0.583932	1	1	1	0.435659	0.407978
HF2329	184	32747	14033	580	2959.377	307.3716	6.210822	0.551977	1	1	1	0.001602	0.006145
HF2330	185	22447	11399	1041	2028.556	210.6932	7.148556	0.560992	1	1	1	0.00277	0.023457
HF2353	186	10906	11399	1041	985.5854	102.3665	7.874321	0.572233	1	1	1	0.011735	0.09937
HF2375	187	7819	0	0	706.6103	73.39111	-3.58754	3.315494	0.139614	0	0	1.98E-05	3.53E-04
HF2376	188	20273	1861	724	1832.09	190.2875	6.829611	0.561373	1	1	1	5.55E-04	0.019991
HF2400	189	34572	0	0	3124.304	324.5015	-4.73698	3.037451	0.059436	0	0	4.98E-07	1.06E-05
HF2410	190	11325	0	0	1023.451	106.2993	-3.82645	3.203121	0.116121	0	0	7.01E-06	1.34E-04
HF2428	191	12476	0	0	1127.468	117.1029	-4.0049	3.270975	0.110405	0	0	6.95E-06	1.27E-04
HF2445	192	107529	334	724	9717.496	1009.294	5.086082	0.563928	1	1	1	3.55E-06	7.10E-04
HF2448	193	30891	0	0	2791.648	289.9507	-4.53717	2.972221	0.063439	0	0	5.40E-07	1.19E-05
HF2457	194	18179	2482	272	1642.853	170.6327	6.002869	0.550861	1	1	1	9.20E-04	0.009351
HF2475	195	9079	0	0	820.4777	85.21779	-3.7339	3.308492	0.129538	0	0	1.45E-05	2.58E-04
HF2481	196	12101	1665	206	1093.579	113.583	6.12819	0.553024	1	1	1	0.001393	0.015986
HF2490	197	17047	13471	1315	1540.553	160.0075	7.657097	0.568441	1	1	1	0.005676	0.051373
HF2494	198	17538	0	0	1584.925	164.6161	-4.17218	3.142471	0.092143	0	0	2.48E-06	5.07E-05
HF2501	199	4903	0	0	443.0887	46.0208	-3.20935	3.390867	0.171955	0	0	6.24E-05	0.001055
HF2508	200	9958	0	0	899.9137	93.46831	-3.76225	3.246143	0.12323	0	0	1.02E-05	1.87E-04

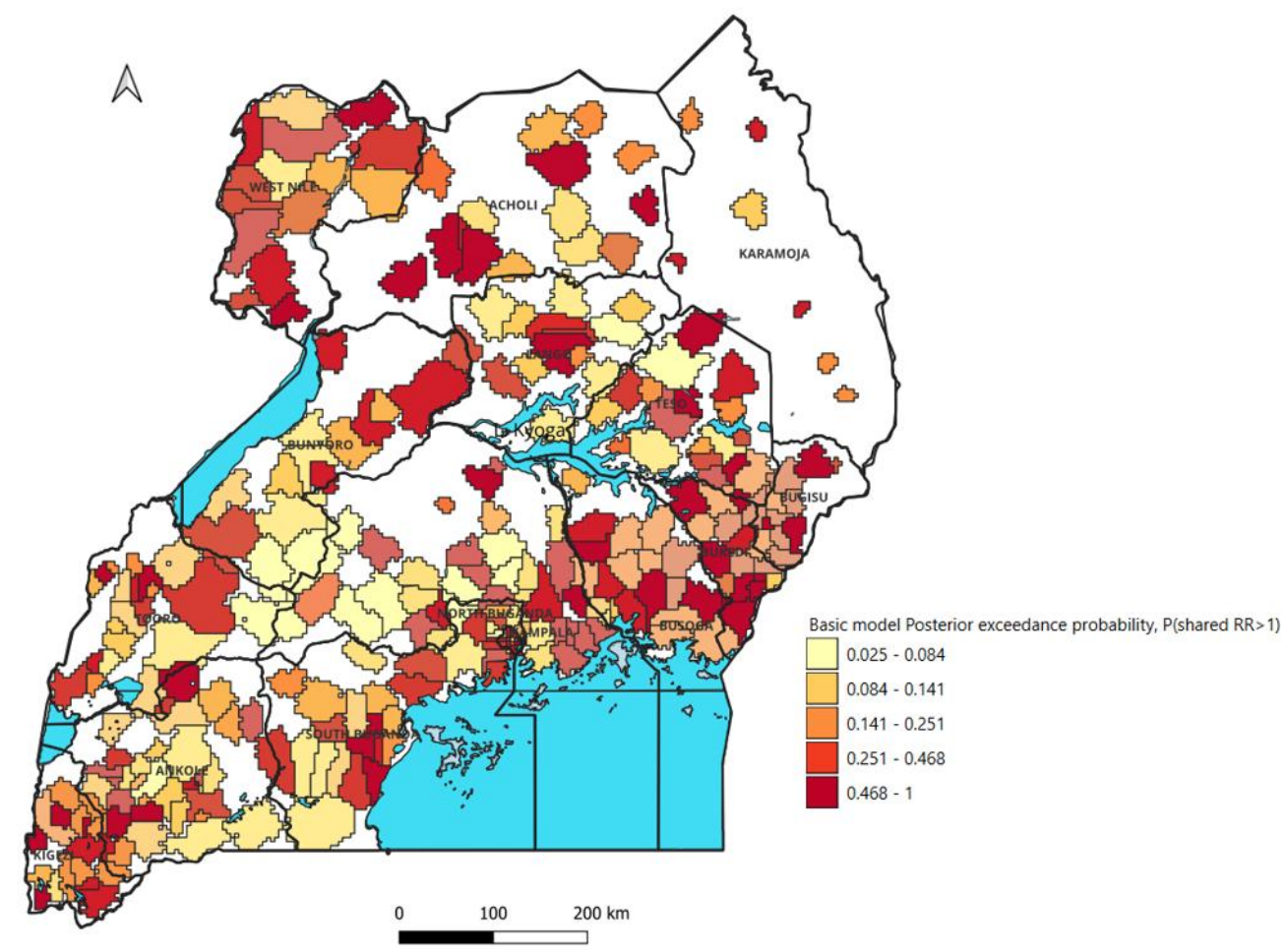
HF2512	201	15960	745	43	1442.32	149.8046	4.305261	0.561441	1	1	1	3.58E-04	0.001939
HF2513	202	6376	1281	16	576.2051	59.84675	4.321528	0.596966	1	1	1	0.003858	0.004745
HF2521	203	10825	0	0	978.2653	101.6062	-3.85339	3.255022	0.11824	0	0	8.84E-06	1.61E-04
HF2522	204	11275	2092	100	1018.932	105.83	5.510131	0.553452	1	1	1	0.002015	0.008977
HF2528	205	5756	0	0	520.1751	54.02727	-3.37671	3.388501	0.159499	0	0	4.52E-05	7.56E-04
HF2565	206	33385	0	0	3017.034	313.3601	-4.70929	3.043108	0.060868	0	0	5.42E-07	1.15E-05
HF2566	207	18722	14128	1777	1691.925	175.7294	7.856403	0.571799	1	1	1	0.004936	0.05755
HF2570	208	25853	0	0	2336.36	242.6628	-4.59523	3.164295	0.073221	0	0	1.22E-06	2.44E-05
HF2578	209	19268	0	0	1741.267	180.8543	-4.52995	3.369181	0.089389	0	0	3.99E-06	6.94E-05
HF2586	210	27097	0	0	2448.781	254.3393	-4.75712	3.265991	0.072619	0	0	1.48E-06	2.77E-05
HF2594	211	9482	4306	137	856.8972	89.00045	6.017312	0.555895	1	1	1	0.005865	0.01738
HF2601	212	29704	4936	1	2684.378	278.8093	1.036962	0.834634	0.892958	1	0	6.85E-04	3.97E-05
HF2622	213	33949	0	0	3068.003	318.6539	-4.73312	3.047954	0.060225	0	0	5.31E-07	1.13E-05
HF2652	214	49953	0	0	4514.299	468.8715	-5.0303	2.979999	0.045703	0	0	2.09E-07	4.64E-06
HF2673	215	3990	0	0	360.58	37.45115	-3.03838	3.439107	0.188489	0	0	1.07E-04	0.001806
HF2677	216	39320	0	0	3553.385	369.0675	-4.8392	3.016925	0.054355	0	0	3.67E-07	7.95E-06
HF2695	217	13838	0	0	1250.553	129.887	-4.01933	3.195218	0.10421	0	0	4.60E-06	8.83E-05
HF2705	218	36682	1268	34	3314.986	344.3065	3.269323	0.572075	1	1	1	1.15E-04	2.93E-04
HF2708	219	10418	0	19	941.4844	97.78599	3.618598	0.635632	1	1	1	2.29E-06	0.001867
HF2734	220	18187	0	0	1643.576	170.7078	-4.22086	3.14558	0.089824	0	0	2.34E-06	4.70E-05
HF2749	221	18606	0	0	1681.442	174.6406	-4.24727	3.139199	0.088031	0	0	2.21E-06	4.39E-05
HF2753	222	22905	0	0	2069.946	214.9921	-4.37957	3.072202	0.076999	0	0	1.24E-06	2.55E-05
HF2761	223	3648	0	0	329.6732	34.24105	-2.97041	3.441417	0.194032	0	0	1.31E-04	0.002132
HF2778	224	33303	0	0	3009.623	312.5904	-4.68168	3.015279	0.060253	0	0	5.13E-07	1.09E-05
HF2791	225	8402	3771	31	759.2966	78.8633	4.71491	0.578166	1	1	1	0.00654	0.005165
HF2801	226	15455	16513	179	1396.683	145.0645	5.837842	0.559525	1	1	1	0.008465	0.008554
HF2806	227	16826	9429	1200	1520.581	157.9331	7.568183	0.567275	1	1	1	0.004078	0.048117
HF2819	228	52231	0	0	4720.164	490.2534	-5.10472	3.014803	0.045207	0	0	2.06E-07	4.58E-06
HF2830	229	30916	0	0	2793.908	290.1854	-4.6532	3.060162	0.064183	0	0	6.58E-07	1.39E-05
HF2836	230	13918	6342	142	1257.783	130.6379	5.683464	0.555137	1	1	1	0.004009	0.008367
HF2837	231	7996	1828	30	722.606	75.05248	4.70096	0.57431	1	1	1	0.003501	0.005489
HF2843	232	27369	0	0	2473.362	256.8924	-4.41953	2.986995	0.069491	0	0	7.06E-07	1.58E-05
HF2874	233	10419	2844	100	941.5747	97.79537	5.602668	0.554896	1	1	1	0.003208	0.010523
HF2879	234	17100	0	0	1545.343	160.5049	-4.19527	3.163826	0.092418	0	0	2.78E-06	5.44E-05
HF2886	235	24853	0	0	2245.989	233.2765	-4.45206	3.07306	0.073705	0	0	1.05E-06	2.19E-05
HF2891	236	53557	5434	387	4839.996	502.6996	5.288709	0.54559	1	1	1	2.32E-04	0.001533
HF2900	237	11614	0	0	1049.568	109.0119	-3.90043	3.237109	0.114119	0	0	7.31E-06	1.35E-04
HF2916	238	2703	0	0	244.2726	25.37104	-2.7776	3.551122	0.217056	0	0	3.27E-04	0.005076
HF2927	239	103509	5434	387	9354.205	971.5617	4.623224	0.543946	1	1	1	6.21E-05	4.10E-04
HF2931	240	72717	0	0	6571.503	682.5402	-5.51753	3.067435	0.03603	0	0	1.23E-07	2.58E-06
HF2932	241	47889	0	0	4327.774	449.4983	-5.16927	3.118512	0.048698	0	0	3.22E-07	6.47E-06
HF2937	242	36257	0	0	3276.579	340.3174	-4.96548	3.182678	0.059361	0	0	6.63E-07	1.28E-05
HF2983	243	12289	0	0	1110.568	115.3477	-3.93047	3.222009	0.111255	0	0	6.24E-06	1.18E-04
HF2984	244	4819	1527	4	435.4975	45.23235	3.447303	0.70378	1	1	1	0.008048	0.002665
HF2991	245	4409	0	0	398.4454	41.38399	-3.1707	3.487125	0.181607	0	0	1.00E-04	0.001693
HF3019	246	45503	11157	727	4112.148	427.1027	6.090891	0.549757	1	1	1	6.60E-04	0.003988
HF3033	247	10245	0	0	925.8502	96.16216	-3.80375	3.26784	0.122213	0	0	1.02E-05	1.87E-04
HF3034	248	23176	2628	711	2094.437	217.5358	6.691295	0.557907	1	1	1	5.99E-04	0.015024
HF3036	249	16320	0	0	1474.854	153.1837	-4.16185	3.177404	0.095127	0	0	3.16E-06	6.15E-05
HF3047	250	70531	0	0	6373.952	662.0218	-5.47634	3.056718	0.0366	0	0	1.27E-07	2.68E-06
HF3083	251	36153	0	0	3267.18	339.3412	-4.95408	3.182885	0.059797	0	0	6.63E-07	1.30E-05
HF3085	252	84598	10855	217	7645.2	794.0582	4.295008	0.549448	1	1	1	1.86E-04	3.45E-04
HF3103	253	17367	0	0	1569.472	163.0111	-4.23348	3.181663	0.091662	0	0	2.83E-06	5.46E-05
HF3109	254	26630	4401	661	2406.578	249.9559	6.500356	0.554081	1	1	1	7.60E-04	0.010582
HF3118	255	10598	0	0	957.7511	99.47551	-3.72577	3.173864	0.120219	0	0	7.46E-06	1.44E-04
HF3137	256	13976	0	0	1263.024	131.1823	-3.99271	3.161283	0.103294	0	0	4.15E-06	8.07E-05
HF3141	257	439	1447	424	39.67284	4.120565	10.1554	0.625522	1	1	1	0.919924	24.96335
HF3145	258	13500	0	0	1220.008	126.7144	-3.9084	3.13221	0.106051	0	0	4.12E-06	8.28E-05
HF3146	259	2071	78	8	187.1582	19.43893	4.626531	0.637701	1	1	1	0.002243	0.02206
HF3149	260	5362	189	295	484.5689	50.32909	7.208068	0.579138	1	1	1	8.11E-04	0.116223
HF3153	261	8864	0	0	801.0479	83.19975	-3.57557	3.19984	0.131907	0	0	1.14E-05	2.17E-04
HF3171	262	27126	3398	101	2451.402	254.6115	4.657928	0.552256	1	1	1	5.65E-04	0.001569
HF3186	263	17676	0	0	1597.397	165.9114	-4.17786	3.120906	0.090339	0	0	2.34E-06	4.68E-05
HF3188	264	43122	0	0	3896.975	404.754	-4.89747	2.984904	0.050425	0	0	2.85E-07	6.23E-06
HF3207	265	26197	0	0	2367.447	245.8917	-4.507	3.06627	0.070799	0	0	9.35E-07	1.93E-05
HF3214	266	6196	0	0	559.9383	58.15722	-3.39268	3.350866	0.155654	0	0	3.48E-05	6.09E-04
HF3222	267	11456	15823	629	1035.289	107.5289	7.350469	0.565666	1	1	1	0.014763	0.054449
HF3226	268	31784	0	0	2872.35	298.3327	-4.63434	3.017186	0.062272	0	0	5.65E-07	1.21E-05
HF3236	269	24292	957	2	2195.291	228.0109	1.323789	0.785157	0.954104	1	1	1.98E-04	6.75E-05
HF3241	270	12615	0	0	1140.029	118.4076	-3.94295	3.203187	0.109171	0	0	5.67E-06	1.07E-04
HF3255	271	21068	0	0	1903.935	197.7496	-4.35056	3.125101	0.081941	0	0	1.66E-06	3.36E-05
HF3264	272	14470	0	0	1307.667	135.8191	-4.03109	3.168764	0.101663	0	0	3.93E-06	7.67E-05
HF3271	273	16976	0	0	1534.137	159.341	-4.17608	3.147566	0.092293	0	0	2.72E-06	5.31E-05
HF3291	274	12078	0	0	1091.5	113.3672	-3.9291	3.230678	0.111958	0	0	6.64E-06	1.24E-04
HF3308	275	26761	672	9	2418.416	251.1855	2.32254	0.637855	0.999864	1	1	1.15E-04	1.59E-04
HF3330	276	15918	0	0	1438.524	149.4104	-4.04787	3.111829	0.096663	0	0	2.81E-06	5.74E-05
HF3331	277	5154	828	4	465.7718	48.37675	3.323101	0.703059	0.999999	1	1	0.003815	0.002256

HF3339	278	29106	0	0	2630.336	273.1963	-4.60427	3.058105	0.066086	0	0	7.44E-07	1.54E-05
HF3353	279	16796	2124	5	1517.87	157.6515	2.393201	0.689605	0.99974	1	1	9.22E-04	2.62E-04
HF3362	280	3052	0	0	275.8121	28.64684	-2.91161	3.548774	0.205978	0	0	2.57E-04	0.003918

Supplementary Figure S3. Baseline Bayesian shared-component model outputs for malaria and SCD cases.



(A) Posterior mean shared latent spatial effect for malaria and sickle cell disease across health facility catchments in Uganda.



(B) Posterior exceedance probability that the shared relative risk exceeds 1 across health facility catchments in Uganda.

Supplementary Table S4. Catchment-level posterior summaries from the extended Bayesian shared-component model. Catchment-specific posterior summaries from the extended shared-component model, including the shared spatial effect, disease-specific residual spatial effects, exceedance probability, hotspot status, and fitted relative risks.

Catchment id	Area index	Catchment population	Malaria spatial mean	Malaria spatial_sd	Scd spatial mean	Scd spatial_sd	Shared mean	Shared sd	Shared q025	Shared q50	Shared q975	Exceed prob_rr gt1	Hotspot 080	Hotspot 090	rr_malaria	rr_scd
HF0002	1	20563	0.003717	0.171313	-2.36E-04	0.164184	6.240171	1.523029	3.266868	6.233874	9.249132	0.999979	1	1	8.52E-04	0.019337

HF0007	2	9446	0.00549	0.182269	-0.00165	0.175666	6.907144	1.529013	3.923671	6.9003	9.929323	0.999997	1	1	0.003607	0.090488
HF0008	3	20547	0.008481	0.168641	-0.00605	0.166797	4.310788	1.507101	1.36363	4.306423	7.282671	0.997884	1	1	0.002358	7.20E-04
HF0019	4	17421	0.006775	0.172223	-0.00469	0.171119	3.507559	1.497674	0.578878	3.503174	6.461168	0.99041	1	1	2.51E-04	5.87E-04
HF0022	5	26312	0.00136	0.144905	0.001076	0.140975	4.440116	1.459837	1.57813	4.438454	7.311531	0.998823	1	1	6.67E-04	0.003268
HF0039	6	30712	-0.01115	0.159938	0.010098	0.157933	-3.21034	2.403712	-7.93744	-3.20672	1.496164	0.090844	0	0	1.94E-07	1.82E-05
HF0052	7	36720	0.017496	0.199022	-0.01548	0.193616	4.155084	1.539683	1.141127	4.151504	7.189356	0.996519	1	1	1.90E-04	8.46E-04
HF0083	8	8716	0.033692	0.302709	-0.03643	0.293797	-5.34855	5.568176	-16.291	-5.34415	5.56881	0.168387	0	0	0.061329	0.744236
HF0092	9	16039	-0.01626	0.238238	0.014687	0.235271	-4.65719	3.673249	-11.8826	-4.6517	2.536983	0.102423	0	0	1.43E-05	7.07E-04
HF0093	10	10903	-0.01362	0.187743	0.01732	0.185275	6.17624	1.541439	3.163332	6.171081	9.218325	0.999969	1	1	0.002359	0.035268
HF0103	11	17771	-0.01943	0.231381	0.018722	0.227362	-3.35179	2.678334	-8.61756	-3.34826	1.893941	0.105386	0	0	8.82E-07	8.10E-05
HF0118	12	35945	0.004527	0.14652	-0.00159	0.140387	4.863159	1.437168	2.047574	4.860878	7.691682	0.999643	1	1	2.49E-04	0.006248
HF0202	13	21561	0.003584	0.142527	-0.00211	0.138229	1.852916	1.396556	-0.88331	1.850881	4.600745	0.907708	1	1	6.23E-06	2.32E-04
HF0203	14	7544	0.005498	0.1456	-0.00286	0.141729	4.310532	1.42036	1.533063	4.306546	7.110652	0.998797	1	1	0.002974	0.003518
HF0204	15	25095	0.005053	0.14927	-0.0017	0.14292	5.700223	1.44277	2.877743	5.696521	8.543709	0.999961	1	1	0.002246	0.010838
HF0216	16	37804	0.004429	0.153062	-0.00108	0.145943	5.678205	1.482447	2.774747	5.675458	8.597164	0.999936	1	1	3.02E-04	0.017402
HF0252	17	28974	0.004003	0.144826	-0.00112	0.138089	4.742541	1.434952	1.932915	4.739705	7.568272	0.999525	1	1	5.50E-04	0.003696
HF0253	18	5788	0.007958	0.172799	-0.00677	0.171228	1.617947	1.701662	-1.72642	1.619129	4.955683	0.829148	1	0	2.71E-04	3.53E-04
HF0262	19	13670	0.005317	0.15048	-0.0039	0.148603	1.889626	1.485782	-1.02306	1.888037	4.811384	0.898279	1	0	9.91E-05	1.68E-04
HF0263	20	10779	0.00188	0.157245	0.001293	0.151741	5.191131	1.499445	2.254514	5.188258	8.143986	0.999732	1	1	3.16E-04	0.03645
HF0264	21	8264	0.003092	0.148815	-0.00298	0.146602	-0.93494	1.939109	-4.74764	-0.9324	2.863331	0.314849	0	0	2.01E-06	1.61E-04
HF0281	22	94308	0.004655	0.152757	-0.00234	0.148055	3.576143	1.448051	0.738323	3.574121	6.425464	0.993237	1	1	1.42E-05	4.89E-04
HF0282	23	46581	0.005578	0.150379	-0.00264	0.145884	4.915051	1.453033	2.071602	4.911571	7.778235	0.999641	1	1	1.05E-04	0.003789
HF0283	24	102942	0.003022	0.151351	-0.0042	0.15103	-3.49521	1.933031	-7.3014	-3.49071	0.285381	0.035292	0	0	1.29E-08	1.05E-06
HF0290	25	7616	0.001575	0.193374	-0.00239	0.19068	-2.27559	2.630781	-7.4454	-2.27298	2.879363	0.193523	0	0	4.18E-06	4.12E-04
HF0298	26	25610	0.00359	0.143144	-0.00115	0.138069	4.400563	1.475038	1.514423	4.396883	7.307588	0.998574	1	1	7.67E-04	0.001239
HF0304	27	45037	0.003724	0.151329	-5.63E-04	0.144999	5.840457	1.479589	2.947985	5.835844	8.759066	0.99996	1	1	6.88E-04	0.005828
HF0316	28	15892	0.019853	0.180246	-0.02145	0.173577	-3.77988	2.703064	-9.09733	-3.77555	1.512967	0.081001	0	0	1.44E-06	9.48E-05
HF0327	29	25071	0.036049	0.24378	-0.0347	0.23671	4.207371	1.55084	1.168346	4.204901	7.260354	0.996666	1	1	7.35E-04	0.001184
HF0354	30	22487	0.036766	0.266938	-0.0398	0.260405	-4.21567	3.051085	-10.2118	-4.21258	1.76282	0.083533	0	0	1.15E-06	9.76E-05
HF0362	31	28783	0.016379	0.17892	-0.01413	0.173708	4.608081	1.522688	1.624023	4.605792	7.605086	0.998762	1	1	9.32E-04	0.002034
HF0376	32	11582	0.006566	0.15598	-0.00753	0.154893	-2.74144	2.43812	-7.53526	-2.73818	2.033803	0.13042	0	0	1.59E-06	1.31E-04
HF0401	33	29137	0.012511	0.23361	-0.01429	0.231431	-4.05165	2.813701	-9.58274	-4.04825	1.460109	0.074938	0	0	4.07E-07	3.82E-05
HF0402	34	14744	0.008391	0.202027	-0.01035	0.20169	-4.6777	3.679414	-11.9151	-4.67245	2.529811	0.101808	0	0	1.87E-05	8.21E-04
HF0411	35	33798	0.011573	0.204493	-0.01301	0.201741	-3.43031	2.3089	-7.97109	-3.42681	1.090542	0.06868	0	0	1.56E-07	1.34E-05
HF0429	36	24780	-0.0057	0.162455	0.004039	0.163033	-3.7945	2.722067	-9.14653	-3.79083	1.536618	0.081662	0	0	4.94E-07	4.44E-05
HF0438	37	38773	0.006655	0.174097	-0.00906	0.175503	-4.22525	2.616705	-9.37084	-4.22148	0.898858	0.053185	0	0	1.81E-07	1.52E-05
HF0446	38	27853	-0.00241	0.167365	2.75E-04	0.161956	-4.43167	2.980477	-10.2929	-4.42738	1.405115	0.068521	0	0	7.05E-07	5.37E-05
HF0465	39	25852	0.035384	0.264149	-0.03392	0.257059	4.907947	1.56333	1.853205	4.90239	7.994187	0.999154	1	1	0.001447	0.001117
HF0494	40	29288	-0.00638	0.161453	0.010134	0.162591	6.972955	1.531709	3.982322	6.966735	9.998726	0.999997	1	1	0.001341	0.034448
HF0499	41	13401	0.004431	0.181052	-0.00336	0.180587	2.513943	1.683902	-0.79595	2.515161	5.816862	0.932272	1	1	0.001806	8.22E-05
HF0505	42	8760	-0.00952	0.19712	0.008496	0.196995	-2.89938	2.807851	-8.41816	-2.89631	2.601912	0.150896	0	0	4.64E-06	4.07E-04
HF0506	43	17879	-0.00969	0.186699	0.01325	0.186219	6.274106	1.541121	3.265742	6.267591	9.319377	0.999977	1	1	0.00179	0.015428
HF0537	44	14188	0.016578	0.174208	-0.01794	0.168584	-3.00705	2.448972	-7.82293	-3.00357	1.788978	0.109745	0	0	1.09E-06	8.79E-05
HF0540	45	14995	0.022427	0.195479	-0.02379	0.188537	-3.15295	2.586582	-8.23725	-3.14997	1.914375	0.111429	0	0	1.12E-06	9.75E-05
HF0550	46	29277	0.022348	0.178565	-0.02007	0.171602	4.360054	1.479126	1.45921	4.358687	7.268643	0.998399	1	1	3.85E-04	0.002997
HF0560	47	10014	0.023517	0.172297	-0.02265	0.165878	1.798002	1.617273	-1.39009	1.802624	4.959731	0.866877	1	0	0.002984	1.27E-04

HF0583	48	44464	0.008455	0.156626	-0.00943	0.153801	-2.74986	1.952297	-6.5926	-2.74581	1.069834	0.079488	0	0	6.98E-08	5.62E-06
HF0588	49	27583	-0.02074	0.317257	0.023935	0.313553	4.567022	1.575231	1.483927	4.563085	7.672454	0.99813	1	1	1.72E-04	0.002349
HF0608	50	45130	0.025917	0.189558	-0.02828	0.184109	-3.82767	2.256013	-8.26553	-3.8239	0.588727	0.044881	0	0	8.63E-08	6.98E-06
HF0615	51	28603	0.034211	0.241955	-0.03759	0.236251	-4.91321	3.174794	-11.1543	-4.90952	1.306909	0.060863	0	0	1.22E-06	7.04E-05
HF0650	52	42103	0.044752	0.26453	-0.04679	0.258703	-0.5368	1.731398	-3.95931	-0.52871	2.839682	0.378267	0	0	1.97E-05	6.70E-06
HF0678	53	45197	0.002795	0.140752	-0.00384	0.140487	-3.15443	2.023751	-7.13888	-3.14985	0.804029	0.059533	0	0	7.06E-08	5.64E-06
HF0680	54	49954	0.003034	0.131602	-1.45E-04	0.125132	4.917712	1.460591	2.052806	4.916548	7.789173	0.99962	1	1	1.00E-04	0.009193
HF0698	55	15464	0.003945	0.159613	-0.00461	0.159918	-2.33885	2.217424	-6.69992	-2.33545	2.002895	0.145767	0	0	6.72E-07	5.76E-05
HF0715	56	11149	0.013983	0.214815	-0.01065	0.210749	6.333204	1.554374	3.291482	6.329218	9.397368	0.999977	1	1	0.002589	0.064946
HF0718	57	14181	0.017107	0.21189	-0.01442	0.206631	5.455269	1.532188	2.453984	5.452484	8.472283	0.999815	1	1	0.005511	0.00979
HF0720	58	376	-8.57E-04	0.212116	-1.80E-04	0.206014	-2.15398	5.96545	-13.8737	-2.15089	9.548184	0.359021	0	0	131.7412	1538.327
HF0724	59	24745	0.016517	0.172669	-0.01756	0.167049	-2.33972	1.990434	-6.25546	-2.33632	1.556671	0.119901	0	0	2.30E-07	1.85E-05
HF0732	60	5972	-0.0016	0.212219	-4.38E-04	0.206102	-4.21213	4.798602	-13.6445	-4.20756	5.194208	0.190031	0	0	0.003779	0.136053
HF0741	61	3728	0.010346	0.215881	-0.01129	0.215744	-0.93711	1.123948	-3.14667	-0.93652	1.269107	0.202206	0	0	0.013552	0.009963
HF0752	62	7103	-0.01035	0.215881	0.011288	0.215744	0.937111	1.123948	-1.26911	0.936521	3.146673	0.797794	0	0	0.013264	0.068157
HF0773	63	46103	0.003566	0.27426	-0.00676	0.278878	-5.98452	4.143814	-14.1373	-5.97786	2.130123	0.07434	0	0	7.26E-06	3.14E-04
HF0777	64	33538	-0.00467	0.169353	0.006867	0.172015	3.992346	1.529996	0.996248	3.98917	7.006489	0.995465	1	1	4.84E-05	0.001507
HF0854	65	20865	-0.0025	0.135122	0.004537	0.137916	3.991321	1.501376	1.045031	3.990482	6.942343	0.996075	1	1	4.46E-04	0.002252
HF0859	66	5596	-0.0128	0.329217	0.01115	0.331426	-4.20506	4.564138	-13.183	-4.19875	4.736984	0.17844	0	0	0.002475	0.068981
HF0860	67	19756	-0.01031	0.247375	0.008582	0.248462	-4.19303	3.166452	-10.4174	-4.18933	2.010174	0.092718	0	0	1.97E-06	1.60E-04
HF0873	68	14170	0.00586	0.167757	-0.00698	0.167531	-3.16558	2.612709	-8.30236	-3.1622	1.951948	0.112831	0	0	1.32E-06	1.13E-04
HF0883	69	30893	0.012846	0.162505	-0.01139	0.157816	2.617983	1.50919	-0.34127	2.616426	5.586028	0.958603	1	1	2.92E-04	1.16E-04
HF0887	70	60614	0.004799	0.189857	-0.00266	0.185107	3.172036	1.535727	0.159333	3.170731	6.192158	0.980562	1	1	1.48E-05	4.61E-04
HF0889	71	71727	5.51E-04	0.141614	-7.83E-04	0.135688	-1.70347	1.587265	-4.83496	-1.69819	1.398112	0.141589	0	0	2.63E-08	5.26E-06
HF0919	72	22924	0.016602	0.164836	-0.01783	0.158861	-2.94993	2.13794	-7.15611	-2.94619	1.23503	0.083824	0	0	3.03E-07	2.37E-05
HF0921	73	35279	0.00577	0.158298	-0.00435	0.156675	1.954829	1.478013	-0.94851	1.955144	4.856386	0.907017	1	1	5.03E-05	1.32E-04
HF0926	74	51320	0.003009	0.135274	-0.00392	0.134589	-2.88891	1.821595	-6.47731	-2.88414	0.672417	0.056378	0	0	5.05E-08	3.86E-06
HF0940	75	65003	0.003011	0.153293	-0.00189	0.148325	1.060748	1.453921	-1.79427	1.060782	3.91567	0.767176	0	0	3.43E-07	8.39E-05
HF0948	76	10195	0.002274	0.172377	-0.00272	0.1694	-2.17788	2.376444	-6.84989	-2.17484	2.476822	0.179717	0	0	1.76E-06	1.59E-04
HF0957	77	23314	0.003054	0.190718	-0.00462	0.191303	-4.29397	3.070326	-10.3308	-4.28996	1.720007	0.080976	0	0	1.17E-06	9.28E-05
HF0966	78	39980	-0.00539	0.156266	-0.00242	0.153975	-16.1655	2.686271	-21.4696	-16.1553	-10.9197	8.84E-10	0	0	1.13E-10	1.20E-09
HF0975	79	8327	0.009224	0.294746	-0.00649	0.294457	6.142325	1.582879	3.044742	6.138277	9.262748	0.999948	1	1	0.013811	0.031505
HF0998	80	41002	-0.02369	0.220382	0.021896	0.212622	-3.56702	2.365235	-8.21766	-3.56374	1.064916	0.065764	0	0	1.09E-07	9.83E-06
HF1015	81	71146	-0.01412	0.275102	0.015187	0.271044	3.481987	1.547641	0.443821	3.481389	6.523525	0.987771	1	1	4.03E-05	4.63E-04
HF1032	82	51046	-0.01403	0.187756	0.016109	0.183202	4.476852	1.487005	1.556502	4.476907	7.396881	0.998697	1	1	5.46E-04	0.002401
HF1063	83	42335	-0.01518	0.200041	0.012996	0.195615	-4.44736	2.684407	-9.72682	-4.4433	0.809019	0.048786	0	0	1.74E-07	1.42E-05
HF1065	84	22892	-0.02133	0.245122	0.019077	0.236872	-4.00985	2.786312	-9.48723	-4.0065	1.448441	0.075058	0	0	7.25E-07	5.74E-05
HF1092	85	45814	-0.01616	0.262879	0.013266	0.25686	-4.66273	2.752246	-10.0746	-4.65893	0.727439	0.045118	0	0	1.70E-07	1.35E-05
HF1113	86	54227	-0.01449	0.206905	0.016668	0.202106	4.785121	1.520486	1.8067	4.782355	7.779212	0.999176	1	1	1.64E-04	0.002147
HF1125	87	37541	-0.01593	0.201118	0.018391	0.195181	5.43651	1.509611	2.477402	5.434541	8.40675	0.999842	1	1	0.001938	0.005471
HF1130	88	31671	-0.00961	0.2278	0.010144	0.225133	2.477404	1.535613	-0.53723	2.476996	5.494321	0.94666	1	1	2.26E-04	1.32E-04
HF1147	89	16454	-0.00187	0.212276	-5.67E-04	0.206188	-5.03768	4.512682	-13.9099	-5.03261	3.805689	0.132139	0	0	1.66E-04	0.007606
HF1165	90	28965	-0.01548	0.194124	0.013825	0.189507	-3.39062	2.418901	-8.1467	-3.3873	1.346568	0.080499	0	0	2.34E-07	2.10E-05
HF1166	91	82312	-0.00901	0.172577	0.01132	0.171416	4.801438	1.501955	1.855791	4.799998	7.75522	0.999305	1	1	2.06E-04	0.001744
HF1180	92	21313	-0.00865	0.200852	0.006721	0.201215	-4.14474	3.056556	-10.1513	-4.14171	1.844633	0.087547	0	0	1.32E-06	1.12E-04
HF1207	93	22757	-5.59E-05	0.157558	-9.80E-04	0.153024	-2.76825	2.224826	-7.14346	-2.76497	1.588265	0.106703	0	0	3.14E-07	2.69E-05

HF1209	94	19238	-0.00367	0.163552	0.002367	0.16167	-3.07509	2.462338	-7.91573	-3.072	1.748018	0.10586	0	0	5.58E-07	5.03E-05
HF1234	95	43021	0.00292	0.172519	-0.00102	0.16684	3.389077	1.496577	0.452824	3.388061	6.331087	0.98823	1	1	1.42E-04	4.81E-04
HF1246	96	4118	-0.00205	0.172167	9.76E-04	0.167198	-2.75958	3.042052	-8.73852	-2.75651	3.201916	0.182166	0	0	4.14E-05	0.002564
HF1249	97	27547	-0.00154	0.172364	-6.36E-05	0.16649	-3.91685	2.651401	-9.13057	-3.91311	1.275602	0.069801	0	0	3.77E-07	3.20E-05
HF1250	98	38980	-0.00523	0.169717	0.006968	0.171494	3.668589	1.51932	0.69157	3.666161	6.65937	0.992124	1	1	2.60E-04	4.08E-04
HF1269	99	14944	-0.00184	0.21227	-5.54E-04	0.206178	-4.95808	4.537577	-13.8791	-4.95305	3.934319	0.137269	0	0	2.21E-04	0.009921
HF1276	100	20124	0.002142	0.171315	0.001325	0.167404	7.053694	1.492548	4.144918	7.045814	10.00712	0.999999	1	1	0.001842	0.024231
HF1281	101	43229	-0.00552	0.201056	0.002868	0.198197	-5.72766	3.654187	-12.9152	-5.72235	1.42966	0.058508	0	0	2.03E-06	9.36E-05
HF1312	102	38659	0.004214	0.169998	-0.00266	0.167713	2.799761	1.499574	-0.13828	2.797348	5.751502	0.969052	1	1	1.13E-04	1.35E-04
HF1313	103	32937	0.002932	0.159497	-5.73E-04	0.153529	4.329547	1.473537	1.443032	4.327005	7.230484	0.998349	1	1	2.36E-04	0.001992
HF1322	104	35682	7.95E-04	0.170911	-0.00198	0.168063	-3.06235	2.248297	-7.48381	-3.05897	1.339898	0.086587	0	0	1.26E-07	1.12E-05
HF1340	105	39080	-0.00101	0.176496	-6.94E-04	0.171115	-3.95575	2.519543	-8.90937	-3.95241	0.978833	0.058204	0	0	1.49E-07	1.32E-05
HF1354	106	26873	-0.00754	0.308577	0.008489	0.308161	3.821992	1.560942	0.762711	3.819662	6.894462	0.992828	1	1	5.64E-04	6.23E-04
HF1375	107	16880	-0.03963	0.436059	0.043399	0.422491	7.308026	1.608999	4.16791	7.300818	10.48887	0.999997	1	1	0.009839	0.038143
HF1390	108	7587	-0.04449	0.399614	0.043498	0.384445	-1.83936	2.473172	-6.7	-1.83672	3.006308	0.228521	0	0	3.35E-06	3.29E-04
HF1405	109	32289	-0.02158	0.231504	0.018975	0.224937	-5.41591	3.633862	-12.566	-5.40995	1.700224	0.06806	0	0	3.47E-06	1.58E-04
HF1408	110	14010	-0.04758	0.306903	0.052309	0.297218	7.326053	1.52986	4.335369	7.32116	10.34414	0.999999	1	1	0.003044	0.253695
HF1419	111	24695	-0.04866	0.316773	0.048685	0.305662	-1.88756	2.016768	-5.85414	-1.88446	2.061476	0.174654	0	0	2.29E-07	1.92E-05
HF1421	112	18583	-0.04605	0.302945	0.049567	0.292903	5.167701	1.489408	2.255212	5.16334	8.104952	0.999739	1	1	0.001337	0.005111
HF1431	113	3748	-0.04361	0.359466	0.046512	0.345253	6.113565	1.568643	3.045646	6.10894	9.20764	0.999951	1	1	0.015841	0.0747
HF1438	114	2355	-0.00998	0.355624	0.008028	0.353559	-2.13825	3.451648	-8.91712	-2.13631	4.629515	0.267797	0	0	2.27E-04	0.022119
HF1475	115	47838	-1.18E-04	0.244651	-0.00195	0.239225	-4.83913	3.140151	-11.0097	-4.83604	1.313896	0.061652	0	0	2.68E-07	2.74E-05
HF1483	116	35867	-0.01526	0.27507	0.012379	0.270348	-4.57457	2.985353	-10.4427	-4.57112	1.273895	0.06272	0	0	4.00E-07	3.44E-05
HF1492	117	28252	-0.01343	0.243707	0.010482	0.238828	-4.63857	2.841004	-10.2261	-4.63442	0.925366	0.051264	0	0	6.45E-07	3.74E-05
HF1515	118	40545	-0.01251	0.18074	0.01066	0.177414	-3.78492	2.349663	-8.40672	-3.78107	0.815036	0.053608	0	0	1.16E-07	9.72E-06
HF1524	119	29019	-0.01526	0.24297	0.01195	0.236957	-5.49776	3.270959	-11.9349	-5.49215	0.907356	0.046403	0	0	2.22E-06	6.77E-05
HF1530	120	29079	-0.01402	0.22824	0.011312	0.222689	-4.36644	2.695628	-9.66901	-4.36213	0.911651	0.052635	0	0	4.30E-07	2.87E-05
HF1532	121	23806	-0.00676	0.255823	0.006901	0.254201	2.098718	1.571285	-0.98696	2.098627	5.184897	0.909171	1	1	1.64E-04	1.15E-04
HF1533	122	56800	-0.01335	0.232082	0.010485	0.226808	-4.50729	2.396204	-9.22057	-4.50345	0.184075	0.029985	0	0	6.80E-08	5.11E-06
HF1547	123	63519	-0.01547	0.192681	0.013468	0.187858	-4.03882	2.285695	-8.53547	-4.03484	0.435152	0.038614	0	0	4.35E-08	3.70E-06
HF1583	124	52652	-0.01813	0.198256	0.01622	0.192401	-3.89427	2.304236	-8.42636	-3.8906	0.616985	0.04551	0	0	6.46E-08	5.51E-06
HF1591	125	18565	-0.04775	0.317799	0.051391	0.307499	5.336678	1.513276	2.375695	5.332807	8.319585	0.99979	1	1	7.60E-04	0.010122
HF1595	126	6851	-0.05065	0.366358	0.049678	0.353527	-2.82917	3.155804	-9.02907	-2.82661	3.356162	0.184994	0	0	1.30E-05	0.001385
HF1600	127	44898	-0.04476	0.283145	0.044513	0.273238	-2.39621	1.994315	-6.31983	-2.39273	1.507652	0.114774	0	0	6.89E-08	5.76E-06
HF1609	128	12997	-0.02788	0.237103	0.030838	0.227382	6.634904	1.540301	3.618794	6.631704	9.669003	0.999992	1	1	0.008263	0.069965
HF1617	129	54737	-0.04775	0.313766	0.051001	0.303956	4.440673	1.551686	1.401344	4.4377	7.496867	0.997894	1	1	3.31E-05	0.002194
HF1629	130	6620	-2.87E-04	0.210827	7.80E-04	0.204902	0.932798	1.399446	-1.81245	0.93194	3.682936	0.74747	0	0	3.10E-06	2.36E-04
HF1643	131	40788	2.87E-04	0.210827	-7.80E-04	0.204902	-0.9328	1.399446	-3.68294	-0.93194	1.812445	0.25253	0	0	8.13E-08	6.32E-06
HF1647	132	35229	-0.01624	0.24933	0.013334	0.242902	-4.73843	2.791471	-10.2283	-4.7344	0.728417	0.044804	0	0	3.62E-07	2.28E-05
HF1667	133	25438	-0.01684	0.241493	0.01367	0.234617	-5.36804	3.273999	-11.8111	-5.36243	1.043127	0.050545	0	0	2.91E-06	8.83E-05
HF1688	134	8949	-6.76E-04	0.261092	-2.18E-04	0.253333	-1.92822	2.208067	-6.26426	-1.92722	2.402128	0.191261	0	0	2.04E-06	1.72E-04
HF1697	135	13885	0.002661	0.210187	-5.29E-04	0.20339	4.543148	1.561503	1.481878	4.541108	7.615923	0.99819	1	1	0.00184	0.003973
HF1698	136	44836	-0.00646	0.212388	0.009465	0.215222	5.480015	1.562257	2.418212	5.477612	8.555333	0.999774	1	1	4.56E-04	0.006591
HF1722	137	20993	-0.01671	0.216806	0.01502	0.210229	-2.85412	2.205249	-7.19275	-2.85023	1.462457	0.097792	0	0	3.75E-07	3.07E-05
HF1726	138	10796	-0.01656	0.200236	0.015708	0.194351	-1.50001	2.102925	-5.63418	-1.49741	2.619404	0.23783	0	0	1.26E-06	1.07E-04
HF1734	139	26861	-0.01605	0.209366	0.014669	0.202924	-2.21482	2.000851	-6.15076	-2.2115	1.702258	0.13416	0	0	1.94E-07	1.60E-05

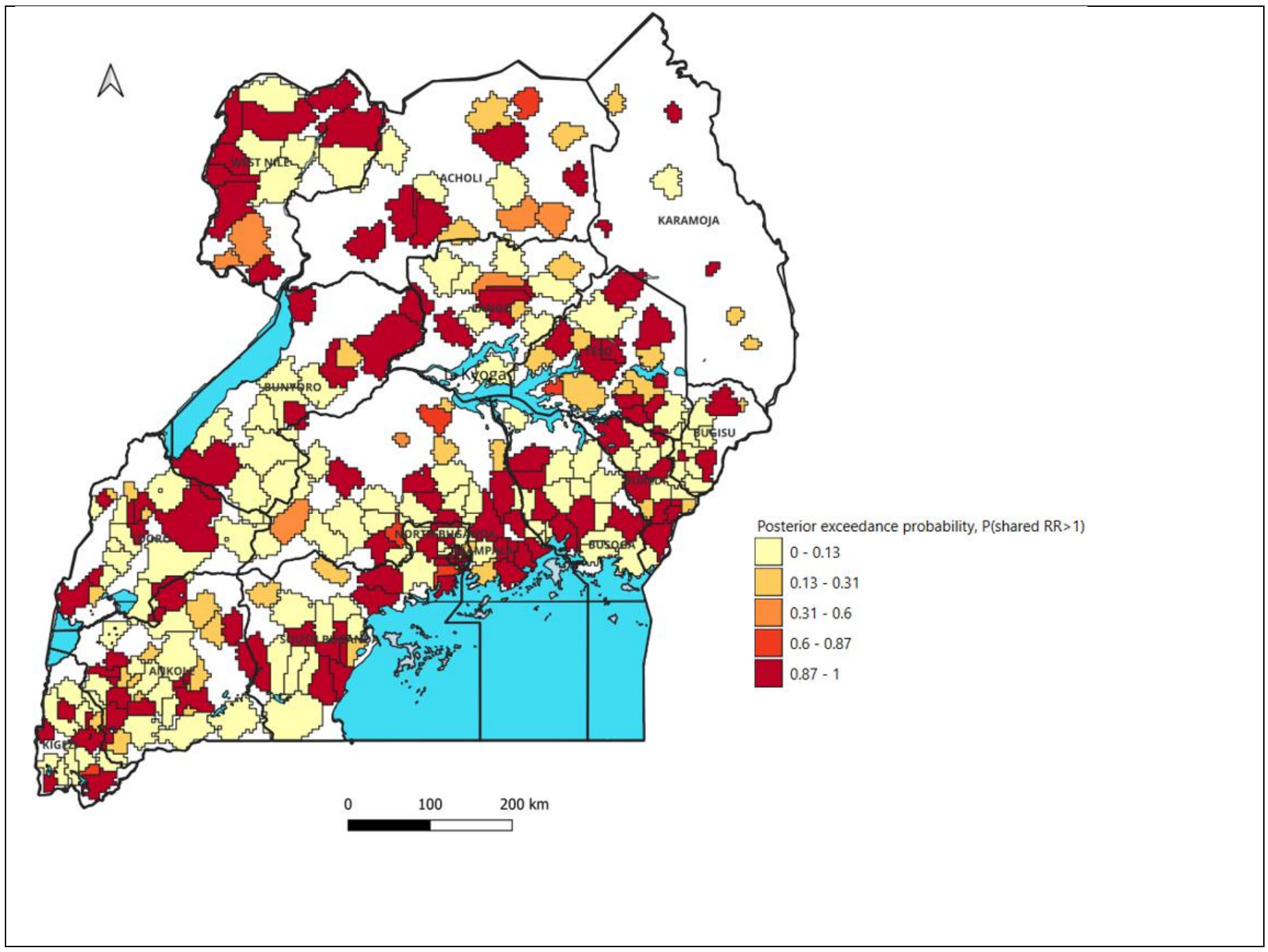
HF1747	140	18869	-0.01368	0.205083	0.015557	0.199891	4.472489	1.481839	1.572462	4.469004	7.392307	0.998729	1	1	0.001496	0.001836
HF1751	141	10610	-0.01842	0.241395	0.017158	0.234884	-2.20301	2.376499	-6.87552	-2.19984	2.451505	0.176964	0	0	1.62E-06	1.48E-04
HF1752	142	8639	-0.01542	0.221196	0.01889	0.215517	7.597376	1.506105	4.652323	7.59293	10.56744	1	1	1	0.027739	0.375977
HF1754	143	13722	-0.01733	0.256141	0.019906	0.25056	5.633562	1.550377	2.602924	5.628454	8.693123	0.99986	1	1	8.94E-04	0.015757
HF1769	144	3752	-9.99E-04	0.211206	0.005056	0.214915	7.900338	1.60043	4.78101	7.891798	11.06789	1	1	1	0.034257	0.448613
HF1785	145	39520	0.033359	0.260275	-0.0319	0.253184	4.983534	1.540695	1.964096	4.981229	8.015975	0.999391	1	1	0.001097	0.002387
HF1805	146	9776	0.034023	0.313491	-0.03842	0.305841	-6.9126	5.660369	-18.0786	-6.89655	4.16244	0.111	0	0	0.093049	0.246208
HF1831	147	8783	0.001435	0.211283	0.002652	0.209393	8.1888	1.602773	5.066423	8.179737	11.36238	1	1	1	0.026089	0.207238
HF1856	148	13211	2.22E-05	0.210873	-6.37E-05	0.204489	-0.07889	1.428413	-2.88348	-0.07882	2.725307	0.477977	0	0	7.60E-07	5.74E-05
HF1868	149	20616	0.001397	0.274499	-0.00429	0.274447	-5.69669	4.710646	-14.9784	-5.68638	3.526317	0.113269	0	0	3.68E-04	0.007575
HF1882	150	22673	-0.00195	0.212296	-6.12E-04	0.206225	-5.30446	4.432798	-14.0202	-5.29925	3.381588	0.115724	0	0	6.49E-05	0.003173
HF1903	151	5663	0.034058	0.313503	-0.03841	0.305842	-6.82108	5.832394	-18.3225	-6.80544	4.591714	0.121098	0	0	0.437646	0.949991
HF1927	152	16106	-0.01135	0.22417	0.009833	0.223636	-4.15938	3.292668	-10.6387	-4.15359	2.287008	0.103254	0	0	4.15E-06	2.99E-04
HF1934	153	21637	-0.01561	0.294022	0.019243	0.292771	5.807634	1.581736	2.71252	5.803449	8.926358	0.99988	1	1	6.49E-04	0.017875
HF1965	154	65633	-0.0075	0.208752	0.010449	0.210801	5.172091	1.564108	2.110725	5.168248	8.255207	0.999528	1	1	2.08E-04	0.002114
HF1970	155	48979	-0.012	0.185938	0.014932	0.184148	4.698036	1.546896	1.668341	4.694977	7.745047	0.998806	1	1	9.43E-05	0.002632
HF1994	156	37474	-0.00813	0.148251	0.006134	0.147546	-4.82517	3.008149	-10.7462	-4.81932	1.062511	0.054353	0	0	4.32E-07	3.02E-05
HF1999	157	16857	-0.01008	0.216473	0.008559	0.216583	-3.84609	1.439891	-6.68691	-3.84148	-1.03151	0.00378	0	0	4.24E-07	1.38E-05
HF2009	158	44937	0.010076	0.216473	-0.00856	0.216583	3.846094	1.439891	1.031511	3.841481	6.686908	0.99622	1	1	1.09E-04	2.85E-04
HF2019	159	28213	-8.12E-04	0.17123	-6.83E-04	0.166598	-3.03998	2.126342	-7.21713	-3.0384	1.128176	0.076405	0	0	1.97E-07	1.58E-05
HF2027	160	8475	-0.00156	0.171225	-0.00722	0.175767	-17.6604	2.693514	-22.9798	-17.6496	-12.4015	2.75E-11	0	0	1.20E-10	1.47E-09
HF2028	161	8613	0.006029	0.173367	-0.00413	0.173541	4.341691	1.545511	1.309629	4.340477	7.380578	0.997517	1	1	0.003006	0.006037
HF2029	162	28907	-0.00449	0.260497	0.008349	0.265384	7.165916	1.591089	4.054482	7.161017	10.30483	0.999997	1	1	0.002174	0.062283
HF2057	163	12103	-0.00179	0.212258	-5.26E-04	0.206158	-4.78453	4.593664	-13.8154	-4.77961	4.218307	0.14881	0	0	4.16E-04	0.017857
HF2067	164	42088	8.11E-06	0.25904	0.002988	0.255599	6.152704	1.567877	3.080192	6.150199	9.239251	0.999956	1	1	0.001228	0.015194
HF2099	165	7239	0.002593	0.211616	9.10E-04	0.205425	7.137646	1.594429	4.027987	7.129848	10.29143	0.999996	1	1	0.012698	0.083928
HF2108	166	3020	-0.00141	0.212185	-3.62E-04	0.206064	-3.67444	5.025322	-13.5511	-3.67021	6.178186	0.232333	0	0	0.036749	1.075023
HF2124	167	29361	0.00587	0.213409	-0.0028	0.209755	6.464768	1.570953	3.400515	6.457235	9.571702	0.999981	1	1	0.002248	0.008289
HF2140	168	4459	-0.00152	0.212204	-4.04E-04	0.206084	-3.97981	4.891941	-13.5951	-3.97538	5.610277	0.207953	0	0	0.009827	0.325224
HF2145	169	44637	-0.01885	0.19861	0.021993	0.194307	5.044706	1.549923	2.009657	5.041438	8.098235	0.999433	1	1	1.62E-04	0.004052
HF2152	170	18498	0.009487	0.23915	-0.01459	0.241163	-9.26357	5.799049	-20.7024	-9.24515	2.070743	0.055085	0	0	0.011941	0.02902
HF2167	171	17234	-0.00188	0.212279	-5.74E-04	0.206193	-5.07607	4.500856	-13.9251	-5.07098	3.744017	0.129702	0	0	1.45E-04	0.006697
HF2186	172	1735	-1.26E-04	0.21113	5.61E-04	0.205064	0.838663	1.593906	-2.28829	0.838108	3.968797	0.700614	0	0	4.21E-05	0.002973
HF2190	173	9064	1.26E-04	0.21113	-5.61E-04	0.205064	-0.83866	1.593906	-3.9688	-0.83811	2.288295	0.299386	0	0	1.54E-06	1.11E-04
HF2205	174	28991	0.011754	0.147381	-0.01237	0.145207	0.033009	1.698752	-3.31531	0.037617	3.355183	0.507752	0	0	2.62E-05	1.43E-05
HF2206	175	20149	0.005143	0.20277	-0.00167	0.196803	7.388337	1.568647	4.322809	7.382868	10.48461	0.999999	1	1	0.004044	0.103395
HF2211	176	7013	-6.54E-04	0.260944	-4.22E-04	0.253233	-2.08543	2.274975	-6.55514	-2.08366	2.374184	0.179655	0	0	3.74E-06	2.93E-04
HF2222	177	47543	-9.18E-04	0.163873	0.003902	0.166733	5.890098	1.555614	2.845426	5.886276	8.95633	0.999924	1	1	4.06E-04	0.009131
HF2228	178	29883	-0.00203	0.212314	-6.53E-04	0.206261	-5.53616	4.367526	-14.1241	-5.53082	3.021341	0.102475	0	0	2.95E-05	0.001513
HF2249	179	12609	0.012893	0.198058	-0.01669	0.196327	-6.35049	5.713664	-17.6182	-6.33664	4.838713	0.133186	0	0	0.083307	0.349824
HF2278	180	12616	-0.02357	0.242225	0.027366	0.236702	6.201172	1.574102	3.121989	6.196682	9.305662	0.999999	1	1	0.001432	0.051647
HF2280	181	3782	-0.00147	0.212196	-3.86E-04	0.206075	-3.85011	4.947013	-13.5733	-3.84577	5.84838	0.218205	0	0	0.017048	0.53641
HF2292	182	5048	-0.00155	0.21221	-4.18E-04	0.206091	-4.07813	4.851639	-13.6145	-4.07364	5.432622	0.200295	0	0	0.006527	0.224072
HF2305	183	2347	0.012072	0.218218	-0.00835	0.216882	8.415158	1.609583	5.282417	8.405136	11.60469	1	1	1	0.435668	0.407331
HF2329	184	32747	0.004602	0.211596	-0.00463	0.212893	0.294156	1.118144	-1.90068	0.293548	2.492431	0.603754	0	0	0.001602	0.006129
HF2330	185	22447	0.001404	0.211269	0.002017	0.207684	6.888322	1.592249	3.781767	6.880924	10.03674	0.999992	1	1	0.00277	0.023461

HF2353	186	10906	0.00124	0.211241	0.002569	0.20916	7.629697	1.59798	4.514601	7.621365	10.7919	0.999999	1	1	0.011735	0.099385
HF2375	187	7819	-0.00167	0.212233	-4.70E-04	0.206121	-4.42879	4.717137	-13.7016	-4.42408	4.817166	0.173898	0	0	0.001602	0.061952
HF2376	188	20273	-0.00565	0.213297	0.008952	0.216937	6.017661	1.587435	2.915793	6.011899	9.152115	0.999925	1	1	5.55E-04	0.020003
HF2400	189	34572	-0.00207	0.212324	-6.74E-04	0.206281	-5.65915	4.334307	-14.182	-5.65375	2.832917	0.095833	0	0	1.95E-05	0.001028
HF2410	190	11325	-2.22E-05	0.210873	6.37E-05	0.204489	0.078892	1.428413	-2.72531	0.078823	2.883484	0.522023	0	0	1.03E-06	7.80E-05
HF2428	191	12476	-0.00587	0.213409	0.002797	0.209755	-6.46477	1.570953	-9.5717	-6.45724	-3.40051	1.93E-05	0	0	3.63E-07	3.21E-06
HF2445	192	107529	-0.02153	0.185334	0.023855	0.181078	3.266164	1.533837	0.25428	3.265924	6.279419	0.98339	1	1	3.54E-06	7.11E-04
HF2448	193	30891	-0.02287	0.217113	0.021838	0.212973	-4.4572	3.027538	-10.4142	-4.45191	1.469653	0.070481	0	0	5.82E-07	4.85E-05
HF2457	194	18179	-0.0046	0.211596	0.004631	0.212893	-0.29416	1.118144	-2.49243	-0.29355	1.900685	0.396246	0	0	9.19E-04	0.009305
HF2475	195	9079	-0.00433	0.194431	0.00271	0.19233	-3.3689	3.181324	-9.61805	-3.36661	2.867222	0.144809	0	0	9.16E-06	8.12E-04
HF2481	196	12101	1.43E-04	0.211121	0.002739	0.209622	5.732685	1.585164	2.635532	5.726846	8.862916	0.999851	1	1	0.001393	0.016002
HF2490	197	17047	-0.00703	0.26979	0.010827	0.273685	7.014258	1.560198	3.964951	7.008937	10.09357	0.999997	1	1	0.005676	0.051376
HF2494	198	17538	-0.01147	0.327611	0.010115	0.329721	-3.70757	3.036901	-9.676	-3.70434	2.242481	0.111073	0	0	1.60E-06	1.61E-04
HF2501	199	4903	-0.00591	0.212887	0.004846	0.213594	-2.62825	3.010678	-8.54549	-2.62509	3.270996	0.191338	0	0	2.25E-05	0.001822
HF2508	200	9958	0.002349	0.129462	-0.0034	0.13138	-1.76966	2.111621	-5.91984	-1.76739	2.367625	0.200999	0	0	1.52E-06	1.25E-04
HF2512	201	15960	0.007509	0.199326	-0.0057	0.20002	4.390536	1.551076	1.357891	4.385634	7.451018	0.997677	1	1	3.58E-04	0.001943
HF2513	202	6376	0.008735	0.173093	-0.00682	0.172549	4.767014	1.542762	1.751003	4.762103	7.810874	0.998999	1	1	0.003859	0.00467
HF2521	203	10825	-0.00239	0.156178	7.42E-04	0.151553	-3.29945	2.794497	-8.78978	-3.29708	2.177386	0.118862	0	0	3.31E-06	2.56E-04
HF2522	204	11275	0.010446	0.212889	-0.00823	0.212277	5.481833	1.556459	2.439265	5.476702	8.553472	0.999786	1	1	0.002015	0.008978
HF2528	205	5756	-5.72E-04	0.146634	-7.07E-04	0.142782	-2.44432	2.701618	-7.7509	-2.44246	2.851683	0.182796	0	0	9.45E-06	7.83E-04
HF2565	206	33385	-0.00206	0.212321	-6.69E-04	0.206276	-5.62962	4.342194	-14.1679	-5.62424	2.877977	0.097403	0	0	2.16E-05	0.001127
HF2566	207	18722	-3.31E-04	0.211136	0.004143	0.213232	7.490894	1.596797	4.377282	7.482843	10.65	0.999999	1	1	0.004936	0.057557
HF2570	208	25853	-0.00319	0.271138	5.88E-04	0.262662	-5.22556	3.159803	-11.4429	-5.22009	0.960573	0.049088	0	0	2.01E-06	6.86E-05
HF2578	209	19268	0.039715	0.353312	-0.04286	0.34254	-5.99091	5.311279	-16.4311	-5.98586	4.42046	0.129668	0	0	0.004223	0.065441
HF2586	210	27097	0.04064	0.356885	-0.04292	0.345538	-4.33348	3.198949	-10.6185	-4.33073	1.935806	0.087763	0	0	9.12E-07	9.68E-05
HF2594	211	9482	0.025864	0.337326	-0.02391	0.331206	6.320292	1.590498	3.216746	6.313056	9.464839	0.999965	1	1	0.005865	0.017375
HF2601	212	29704	0.029636	0.260371	-0.02948	0.253541	2.354632	1.642577	-0.86938	2.354095	5.581484	0.924142	1	1	6.85E-04	2.83E-05
HF2622	213	33949	0.008525	0.227627	-0.01102	0.229285	-4.05452	2.788518	-9.53188	-4.05251	1.411367	0.072972	0	0	2.72E-07	2.76E-05
HF2652	214	49953	-0.00216	0.212349	-7.32E-04	0.206338	-5.97173	4.253987	-14.3373	-5.96616	2.362132	0.08019	0	0	7.04E-06	3.92E-04
HF2673	215	3990	-0.02289	0.301438	0.022982	0.297062	-1.82046	2.896643	-7.50786	-1.81922	3.859874	0.264848	0	0	2.10E-05	0.002452
HF2677	216	39320	-0.0021	0.212332	-6.94E-04	0.2063	-5.76812	4.305654	-14.2349	-5.76266	2.667508	0.090178	0	0	1.36E-05	7.32E-04
HF2695	217	13838	-0.02757	0.231721	0.02672	0.225595	-2.88065	2.589387	-7.9683	-2.87831	2.193693	0.132965	0	0	1.24E-06	1.18E-04
HF2705	218	36682	-0.02568	0.284146	0.028249	0.278549	3.422579	1.558674	0.369316	3.419673	6.492348	0.985948	1	1	1.15E-04	2.92E-04
HF2708	219	10418	-0.04605	0.287005	0.048725	0.28229	1.180232	1.654994	-2.09124	1.187494	4.41064	0.762118	0	0	1.35E-06	0.00198
HF2734	220	18187	-0.03866	0.30157	0.037981	0.293546	-3.05041	2.625705	-8.20912	-3.04813	2.09533	0.122669	0	0	7.16E-07	7.27E-05
HF2749	221	18606	-0.02272	0.231213	0.022133	0.226667	-2.9853	2.507848	-7.91246	-2.98312	1.92946	0.116948	0	0	6.08E-07	5.80E-05
HF2753	222	22905	-8.23E-04	0.146638	-7.67E-04	0.142554	-3.28218	2.379345	-7.96212	-3.27839	1.376215	0.083879	0	0	3.75E-07	3.11E-05
HF2761	223	3648	5.30E-04	0.198072	-0.00131	0.193273	-1.6017	2.658165	-6.82489	-1.59924	3.607542	0.273402	0	0	1.90E-05	0.001841
HF2778	224	33303	2.52E-04	0.134916	-0.00211	0.134811	-3.71378	2.378082	-8.3913	-3.70995	0.941995	0.059183	0	0	1.80E-07	1.47E-05
HF2791	225	8402	0.004729	0.153738	-0.00252	0.151762	4.824445	1.507625	1.873679	4.820943	7.795052	0.999313	1	1	0.006541	0.005086
HF2801	226	15455	0.004216	0.14999	-0.00151	0.145769	5.77392	1.490765	2.858715	5.769602	8.713593	0.999946	1	1	0.008465	0.008533
HF2806	227	16826	0.002395	0.197323	0.001072	0.191563	7.112382	1.552913	4.081678	7.105572	10.18159	0.999998	1	1	0.004078	0.048126
HF2819	228	52231	0.016344	0.334154	-0.01962	0.333628	-4.97845	3.200673	-11.2656	-4.97613	1.295429	0.059921	0	0	2.49E-07	2.71E-05
HF2830	229	30916	0.019301	0.261081	-0.02192	0.256873	-3.38401	2.397602	-8.09536	-3.38166	1.313922	0.079061	0	0	1.97E-07	1.81E-05
HF2836	230	13918	0.058307	0.417276	-0.05639	0.405265	6.123446	1.601029	2.999463	6.116083	9.289123	0.999935	1	1	0.004009	0.00836
HF2837	231	7996	0.060058	0.41753	-0.05899	0.406535	5.228549	1.600372	2.102987	5.22219	8.390128	0.999457	1	1	0.003501	0.005453

HF2843	232	27369	0.050506	0.358993	-0.05352	0.350658	-3.03744	2.398289	-7.7501	-3.03511	1.661934	0.102667	0	0	2.38E-07	2.31E-05
HF2874	233	10419	0.006138	0.28149	-0.00358	0.275957	5.54627	1.55219	2.51225	5.541124	8.609462	0.999824	1	1	0.003208	0.010514
HF2879	234	17100	-4.95E-04	0.202866	-0.00226	0.200166	-5.51973	3.991371	-13.3752	-5.51277	2.296016	0.083345	0	0	5.65E-05	0.001016
HF2886	235	24853	0.001016	0.241654	-0.00312	0.238792	-4.16383	2.763347	-9.59787	-4.15989	1.247805	0.06593	0	0	6.14E-07	4.48E-05
HF2891	236	53557	0.005205	0.292656	-0.00286	0.285264	5.037555	1.548918	2.00845	5.032914	8.092982	0.999428	1	1	2.32E-04	0.001533
HF2900	237	11614	-7.10E-04	0.166677	-0.00191	0.16495	-5.26033	4.101299	-13.3324	-5.25321	2.771189	0.099816	0	0	1.85E-04	0.002795
HF2916	238	2703	0.001643	0.280567	-0.00294	0.275387	-2.51701	3.306997	-9.01572	-2.514	3.964524	0.223293	0	0	1.67E-04	0.010204
HF2927	239	103509	0.043824	0.30773	-0.04297	0.298359	3.491068	1.49208	0.560077	3.49133	6.420554	0.990351	1	1	6.21E-05	4.10E-04
HF2931	240	72717	0.040307	0.302922	-0.04393	0.295106	-5.30105	2.85917	-10.9213	-5.29767	0.299964	0.031866	0	0	8.17E-08	6.49E-06
HF2932	241	47889	0.040428	0.267703	-0.04429	0.261344	-5.26568	2.742464	-10.6607	-5.26128	0.104255	0.027425	0	0	2.01E-07	1.02E-05
HF2937	242	36257	0.040937	0.28895	-0.04448	0.281888	-4.78017	2.929506	-10.539	-4.77668	0.958782	0.051368	0	0	3.98E-07	2.83E-05
HF2983	243	12289	-0.00179	0.212259	-5.28E-04	0.206159	-4.79705	4.589534	-13.8198	-4.79212	4.197661	0.147962	0	0	3.97E-04	0.017109
HF2984	244	4819	0.049989	0.336574	-0.04998	0.329758	4.351959	1.597639	1.228154	4.347081	7.503389	0.996775	1	1	0.008052	0.002397
HF2991	245	4409	0.044565	0.402726	-0.04698	0.392409	-2.7954	3.44807	-9.56974	-2.79261	3.963048	0.208765	0	0	6.29E-05	0.006084
HF3019	246	45503	0.006356	0.27171	-0.00346	0.267213	6.048353	1.547025	3.028227	6.04186	9.105299	0.999954	1	1	6.60E-04	0.003988
HF3033	247	10245	6.77E-04	0.211602	-0.00221	0.208805	-2.96885	2.2987	-7.48679	-2.96649	1.535674	0.09826	0	0	2.03E-06	1.28E-04
HF3034	248	23176	-0.00128	0.211108	0.0046	0.213971	6.38895	1.497641	3.468008	6.381785	9.350518	0.99999	1	1	5.99E-04	0.015034
HF3036	249	16320	5.99E-04	0.211593	-0.00236	0.209199	-3.42006	2.295889	-7.933	-3.41752	1.078398	0.068159	0	0	7.96E-07	5.07E-05
HF3047	250	70531	0.041299	0.280685	-0.04536	0.274216	-5.60159	3.000505	-11.5015	-5.59755	0.275377	0.030959	0	0	1.29E-07	8.40E-06
HF3083	251	36153	0.042558	0.34031	-0.04486	0.329927	-3.66851	2.546011	-8.67325	-3.66542	1.318622	0.074808	0	0	1.63E-07	1.62E-05
HF3085	252	84598	0.046854	0.340345	-0.04535	0.329553	4.390881	1.557776	1.342384	4.386936	7.461725	0.997589	1	1	1.86E-04	3.45E-04
HF3103	253	17367	-0.00309	0.19206	0.001148	0.186689	-3.99006	2.753085	-9.4035	-3.98634	1.402158	0.073626	0	0	1.38E-06	8.48E-05
HF3109	254	26630	-2.35E-04	0.156495	0.003233	0.158469	5.985483	1.518712	3.017612	5.980203	8.98326	0.999999	1	1	7.60E-04	0.010586
HF3118	255	10598	1.68E-04	0.19164	-0.0015	0.187843	-2.81823	2.588342	-7.90556	-2.81539	2.252887	0.138117	0	0	2.26E-06	1.93E-04
HF3137	256	13976	-0.01674	0.181524	0.01599	0.178379	-3.03303	2.64722	-8.23362	-3.03087	2.155248	0.125951	0	0	1.35E-06	1.27E-04
HF3141	257	439	-0.02852	0.229692	0.03463	0.226203	9.544626	1.574339	6.480709	9.534751	12.66397	1	1	1	0.919734	24.98959
HF3145	258	13500	-0.03189	0.264531	0.031252	0.257582	-2.89503	2.694323	-8.18773	-2.89298	2.385929	0.141301	0	0	1.44E-06	1.48E-04
HF3146	259	2071	-0.02822	0.226314	0.03152	0.22174	4.488995	1.530948	1.495474	4.484374	7.508763	0.998317	1	1	0.002241	0.022631
HF3149	260	5362	-0.02377	0.17678	0.027727	0.174078	5.726714	1.496134	2.79946	5.72285	8.675774	0.999935	1	1	8.08E-04	0.11651
HF3153	261	8864	-0.03048	0.22683	0.030798	0.221657	-1.509	2.224031	-5.87954	-1.5068	2.849075	0.248729	0	0	1.99E-06	1.78E-04
HF3171	262	27126	-0.01812	0.210211	0.021187	0.206453	4.823146	1.536989	1.817819	4.818421	7.855269	0.99915	1	1	5.66E-04	0.001568
HF3186	263	17676	-0.01947	0.183664	0.01874	0.180087	-3.11369	2.57809	-8.17892	-3.11143	1.938671	0.113571	0	0	7.52E-07	7.16E-05
HF3188	264	43122	-0.00827	0.169217	0.006541	0.168918	-3.99745	2.558001	-9.02408	-3.99489	1.014613	0.059059	0	0	1.24E-07	1.17E-05
HF3207	265	26197	0.003155	0.141127	-0.00481	0.143762	-2.91663	2.297053	-7.43094	-2.91422	1.583906	0.102091	0	0	2.46E-07	2.21E-05
HF3214	266	6196	3.50E-04	0.23379	-0.00181	0.228911	-2.91472	3.212177	-9.2259	-2.91198	3.380849	0.182098	0	0	1.92E-05	0.001792
HF3222	267	11456	-0.00215	0.25976	0.005994	0.263401	7.596654	1.591037	4.496225	7.587994	10.7461	0.999999	1	1	0.014763	0.054451
HF3226	268	31784	-0.00778	0.162032	0.00644	0.161993	-3.22068	2.328952	-7.79809	-3.21808	1.3419	0.08335	0	0	1.73E-07	1.56E-05
HF3236	269	24292	0.002216	0.125171	-0.00143	0.123068	1.954477	1.528415	-1.04483	1.953853	4.957301	0.899509	1	0	1.99E-04	5.64E-05
HF3241	270	12615	-0.00668	0.167653	0.005467	0.168509	-2.77408	2.582169	-7.84645	-2.77213	2.287149	0.141339	0	0	1.47E-06	1.40E-04
HF3255	271	21068	0.012185	0.248264	-0.01444	0.247783	-3.36307	2.696326	-8.65936	-3.36111	1.922018	0.106147	0	0	5.90E-07	6.12E-05
HF3264	272	14470	0.021227	0.233146	-0.02372	0.228586	-3.70646	3.088957	-9.77389	-3.70429	2.348598	0.115088	0	0	2.82E-06	2.65E-04
HF3271	273	16976	0.00439	0.135476	-0.0061	0.137233	-3.02249	2.413502	-7.76496	-3.02019	1.70688	0.105226	0	0	7.07E-07	6.01E-05
HF3291	274	12078	-0.00441	0.189389	0.002963	0.188032	-3.06498	2.830139	-8.62344	-3.06317	2.483167	0.139409	0	0	2.30E-06	2.34E-04
HF3308	275	26761	-0.00882	0.164219	0.01025	0.163434	2.452551	1.509655	-0.50909	2.451469	5.420325	0.947874	1	1	1.15E-04	1.54E-04
HF3330	276	15918	0.041202	0.299578	-0.04412	0.293069	-3.22786	2.681216	-8.49462	-3.22586	2.02754	0.114319	0	0	1.07E-06	1.03E-04
HF3331	277	5154	0.056925	0.369322	-0.05696	0.361484	4.160049	1.616589	0.998376	4.155311	7.348549	0.994964	1	1	0.003818	0.002083

HF3339	278	29106	-0.00376	0.125982	0.002039	0.124634	-3.56145	2.2832	-8.05006	-3.55855	0.910655	0.059398	0	0	2.21E-07	1.72E-05
HF3353	279	16796	0.005487	0.122001	-0.00437	0.12211	2.922741	1.50879	-0.03558	2.921192	5.889791	0.973636	1	1	9.22E-04	2.35E-04
HF3362	280	3052	-0.01267	0.19392	0.011928	0.191921	-2.45938	3.329776	-8.9998	-2.45718	4.068462	0.230074	0	0	1.19E-04	0.009671

Supplementary Figure S4. Posterior exceedance probability map from the extended Bayesian shared-component model, showing P(shared RR>1) across health-facility catchments in Uganda. Higher values indicate greater posterior support that the shared malaria-SCD relative risk exceeded the national baseline. The map shows broad areas of elevated posterior support across several northern and eastern catchments.



Supplementary Table S5. Catchment-level posterior exceedance probabilities from the fully adjusted shared-component model. Catchment-specific posterior mean adjusted shared effect, posterior standard deviation, credible interval bounds, exceedance probability, and hotspot status under the fully adjusted Bayesian shared-component model.

catchment_id	Area index	Shared mean_adj	Shared sd_adj	Shared q025_adj	Shared q50_adj	Shared q975_adj	exceed_prob rr_gt1_adj	Hotspot 080_adj	Hotspot 090_adj
HF0002	1	-3.77431	1.011587	-5.73532	-3.79078	-1.71142	9.53E-05	0	0
HF0007	2	-2.83827	0.846858	-4.47997	-2.8641	-1.05028	4.02E-04	0	0
HF0008	3	-6.48856	1.406036	-9.16494	-6.52316	-3.62852	1.97E-06	0	0
HF0019	4	-6.8764	1.126084	-8.95177	-6.93168	-4.50438	5.09E-10	0	0
HF0022	5	-4.73304	1.111539	-6.73834	-4.80684	-2.35071	1.03E-05	0	0
HF0039	6	-0.01093	2.736492	-5.39208	-0.00958	5.36262	0.498407	0	0
HF0052	7	-0.50234	1.159951	-2.78538	-0.50335	1.786598	0.332481	0	0
HF0083	8	-6.09715	6.126845	-18.1968	-6.07931	5.901711	0.159831	0	0
HF0092	9	-0.05199	3.508517	-6.95051	-0.05096	6.840698	0.494089	0	0
HF0093	10	1.529033	1.132274	-0.71937	1.53436	3.745294	0.911558	1	1
HF0103	11	-0.16714	2.989923	-6.04601	-0.16628	5.706901	0.47771	0	0
HF0118	12	1.393006	2.168818	-2.91275	1.409377	5.609341	0.739657	0	0
HF0202	13	-1.86026	2.165495	-6.20013	-1.82878	2.305591	0.195158	0	0
HF0203	14	0.079391	1.89421	-3.70154	0.101107	3.741175	0.516716	0	0
HF0204	15	2.025163	1.88494	-1.73799	2.047323	5.667594	0.858676	1	0
HF0216	16	2.125813	2.150485	-2.15194	2.145497	6.296291	0.838553	1	0
HF0252	17	0.902668	2.156664	-3.39702	0.925681	5.075982	0.662227	0	0
HF0253	18	-1.9439	2.162392	-6.25195	-1.92204	2.24214	0.184338	0	0
HF0262	19	6.244997	2.025763	2.285999	6.238275	10.24474	0.998975	1	1
HF0263	20	1.620647	2.144352	-2.65438	1.643731	5.769575	0.775108	0	0
HF0264	21	-3.9697	2.300756	-8.59328	-3.93121	0.440727	0.042229	0	0
HF0281	22	0.311702	1.897897	-3.48971	0.33883	3.964184	0.565227	0	0

HF0282	23	1.47594	1.866048	-2.25647	1.50075	5.072805	0.785512	0	0
HF0283	24	-5.89394	2.226392	-10.3845	-5.84945	-1.64931	0.004057	0	0
HF0290	25	-2.96459	2.290117	-7.48937	-2.95419	1.50087	0.097744	0	0
HF0298	26	3.788254	0.919114	2.08888	3.742488	5.735539	0.999981	1	1
HF0304	27	-3.42886	0.970689	-5.03887	-3.55522	-1.25166	2.06E-04	0	0
HF0316	28	-3.67072	2.029251	-7.68493	-3.65981	0.281343	0.035233	0	0
HF0327	29	-5.63065	0.98096	-7.31983	-5.72604	-3.45404	4.74E-09	0	0
HF0354	30	-5.74974	3.065234	-11.8218	-5.73247	0.223675	0.030342	0	0
HF0362	31	-4.96247	0.959778	-6.57146	-5.07809	-2.80511	1.17E-07	0	0
HF0376	32	-3.23303	2.010201	-7.2272	-3.21695	0.669559	0.053883	0	0
HF0401	33	-5.19636	2.697708	-10.5451	-5.1787	0.051762	0.027039	0	0
HF0402	34	-4.74753	2.760304	-10.2167	-4.73087	0.626835	0.042723	0	0
HF0411	35	-4.90052	2.168354	-9.1995	-4.88627	-0.68297	0.01191	0	0
HF0429	36	-0.91494	3.215198	-7.24103	-0.91317	5.401204	0.387988	0	0
HF0438	37	-1.22978	2.896545	-6.93182	-1.22723	4.457804	0.335575	0	0
HF0446	38	-1.15449	3.288562	-7.62597	-1.1524	5.305241	0.362771	0	0
HF0465	39	-0.27284	1.193053	-2.57219	-0.29082	2.129507	0.409556	0	0
HF0494	40	1.890341	1.53207	-1.13844	1.895982	4.887553	0.89137	1	0
HF0499	41	6.350265	1.409227	3.657529	6.325012	9.187236	0.999997	1	1
HF0505	42	1.410588	2.684936	-3.86391	1.410562	6.685277	0.700337	0	0
HF0506	43	1.315707	1.135543	-0.92642	1.316755	3.550626	0.876702	1	0
HF0537	44	-3.15284	1.911615	-6.93218	-3.14341	0.572655	0.049543	0	0
HF0540	45	-3.99714	2.240429	-8.42996	-3.98498	0.366392	0.037204	0	0
HF0550	46	-4.64023	0.909538	-6.18776	-4.74151	-2.60034	1.68E-07	0	0
HF0560	47	0.707382	1.203325	-1.57026	0.673643	3.168931	0.721686	0	0
HF0583	48	-0.99861	2.218713	-5.41047	-0.98183	3.317987	0.326325	0	0
HF0588	49	-0.18724	1.182641	-2.54155	-0.17928	2.120689	0.437101	0	0
HF0608	50	-5.399	2.074941	-9.52196	-5.38215	-1.37218	0.004634	0	0
HF0615	51	-7.1461	3.470259	-14.0375	-7.12178	-0.39298	0.019736	0	0
HF0650	52	-10.556	1.649906	-13.7592	-10.5696	-7.27702	7.87E-11	0	0
HF0678	53	-1.00271	2.472679	-5.90729	-0.98751	3.816583	0.342549	0	0
HF0680	54	2.168552	1.835512	-1.48225	2.18588	5.725138	0.881286	1	0
HF0698	55	-2.80804	2.780195	-8.33678	-2.78608	2.596387	0.156244	0	0
HF0715	56	-3.18763	1.02674	-5.17212	-3.20645	-1.08952	9.53E-04	0	0
HF0718	57	-4.189	1.034481	-6.01564	-4.27438	-1.93148	2.57E-05	0	0
HF0720	58	-1.6894	3.895285	-9.34889	-1.68599	5.950237	0.332252	0	0
HF0724	59	-3.78152	1.760844	-7.26503	-3.77213	-0.35187	0.015874	0	0
HF0732	60	-3.7714	3.164273	-10.0112	-3.76172	2.413107	0.116656	0	0
HF0741	61	-5.94915	0.630798	-7.16985	-5.95913	-4.68008	0	0	0
HF0752	62	5.949154	0.630798	4.680079	5.959133	7.169845	1	1	1
HF0773	63	-1.82207	4.718132	-11.1051	-1.81931	7.445443	0.34968	0	0
HF0777	64	-0.15712	1.151917	-2.44916	-0.14955	2.091248	0.445752	0	0
HF0854	65	-0.59332	1.281768	-3.12685	-0.58879	1.915107	0.321721	0	0
HF0859	66	-0.92786	6.276812	-13.2772	-0.92522	11.4067	0.441241	0	0
HF0860	67	-1.01743	4.183074	-9.24873	-1.01487	7.199494	0.403915	0	0
HF0873	68	-0.86897	3.365688	-7.52381	-0.85597	5.712604	0.398132	0	0
HF0883	69	-0.49946	2.948588	-6.31417	-0.49399	5.284256	0.432745	0	0
HF0887	70	-0.19377	1.907312	-4.01565	-0.16573	3.474013	0.459541	0	0
HF0889	71	-4.36845	2.077702	-8.55102	-4.32967	-0.40002	0.017753	0	0
HF0919	72	-2.40051	2.270157	-6.89554	-2.38873	2.027383	0.14516	0	0
HF0921	73	-1.64212	1.894455	-5.42613	-1.61893	2.014237	0.193025	0	0
HF0926	74	-1.32205	2.344374	-5.99026	-1.30088	3.227743	0.286403	0	0
HF0940	75	-1.81926	1.919473	-5.67657	-1.78672	1.858151	0.171618	0	0
HF0948	76	0.376641	3.301621	-6.15943	0.392669	6.822939	0.545412	0	0
HF0957	77	-3.25961	4.042994	-11.3026	-3.2269	4.598772	0.210053	0	0
HF0966	78	-15.122	3.14637	-21.4474	-15.0693	-9.09346	7.69E-07	0	0
HF0975	79	9.10239	1.241565	6.707814	9.087924	11.57916	1	1	1
HF0998	80	4.254528	2.774736	-1.14122	4.235276	9.759647	0.9374	1	1
HF1015	81	6.900578	3.480237	0.250302	6.833676	13.9233	0.976305	1	1
HF1032	82	4.069874	1.793929	0.600694	4.04959	7.651739	0.988356	1	1
HF1063	83	2.94424	2.920331	-2.76326	2.933298	8.714043	0.843318	1	0
HF1065	84	0.894054	1.873205	-2.78303	0.893842	4.572432	0.683421	0	0
HF1092	85	4.493996	3.70853	-2.68816	4.455614	11.89318	0.887206	1	0
HF1113	86	4.253878	1.89009	0.573047	4.241056	8.005587	0.987795	1	1
HF1125	87	5.217067	1.875041	1.593678	5.195014	8.962583	0.997302	1	1
HF1130	88	1.534499	1.856586	-2.06825	1.517777	5.230186	0.795744	0	0
HF1147	89	-0.0691	5.947477	-11.7679	-0.06975	11.63341	0.495365	0	0
HF1165	90	3.450965	2.828252	-2.07345	3.439268	9.041914	0.888801	1	0
HF1166	91	4.120397	1.813786	0.608834	4.100971	7.739496	0.988448	1	1
HF1180	92	3.143997	3.424417	-3.55229	3.132893	9.903506	0.82072	1	0
HF1207	93	0.028472	2.759845	-5.41503	0.034338	5.438783	0.504116	0	0
HF1209	94	-1.45366	3.268339	-7.90974	-1.44327	4.943554	0.328243	0	0
HF1234	95	2.125282	1.81443	-1.39489	2.108346	5.739354	0.879265	1	0
HF1246	96	-2.39066	3.450766	-9.22163	-2.37431	4.34757	0.244221	0	0
HF1249	97	-2.02318	3.386935	-8.72691	-2.00754	4.591919	0.275137	0	0
HF1250	98	1.741102	1.926245	-2.0038	1.725983	5.570317	0.816971	1	0
HF1269	99	-1.89673	3.841185	-9.44478	-1.89825	5.66007	0.310728	0	0
HF1276	100	-0.12547	1.230285	-2.55535	-0.12224	2.285736	0.459384	0	0
HF1281	101	2.491363	3.508799	-4.37269	2.48113	9.413712	0.761158	0	0
HF1312	102	1.101399	1.781829	-2.35554	1.084958	4.649727	0.731755	0	0
HF1313	103	3.018765	1.814098	-0.50222	3.002312	6.631809	0.95195	1	1
HF1322	104	2.801878	2.774215	-2.62194	2.791507	8.284761	0.843746	1	0
HF1340	105	2.213928	3.147946	-3.93766	2.201363	8.436957	0.759063	0	0

HF1354	106	6.559847	3.450131	-0.0375	6.495622	13.51434	0.971371	1	1
HF1375	107	0.455303	1.31294	-2.11654	0.450609	3.053066	0.635622	0	0
HF1390	108	-0.62428	2.833211	-6.17984	-0.62817	4.953303	0.412802	0	0
HF1405	109	3.552261	3.239297	-2.75884	3.533031	9.972658	0.863595	1	0
HF1408	110	1.646223	1.298286	-0.95576	1.662207	4.155947	0.8976	1	0
HF1419	111	-1.06913	2.203231	-5.39029	-1.07168	3.266455	0.313748	0	0
HF1421	112	-1.51437	1.296373	-4.04856	-1.5199	1.050734	0.121371	0	0
HF1431	113	-0.65834	1.294135	-3.2094	-0.65715	1.885498	0.305477	0	0
HF1438	114	7.39492	5.747835	-3.74939	7.340813	18.84512	0.900876	1	1
HF1475	115	2.08546	3.979325	-5.70615	2.075911	9.931451	0.699886	0	0
HF1483	116	6.114408	4.456446	-2.49771	6.062801	15.01827	0.914974	1	1
HF1492	117	4.989316	3.72539	-2.2029	4.943198	12.44188	0.909759	1	1
HF1515	118	2.779804	2.769916	-2.63073	2.767991	8.257565	0.842207	1	0
HF1524	119	4.70828	3.664198	-2.37605	4.666455	12.02919	0.900594	1	1
HF1530	120	3.718569	3.16955	-2.42083	3.686379	10.04019	0.879646	1	0
HF1532	121	4.578508	3.41385	-1.96099	4.518407	11.45264	0.910065	1	1
HF1533	122	3.869	3.110011	-2.14369	3.8336	10.08169	0.893259	1	0
HF1547	123	3.011094	2.539068	-1.94481	2.999459	8.033217	0.88217	1	0
HF1583	124	3.426637	2.442679	-1.33825	3.414679	8.259556	0.919665	1	1
HF1591	125	-1.14314	1.272829	-3.67095	-1.13531	1.339677	0.184563	0	0
HF1595	126	-1.21141	3.747425	-8.56258	-1.21591	6.165415	0.373247	0	0
HF1600	127	2.719438	2.606234	-2.34569	2.699051	7.900294	0.851627	1	0
HF1609	128	6.430905	1.923668	2.71056	6.40923	10.27083	0.999586	1	1
HF1617	129	4.23515	1.932982	0.459692	4.225604	8.062388	0.985774	1	1
HF1629	130	0.832989	1.723603	-2.55102	0.832609	4.21915	0.685553	0	0
HF1643	131	-0.83299	1.723603	-4.21915	-0.83261	2.551022	0.314447	0	0
HF1647	132	5.18901	3.696553	-1.95019	5.144161	12.58167	0.919802	1	1
HF1667	133	3.90672	3.422176	-2.72705	3.87373	10.72733	0.873188	1	0
HF1688	134	0.580365	2.141912	-3.61669	0.577926	4.791322	0.606788	0	0
HF1697	135	7.537947	1.202612	5.202748	7.527674	9.926372	1	1	1
HF1698	136	-1.51916	1.53237	-4.55814	-1.50957	1.467213	0.160751	0	0
HF1722	137	3.512622	2.715827	-1.7635	3.491042	8.911235	0.902062	1	1
HF1726	138	4.27454	2.697138	-0.96197	4.251746	9.640306	0.943499	1	1
HF1734	139	4.063004	2.622351	-1.02705	4.040872	9.278674	0.939354	1	1
HF1747	140	3.715378	1.869935	0.095023	3.69574	7.444537	0.976533	1	1
HF1751	141	-1.36678	2.238591	-5.77394	-1.36363	3.023298	0.270747	0	0
HF1752	142	7.485275	1.959724	3.671545	7.471204	11.37564	0.999933	1	1
HF1754	143	4.782485	1.940834	0.990914	4.77304	8.624579	0.993133	1	1
HF1769	144	1.79032	2.791461	-3.67297	1.780002	7.31149	0.739354	0	0
HF1785	145	-0.02912	0.853203	-1.56174	-0.0919	1.832239	0.486387	0	0
HF1805	146	0.341255	2.62127	-4.81138	0.343472	5.481792	0.551791	0	0
HF1831	147	4.051838	1.218888	1.674851	4.04287	6.475428	0.999557	1	1
HF1856	148	4.64502	1.256702	2.194951	4.639134	7.128238	0.999891	1	1
HF1868	149	-4.98405	4.818359	-14.5163	-4.96549	4.443505	0.150478	0	0
HF1882	150	-2.46339	3.562978	-9.47762	-2.4583	4.521995	0.244662	0	0
HF1903	151	1.399446	3.273275	-5.03178	1.399614	7.830942	0.665506	0	0
HF1927	152	-1.20735	2.513765	-6.16039	-1.20201	3.715403	0.315509	0	0
HF1934	153	-0.78169	1.353259	-3.47269	-0.77135	1.850724	0.281756	0	0
HF1965	154	-0.76201	1.292275	-3.3016	-0.76366	1.787611	0.277709	0	0
HF1970	155	-0.40591	0.772351	-1.97242	-0.41213	1.187617	0.2996	0	0
HF1994	156	-0.75607	2.313982	-5.30418	-0.75475	3.784591	0.371933	0	0
HF1999	157	2.200659	0.819763	0.403291	2.267125	3.660001	0.996368	1	1
HF2009	158	-2.20066	0.819763	-3.66	-2.26712	-0.40329	0.003632	0	0
HF2019	159	-0.78459	1.953551	-4.62732	-0.78173	3.041877	0.343981	0	0
HF2027	160	-7.06255	3.03042	-13.0978	-7.03418	-1.18811	0.009888	0	0
HF2028	161	0.946507	1.169537	-1.33174	0.938056	3.271416	0.790829	0	0
HF2029	162	3.806751	1.249361	1.329877	3.81185	6.250679	0.998844	1	1
HF2057	163	-0.37679	5.268966	-10.7395	-0.3788	9.997352	0.471496	0	0
HF2067	164	2.308779	1.440201	-0.51796	2.306872	5.146176	0.945543	1	1
HF2099	165	0.888177	2.86904	-4.72886	0.878329	6.559953	0.621557	0	0
HF2108	166	-0.74758	4.840577	-10.2663	-0.74793	8.773344	0.438631	0	0
HF2124	167	2.476451	1.134351	0.288075	2.459259	4.759537	0.985487	1	1
HF2140	168	-0.17334	5.683858	-11.3537	-0.17447	11.0134	0.487835	0	0
HF2145	169	0.182227	0.755958	-1.30961	0.162736	1.771014	0.595244	0	0
HF2152	170	-3.66848	4.216792	-11.9993	-3.65559	4.589405	0.192158	0	0
HF2167	171	-1.64028	4.246337	-9.98933	-1.64019	6.708544	0.349644	0	0
HF2186	172	0.228052	5.154921	-9.9235	0.229457	10.37159	0.517643	0	0
HF2190	173	-0.22805	5.154921	-10.3716	-0.22946	9.9235	0.482357	0	0
HF2205	174	1.92398	1.594918	-1.18243	1.915093	5.080465	0.886153	1	0
HF2206	175	0.046693	1.442008	-2.80523	0.053034	2.863715	0.512916	0	0
HF2211	176	0.32957	2.110979	-3.81981	0.331767	4.466499	0.562032	0	0
HF2222	177	8.670235	1.353282	6.045213	8.657379	11.3642	1	1	1
HF2228	178	-2.6506	3.48588	-9.51402	-2.64519	4.182092	0.223513	0	0
HF2249	179	-0.15475	2.832306	-5.71883	-0.15402	5.405341	0.478213	0	0
HF2278	180	1.364812	0.873153	-0.33889	1.348242	3.155331	0.940984	1	1
HF2280	181	-0.84373	4.750863	-10.1858	-0.84404	8.500263	0.42952	0	0
HF2292	182	-0.97667	4.638744	-10.098	-0.97693	8.146283	0.41662	0	0
HF2305	183	1.318569	2.831735	-4.23046	1.310435	6.91197	0.679263	0	0
HF2329	184	0.288049	0.590128	-0.85081	0.273265	1.511028	0.687265	0	0
HF2330	185	1.1027	0.807268	-0.43664	1.062679	2.846239	0.914025	1	1
HF2353	186	3.812684	1.138763	1.598567	3.801393	6.086215	0.999593	1	1
HF2375	187	-1.77629	3.879295	-9.41054	-1.77202	5.833802	0.323516	0	0
HF2376	188	-0.97443	1.288047	-3.55114	-0.95992	1.520325	0.224669	0	0

HF2400	189	-2.75092	3.450083	-9.54428	-2.74537	4.010929	0.212624	0	0
HF2410	190	-4.64502	1.256702	-7.12824	-4.63913	-2.19495	1.09E-04	0	0
HF2428	191	-2.47645	1.134351	-4.75954	-2.45926	-0.28808	0.014513	0	0
HF2445	192	-0.86757	0.781694	-2.54712	-0.8349	0.62494	0.133529	0	0
HF2448	193	-0.31114	2.41697	-5.05972	-0.31014	4.431673	0.448786	0	0
HF2457	194	-0.28805	0.590128	-1.51103	-0.27326	0.850808	0.312735	0	0
HF2475	195	-4.12559	4.949013	-13.9057	-4.10975	5.564851	0.202248	0	0
HF2481	196	0.785904	1.688844	-2.5224	0.781811	4.116984	0.67916	0	0
HF2490	197	2.61855	1.218299	0.237996	2.612358	5.03228	0.984197	1	1
HF2494	198	0.115276	3.264158	-6.29708	0.114625	6.531375	0.514086	0	0
HF2501	199	0.548787	3.367834	-6.0712	0.547947	7.173604	0.564721	0	0
HF2508	200	-3.10628	2.549558	-8.14742	-3.09676	1.880954	0.111543	0	0
HF2512	201	-1.5571	1.249564	-4.02525	-1.55444	0.897646	0.106361	0	0
HF2513	202	-1.40851	1.255099	-3.85706	-1.41663	1.087188	0.130882	0	0
HF2521	203	-4.54801	3.632741	-11.7375	-4.53214	2.551654	0.105294	0	0
HF2522	204	-0.3857	1.245531	-2.83543	-0.38695	2.071867	0.378406	0	0
HF2528	205	-7.26753	2.112145	-11.458	-7.25252	-3.16243	2.90E-04	0	0
HF2565	206	-1.53452	4.087722	-9.57391	-1.53276	6.49493	0.353683	0	0
HF2566	207	1.621552	1.823402	-1.96378	1.622012	5.204538	0.813079	1	0
HF2570	208	-1.37292	3.637187	-8.52568	-1.3715	5.771646	0.352913	0	0
HF2578	209	-1.94815	5.058787	-11.883	-1.94995	7.997138	0.350081	0	0
HF2586	210	-0.91751	3.723616	-8.22989	-0.91791	6.397366	0.402685	0	0
HF2594	211	0.194908	1.274378	-2.29241	0.186905	2.728303	0.560779	0	0
HF2601	212	5.449593	1.480468	2.652099	5.410758	8.465797	0.999884	1	1
HF2622	213	-0.41345	2.792691	-5.9033	-0.41208	5.068858	0.441152	0	0
HF2652	214	-1.8452	3.841138	-9.39862	-1.84406	5.70202	0.315479	0	0
HF2673	215	-1.36137	5.516446	-12.2273	-1.35655	9.477224	0.402538	0	0
HF2677	216	-1.6982	3.921858	-9.4103	-1.69717	6.008296	0.332504	0	0
HF2695	217	1.911003	4.166617	-6.2681	1.905421	10.12153	0.676756	0	0
HF2705	218	1.946589	2.43412	-2.82852	1.942404	6.744234	0.788061	0	0
HF2708	219	2.29568	2.288927	-2.23467	2.305187	6.773982	0.842058	1	0
HF2734	220	2.050929	4.656751	-7.09058	2.045359	11.22383	0.670184	0	0
HF2749	221	1.816138	4.161053	-6.3593	1.813041	10.00865	0.668749	0	0
HF2753	222	-0.36765	2.462953	-5.21698	-0.36557	4.469942	0.44067	0	0
HF2761	223	0.323512	3.482708	-6.52573	0.323574	7.172342	0.537005	0	0
HF2778	224	-0.78665	2.439585	-5.59259	-0.78378	4.002988	0.373555	0	0
HF2791	225	-0.22052	1.077492	-2.30596	-0.2335	1.938363	0.41892	0	0
HF2801	226	0.792391	1.058293	-1.2564	0.779919	2.911232	0.772994	0	0
HF2806	227	1.717771	1.433346	-1.10366	1.718548	4.534928	0.884626	1	0
HF2819	228	-1.4458	4.244232	-9.79675	-1.44357	6.892713	0.366683	0	0
HF2830	229	-0.02589	2.591784	-5.12036	-0.02544	5.066357	0.496016	0	0
HF2836	230	5.252538	2.329679	0.714454	5.237811	9.871226	0.987921	1	1
HF2837	231	4.066712	2.231638	-0.2931	4.056861	8.48009	0.965795	1	1
HF2843	232	3.460676	3.648427	-3.70309	3.457573	10.64231	0.828573	1	0
HF2874	233	0.475669	1.098764	-1.67573	0.470929	2.653449	0.667461	0	0
HF2879	234	-1.7728	3.829909	-9.32339	-1.76771	5.748812	0.321724	0	0
HF2886	235	-1.12303	2.9257	-6.88321	-1.12081	4.624599	0.350545	0	0
HF2891	236	0.125315	1.112655	-2.06588	0.124985	2.317801	0.544837	0	0
HF2900	237	-1.60385	3.912428	-9.31784	-1.59823	6.07822	0.340926	0	0
HF2916	238	-0.73433	3.847337	-8.30603	-0.73259	6.827579	0.424315	0	0
HF2927	239	-0.61984	1.478681	-3.52588	-0.61966	2.286033	0.33754	0	0
HF2931	240	-2.61567	3.31199	-9.13224	-2.6133	3.887565	0.214834	0	0
HF2932	241	-5.62431	2.897436	-11.3656	-5.60842	0.02687	0.026121	0	0
HF2937	242	-3.88198	3.467206	-10.7266	-3.87276	2.910368	0.131436	0	0
HF2983	243	-1.05812	4.393379	-9.69749	-1.05769	7.579046	0.404838	0	0
HF2984	244	-2.06074	1.358394	-4.68242	-2.07923	0.667831	0.064628	0	0
HF2991	245	-1.92788	5.961415	-13.6731	-1.9216	9.781429	0.373199	0	0
HF3019	246	0.824228	1.534528	-2.17902	0.819754	3.852567	0.704408	0	0
HF3033	247	-0.69536	4.467883	-9.49017	-0.69156	8.078014	0.438161	0	0
HF3034	248	5.898601	2.262083	1.492074	5.883712	10.38767	0.995441	1	1
HF3036	249	-5.20321	3.309812	-11.7326	-5.19435	1.275805	0.057969	0	0
HF3047	250	-6.34157	3.53801	-13.3507	-6.32303	0.562437	0.036533	0	0
HF3083	251	-0.7205	2.95878	-6.5311	-0.72173	5.097267	0.403804	0	0
HF3085	252	-0.38712	1.258302	-2.82732	-0.40023	2.128125	0.379172	0	0
HF3103	253	-0.68658	2.997974	-6.5806	-0.68585	5.203241	0.409429	0	0
HF3109	254	1.133765	1.479705	-1.77739	1.134846	4.038858	0.778224	0	0
HF3118	255	0.101521	3.135552	-6.06182	0.102162	6.261265	0.512915	0	0
HF3137	256	-1.12272	3.286429	-7.59145	-1.12064	5.334438	0.366318	0	0
HF3141	257	3.573212	1.552808	0.507352	3.576684	6.619052	0.989308	1	1
HF3145	258	-7.72981	2.254593	-12.1943	-7.71653	-3.34113	3.03E-04	0	0
HF3146	259	-1.32302	1.358998	-4.01608	-1.31608	1.331559	0.165146	0	0
HF3149	260	0.870983	1.312027	-1.77243	0.892217	3.396756	0.746605	0	0
HF3153	261	-1.27705	3.076966	-7.33487	-1.27447	4.766533	0.339058	0	0
HF3171	262	-1.17294	1.244829	-3.61282	-1.17706	1.291038	0.173032	0	0
HF3186	263	-1.27958	3.398213	-7.97158	-1.27629	5.39418	0.353256	0	0
HF3188	264	-1.68513	3.036259	-7.66237	-1.68235	4.276679	0.289447	0	0
HF3207	265	-0.40903	2.48131	-5.28958	-0.40757	4.46348	0.434534	0	0
HF3214	266	-0.10816	4.351162	-8.67445	-0.1056	8.443495	0.490084	0	0
HF3222	267	1.689451	1.303494	-0.83974	1.675842	4.294308	0.902529	1	1
HF3226	268	0.094242	2.320419	-4.46545	0.094638	4.651808	0.516198	0	0
HF3236	269	5.255489	1.348981	2.686768	5.227163	7.981238	0.999951	1	1
HF3241	270	0.710604	2.447985	-4.09702	0.710156	5.520944	0.614199	0	0
HF3255	271	-1.07947	3.366789	-7.70145	-1.07814	5.535303	0.374248	0	0

HF3264	272	1.725603	4.810585	-7.73505	1.725176	11.18855	0.640094	0	0
HF3271	273	1.021901	3.022213	-4.91428	1.019285	6.972719	0.632367	0	0
HF3291	274	1.351676	4.558874	-7.61202	1.350621	10.32097	0.616573	0	0
HF3308	275	1.703107	2.137551	-2.49034	1.699197	5.91716	0.787204	0	0
HF3330	276	2.518137	4.044693	-5.4344	2.517517	10.4742	0.733219	0	0
HF3331	277	2.794685	2.300179	-1.69896	2.784669	7.342659	0.887814	1	0
HF3339	278	-3.73466	2.772882	-9.22377	-3.72218	1.683822	0.089014	0	0
HF3353	279	-2.78794	1.313155	-5.33058	-2.80286	-0.16194	0.016873	0	0
HF3362	280	-2.50676	4.734924	-11.845	-2.4981	6.782496	0.298257	0	0

Supplementary Figure S5. Posterior exceedance probability and hotspot concordance after covariate adjustment.

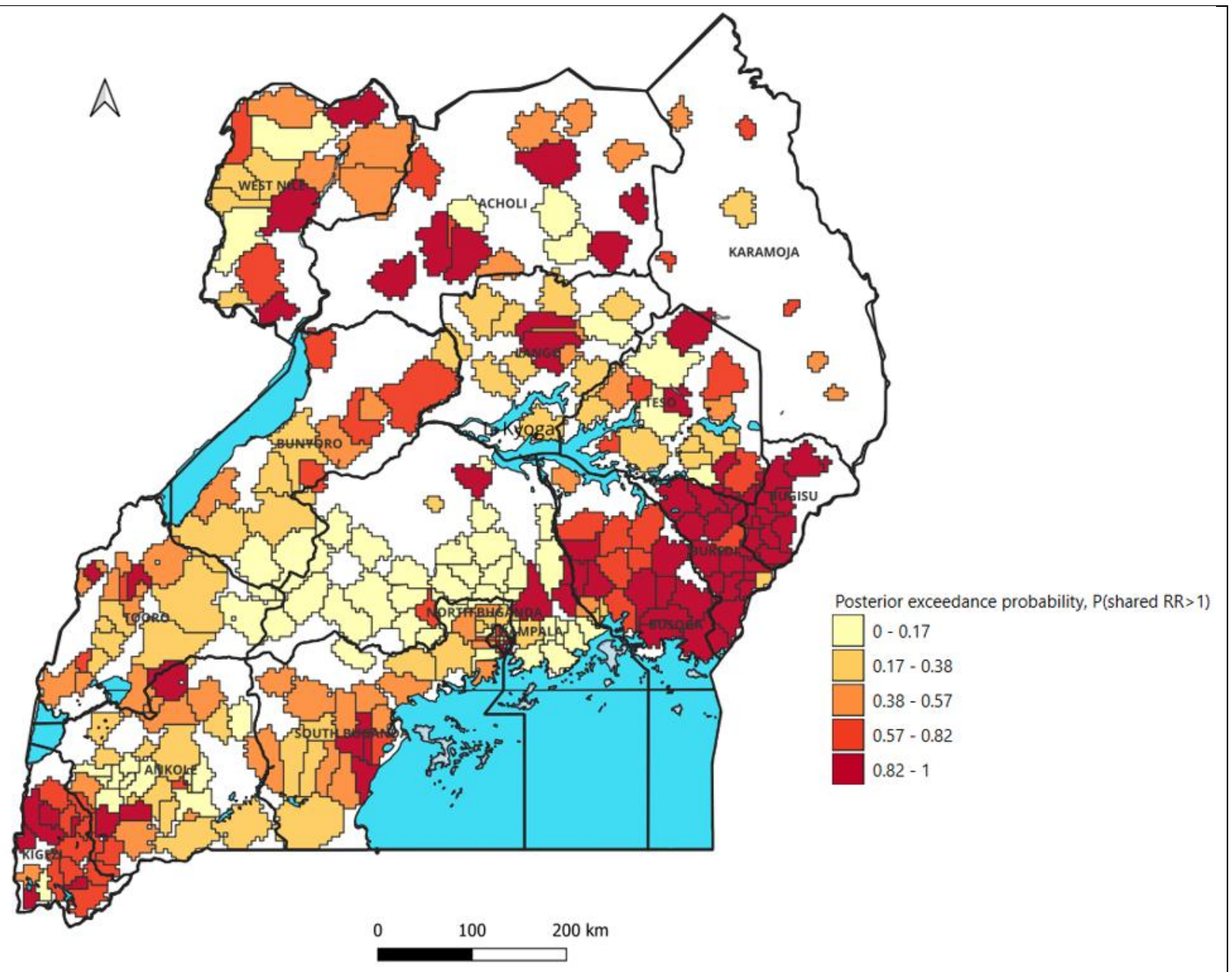


Figure 9. Posterior exceedance probability and hotspot concordance after covariate adjustment.

Supplementary Table S6. Uncertainty class, residual hotspot confidence and adjusted relative risk summaries for all catchments.

Residual priority_rank	catchment_id	Facility name	Facility class	Ownership class	region	Shared mean_adj	Shared sd_adj	Shared q025_adj	Shared q975_adj	Shared ci_width_adj	exceed_prob rr_gt1_adj	Hotspot 080_adj	Hotspot 090_adj	Uncertainty class
1	HF2029	St. Mary's Hospital Lacor	General hospital	PNFP	ACHOLI	3.806751	1.249361	1.329877	6.250679	4.920802	0.998844	1	1	Low uncertainty
2	HF1752	Tororo General HOSPITAL	General hospital	Government	BUKEDI	7.485275	1.959724	3.671545	11.37564	7.704096	0.999933	1	1	Low uncertainty
3	HF1125	Busolwe HOSPITAL	General hospital	Government	BUKEDI	5.217067	1.875041	1.593678	8.962583	7.368905	0.997302	1	1	Low uncertainty
4	HF2067	Gulu Regional Referal Hospital	Hospital	Government	ACHOLI	2.308779	1.440201	-0.51796	5.146176	5.664138	0.945543	1	1	Low uncertainty
5	HF1831	Kalongo Ambrosoli Memorial HOSPITAL	General hospital	PNFP	ACHOLI	4.051838	1.218888	1.674851	6.475428	4.800577	0.999557	1	1	Low uncertainty
6	HF2222	PAG Mission HC IV	HC IV	PNFP	LANGO	8.670235	1.353282	6.045213	11.3642	5.318987	1	1	1	Low uncertainty

7	HF1408	Atutur HOSPITAL	General hospital	Government	TESO	1.646223	1.298286	-0.95576	4.155947	5.111706	0.8976	1	0	Low uncertainty
8	HF1609	Kanginima HOSPITAL	General hospital	Government	BUKEDI	6.430905	1.923668	2.71056	10.27083	7.560268	0.999586	1	1	Low uncertainty
9	HF2490	Bundibugyo Hospital	General hospital	Government	TORO	2.61855	1.218299	0.237996	5.03228	4.794284	0.984197	1	1	Low uncertainty
10	HF2124	Kitgum HOSPITAL	General hospital	Government	ACHOLI	2.476451	1.134351	0.288075	4.759537	4.471461	0.985487	1	1	Low uncertainty
11	HF2353	Anaka HOSPITAL	General hospital	Government	ACHOLI	3.812684	1.138763	1.598567	6.086215	4.487649	0.999593	1	1	Low uncertainty
12	HF0975	Kapelebyong HC IV	HC IV	Government	TESO	9.10239	1.241565	6.707814	11.57916	4.87135	1	1	1	Low uncertainty
13	HF1166	Iganga HOSPITAL	General hospital	Government	BUSOGA	4.120397	1.813786	0.608834	7.739496	7.130662	0.988448	1	1	Low uncertainty
14	HF0494	Masaka REGIONAL REF HOSPITAL	Hospital	Government	SOUTH BUGANDA	1.890341	1.53207	-1.13844	4.887553	6.025993	0.89137	1	0	Low uncertainty
15	HF1032	Bugiri HOSPITAL	General hospital	Government	BUSOGA	4.069874	1.793929	0.600694	7.651739	7.051045	0.988356	1	1	Low uncertainty
16	HF3222	Itojo HOSPITAL	General hospital	Government	ANKOLE	1.689451	1.303494	-0.83974	4.294308	5.134049	0.902529	1	1	Low uncertainty
17	HF2330	Angal St. Luke HOSPITAL	General hospital	PNFP	WEST NILE	1.1027	0.807268	-0.43664	2.846239	3.282875	0.914025	1	1	Low uncertainty
18	HF2566	Hoima REGIONAL REF HOSPITAL	Hospital	Government	BUNYORO	1.621552	1.823402	-1.96378	5.204538	7.168318	0.813079	1	0	Low uncertainty
19	HF0499	Kyannamukaaka HC IV	HC IV	Government	SOUTH BUGANDA	6.350265	1.409227	3.657529	9.187236	5.529707	0.999997	1	1	Low uncertainty
20	HF0298	Kangulumira HC IV	HC IV	Government	NORTH BUGANDA	3.788254	0.919114	2.08888	5.735539	3.646659	0.999981	1	1	Low uncertainty
21	HF0752	Nakasongola HC IV	HC IV	Government	NORTH BUGANDA	5.949154	0.630798	4.680079	7.169845	2.489766	1	1	1	Low uncertainty
22	HF1697	Princes Diana HC IV	HC IV	Government	TESO	7.537947	1.202612	5.202748	9.926372	4.723623	1	1	1	Low uncertainty
23	HF0506	St. Joseph Kitovu HOSPITAL	General hospital	PNFP	SOUTH BUGANDA	1.315707	1.135543	-0.92642	3.550626	4.477048	0.876702	1	0	Low uncertainty
24	HF2806	Fort Portal Regional Referral Hospital	Hospital	Government	TORO	1.717771	1.433346	-1.10366	4.534928	5.638589	0.884626	1	0	Low uncertainty
25	HF1375	Katakwi General Hospital	General hospital	Government	TESO	0.455303	1.31294	-2.11654	3.053066	5.169611	0.635622	0	0	Low uncertainty
26	HF3034	Kisoro HOSPITAL	General hospital	Government	KIGEZI	5.898601	2.262083	1.492074	10.38767	8.895593	0.995441	1	1	Low uncertainty
27	HF2836	Kambuga HOSPITAL	General hospital	Government	KIGEZI	5.252538	2.329679	0.714454	9.871226	9.156772	0.987921	1	1	Moderate uncertainty
28	HF1113	Dabani HOSPITAL	General hospital	PNFP	BUKEDI	4.253878	1.89009	0.573047	8.005587	7.43254	0.987795	1	1	Low uncertainty
29	HF2601	Ishongororo HC IV	HC IV	Government	ANKOLE	5.449593	1.480468	2.652099	8.465797	5.813698	0.999884	1	1	Low uncertainty
30	HF2206	Lira REGIONAL REF HOSPITAL	Hospital	Government	LANGO	0.046693	1.442008	-2.80523	2.863715	5.668949	0.512916	0	0	Low uncertainty

31	HF0204	St. Francis Nsambya HOSPITAL	General hospital	PNFP	KAMPALA	2.025163	1.88494	-1.73799	5.667594	7.405579	0.858676	1	0	Low uncertainty
32	HF2801	Virika HOSPITAL	General hospital	PNFP	TORO	0.792391	1.058293	-1.2564	2.911232	4.167631	0.772994	0	0	Low uncertainty
33	HF0680	St. Francis HOSPITAL Naggalama	General hospital	PNFP	KAMPALA	2.168552	1.835512	-1.48225	5.725138	7.207384	0.881286	1	0	Low uncertainty
34	HF0216	Kiruddu	Hospital	Government	KAMPALA	2.125813	2.150485	-2.15194	6.296291	8.448235	0.838553	1	0	Low uncertainty
35	HF3141	Mbarara Community HOSPITAL	General hospital	PFP	ANKOLE	3.573212	1.552808	0.507352	6.619052	6.1117	0.989308	1	1	Low uncertainty
36	HF1747	Rubongi Military HOSPITAL	General hospital	Government	BUKEDI	3.715378	1.869935	0.095023	7.444537	7.349514	0.976533	1	1	Low uncertainty
37	HF1617	Pallisa HOSPITAL	General hospital	Government	BUKEDI	4.23515	1.932982	0.459692	8.062388	7.602695	0.985774	1	1	Low uncertainty
38	HF1754	St. Anthony'S Tororo HOSPITAL	General hospital	PNFP	BUKEDI	4.782485	1.940834	0.990914	8.624579	7.633665	0.993133	1	1	Low uncertainty
39	HF3236	Rwashamaire HC IV	HC IV	Government	ANKOLE	5.255489	1.348981	2.686768	7.981238	5.294471	0.999951	1	1	Low uncertainty
40	HF2278	Moyo HOSPITAL	General hospital	Government	WEST NILE	1.364812	0.873153	-0.33889	3.155331	3.494221	0.940984	1	1	Low uncertainty
41	HF1354	Kapchorwa HOSPITAL	General hospital	Government	BUGISU	6.559847	3.450131	-0.0375	13.51434	13.55184	0.971371	1	1	High uncertainty
42	HF0262	Benedict Medical centre HC IV	HC IV	PNFP	KAMPALA	6.244997	2.025763	2.285999	10.24474	7.958738	0.998975	1	1	Low uncertainty
43	HF1015	Bududa HOSPITAL	General hospital	Government	BUGISU	6.900578	3.480237	0.250302	13.9233	13.673	0.976305	1	1	High uncertainty
44	HF2305	Matany HOSPITAL	General hospital	PNFP	Other	1.318569	2.831735	-4.23046	6.91197	11.14243	0.679263	0	0	Moderate uncertainty
45	HF1313	Kamuli Mission Hospital	General hospital	PNFP	BUSOGA	3.018765	1.814098	-0.50222	6.631809	7.13403	0.95195	1	1	Low uncertainty
46	HF2837	Bwindi Community HOSPITAL	General hospital	PNFP	KIGEZI	4.066712	2.231638	-0.2931	8.48009	8.773195	0.965795	1	1	Low uncertainty
47	HF0093	Villa Maria HOSPITAL	General hospital	PNFP	SOUTH BUGANDA	1.529033	1.132274	-0.71937	3.745294	4.464664	0.911558	1	1	Low uncertainty
48	HF2329	Nebbi Hospital	General hospital	Government	WEST NILE	0.288049	0.590128	-0.85081	1.511028	2.361836	0.687265	0	0	Low uncertainty
49	HF3019	Kiryandongo HOSPITAL	General hospital	Government	BUNYORO	0.824228	1.534528	-2.17902	3.852567	6.03159	0.704408	0	0	Low uncertainty
50	HF1856	Patongo HC III	HC IV	Government	ACHOLI	4.64502	1.256702	2.194951	7.128238	4.933287	0.999891	1	1	Low uncertainty
51	HF3109	Masindi HOSPITAL	General hospital	Government	BUNYORO	1.133765	1.479705	-1.77739	4.038858	5.816246	0.778224	0	0	Low uncertainty
52	HF1234	Buwenge NGO Hospital	General hospital	PNFP	BUSOGA	2.125282	1.81443	-1.39489	5.739354	7.134248	0.879265	1	0	Low uncertainty
53	HF1999	Rhino Camp HC IV	HC IV	Government	WEST NILE	2.200659	0.819763	0.403291	3.660001	3.256709	0.996368	1	1	Low uncertainty
54	HF1532	Cure Children's Hospital	General hospital	PNFP	BUGISU	4.578508	3.41385	-1.96099	11.45264	13.41363	0.910065	1	1	High uncertainty

55	HF3331	St. Karolii Lwanga Nyakibale HOSPITAL	General hospital	PNFP	KIGEZI	2.794685	2.300179	-1.69896	7.342659	9.041621	0.887814	1	0	Moderate uncertainty
56	HF0282	Mengo HOSPITAL	General hospital	PNFP	KAMPALA	1.47594	1.866048	-2.25647	5.072805	7.329274	0.785512	0	0	Low uncertainty
57	HF1250	Kakira Worker's HOSPITAL	General hospital	PFP	BUSOGA	1.741102	1.926245	-2.0038	5.570317	7.574114	0.816971	1	0	Low uncertainty
58	HF2028	Gulu Military Hospital	General hospital	Government	ACHOLI	0.946507	1.169537	-1.33174	3.271416	4.603156	0.790829	0	0	Low uncertainty
59	HF0118	Kololo HOSPITAL	General hospital	PFP	KAMPALA	1.393006	2.168818	-2.91275	5.609341	8.522087	0.739657	0	0	Low uncertainty
60	HF1785	Adjumani HOSPITAL	General hospital	Government	WEST NILE	-0.02912	0.853203	-1.56174	1.832239	3.393984	0.486387	0	0	Low uncertainty
61	HF1769	Abim HOSPITAL	General hospital	Government	Other	1.79032	2.791461	-3.67297	7.31149	10.98446	0.739354	0	0	Moderate uncertainty
62	HF1130	Kabasa Memorial HOSPITAL	General hospital	PNFP	BUKEDI	1.534499	1.856586	-2.06825	5.230186	7.298435	0.795744	0	0	Low uncertainty
63	HF1726	Mulanda HC IV	HC IV	Government	BUKEDI	4.27454	2.697138	-0.96197	9.640306	10.60228	0.943499	1	1	Moderate uncertainty
64	HF1734	Nagongera HC IV	HC IV	Government	BUKEDI	4.063004	2.622351	-1.02705	9.278674	10.30572	0.939354	1	1	Moderate uncertainty
65	HF0998	Budaka HC IV	HC IV	Government	BUKEDI	4.254528	2.774736	-1.14122	9.759647	10.90087	0.9374	1	1	Moderate uncertainty
66	HF0263	Butabika NATIONAL REFERAL HOSPITAL	Hospital	Government	KAMPALA	1.620647	2.144352	-2.65438	5.769575	8.423958	0.775108	0	0	Low uncertainty
67	HF1583	Nsinze HC IV	HC IV	Government	BUSOGA	3.426637	2.442679	-1.33825	8.259556	9.597809	0.919665	1	1	Moderate uncertainty
68	HF2705	Kabale REGIONAL REF HOSPITAL	Hospital	Government	KIGEZI	1.946589	2.43412	-2.82852	6.744234	9.572752	0.788061	0	0	Moderate uncertainty
69	HF3149	Ruharo Mission HOSPITAL	General hospital	PNFP	ANKOLE	0.870983	1.312027	-1.77243	3.396756	5.169184	0.746605	0	0	Low uncertainty
70	HF3308	Kisiizi NGO HOSPITAL	General hospital	PNFP	KIGEZI	1.703107	2.137551	-2.49034	5.91716	8.407496	0.787204	0	0	Low uncertainty
71	HF2205	Ayira Health Services	HC IV	PFP	LANGO	1.92398	1.594918	-1.18243	5.080465	6.262892	0.886153	1	0	Low uncertainty
72	HF1647	Budadiri HC IV	HC IV	Government	BUGISU	5.18901	3.696553	-1.95019	12.58167	14.53186	0.919802	1	1	High uncertainty
73	HF2708	Rugarama HOSPITAL	General hospital	PNFP	KIGEZI	2.29568	2.288927	-2.23467	6.773982	9.008656	0.842058	1	0	Low uncertainty
74	HF1722	Mukuju HC IV	HC IV	Government	BUKEDI	3.512622	2.715827	-1.7635	8.911235	10.67474	0.902062	1	1	Moderate uncertainty
75	HF1492	Bubulo HC IV	HC IV	Government	BUGISU	4.989316	3.72539	-2.2029	12.44188	14.64479	0.909759	1	1	High uncertainty
76	HF2145	Koboko Hospital	General hospital	Government	WEST NILE	0.182227	0.755958	-1.30961	1.771014	3.080629	0.595244	0	0	Low uncertainty
77	HF0560	Ssekanyonyi HC IV	HC IV	Government	NORTH BUGANDA	0.707382	1.203325	-1.57026	3.168931	4.739192	0.721686	0	0	Low uncertainty
78	HF1312	Kamuli HOSPITAL	General hospital	Government	BUSOGA	1.101399	1.781829	-2.35554	4.649727	7.005271	0.731755	0	0	Low uncertainty
79	HF2874	Kilembe HOSPITAL	General hospital	PNFP	TORO	0.475669	1.098764	-1.67573	2.653449	4.329175	0.667461	0	0	Low uncertainty

80	HF1483	Bugobero HC IV	HC IV	Government	BUGISU	6.114408	4.456446	-2.49771	15.01827	17.51598	0.914974	1	1	High uncertainty
81	HF1165	Busesa HC IV	HC IV	Government	BUSOGA	3.450965	2.828252	-2.07345	9.041914	11.11536	0.888801	1	0	Moderate uncertainty
82	HF1533	Namatala HC IV	HC IV	Government	BUGISU	3.869	3.110011	-2.14369	10.08169	12.22539	0.893259	1	0	Moderate uncertainty
83	HF1547	Buyinja HC IV	HC IV	Government	BUSOGA	3.011094	2.539068	-1.94481	8.033217	9.978028	0.88217	1	0	Moderate uncertainty
84	HF1524	Bufumbo HC IV	HC IV	Government	BUGISU	4.70828	3.664198	-2.37605	12.02919	14.40524	0.900594	1	1	High uncertainty
85	HF0252	Kadic HOSPITAL	General hospital	PFP	KAMPALA	0.902668	2.156664	-3.39702	5.075982	8.472998	0.662227	0	0	Low uncertainty
86	HF1530	Busiu HC IV	HC IV	Government	BUGISU	3.718569	3.16955	-2.42083	10.04019	12.46101	0.879646	1	0	Moderate uncertainty
87	HF2481	Buliisa Hospital	General hospital	Government	BUNYORO	0.785904	1.688844	-2.5224	4.116984	6.63938	0.67916	0	0	Low uncertainty
88	HF1092	Muyembe HC IV	HC IV	Government	BUGISU	4.493996	3.70853	-2.68816	11.89318	14.58134	0.887206	1	0	High uncertainty
89	HF1667	Buwasa HC IV	HC IV	Government	BUGISU	3.90672	3.422176	-2.72705	10.72733	13.45438	0.873188	1	0	High uncertainty
90	HF2891	Kagando HOSPITAL	General hospital	PNFP	TORO	0.125315	1.112655	-2.06588	2.317801	4.383684	0.544837	0	0	Low uncertainty
91	HF1600	Butebo HC IV	HC IV	Government	BUKEDI	2.719438	2.606234	-2.34569	7.900294	10.24598	0.851627	1	0	Moderate uncertainty
92	HF2099	Kaabong HOSPITAL	General hospital	Government	Other	0.888177	2.86904	-4.72886	6.559953	11.28882	0.621557	0	0	Moderate uncertainty
93	HF1405	Kibuku HC IV	HC IV	Government	BUKEDI	3.552261	3.239297	-2.75884	9.972658	12.7315	0.863595	1	0	Moderate uncertainty
94	HF1438	Kaproron HC IV	HC IV	Government	BUGISU	7.39492	5.747835	-3.74939	18.84512	22.59451	0.900876	1	1	High uncertainty
95	HF1322	Nankandulo HC IV	HC IV	Government	BUSOGA	2.801878	2.774215	-2.62194	8.284761	10.9067	0.843746	1	0	Moderate uncertainty
96	HF1515	Kigandalo HC IV	HC IV	Government	BUSOGA	2.779804	2.769916	-2.63073	8.257565	10.8883	0.842207	1	0	Moderate uncertainty
97	HF1063	Nankoma HC IV	HC IV	Government	BUSOGA	2.94424	2.920331	-2.76326	8.714043	11.4773	0.843318	1	0	Moderate uncertainty
98	HF1276	Lwala HOSPITAL	General hospital	PNFP	TESO	-0.12547	1.230285	-2.55535	2.285736	4.841091	0.459384	0	0	Low uncertainty
99	HF2594	Ibanda HOSPITAL	General hospital	PNFP	ANKOLE	0.194908	1.274378	-2.29241	2.728303	5.020709	0.560779	0	0	Low uncertainty
100	HF2843	Kanungu HC IV	HC IV	Government	KIGEZI	3.460676	3.648427	-3.70309	10.64231	14.3454	0.828573	1	0	High uncertainty
101	HF1180	Bugono HC IV	HC IV	Government	BUSOGA	3.143997	3.424417	-3.55229	9.903506	13.45579	0.82072	1	0	High uncertainty
102	HF3085	Kyenjojo Hospital	General hospital	Government	TORO	-0.38712	1.258302	-2.82732	2.128125	4.955447	0.379172	0	0	Low uncertainty
103	HF0281	Lubaga HOSPITAL	General hospital	PNFP	KAMPALA	0.311702	1.897897	-3.48971	3.964184	7.453895	0.565227	0	0	Low uncertainty
104	HF0465	Lyantonde HOSPITAL	General hospital	Government	SOUTH BUGANDA	-0.27284	1.193053	-2.57219	2.129507	4.701702	0.409556	0	0	Low uncertainty
105	HF1340	Namwendwa HC IV	HC IV	Government	BUSOGA	2.213928	3.147946	-3.93766	8.436957	12.37461	0.759063	0	0	Moderate uncertainty
106	HF1281	Bumanya HC IV	HC IV	Government	BUSOGA	2.491363	3.508799	-4.37269	9.413712	13.7864	0.761158	0	0	High uncertainty

107	HF1629	Apapai HC IV	HC IV	Government	TESO	0.832989	1.723603	-2.55102	4.21915	6.770172	0.685553	0	0	Low uncertainty
108	HF1065	Bukedea HC IV	HC IV	Government	TESO	0.894054	1.873205	-2.78303	4.572432	7.355462	0.683421	0	0	Low uncertainty
109	HF0505	Kiyumba HC IV	HC IV	Government	SOUTH BUGANDA	1.410588	2.684936	-3.86391	6.685277	10.54919	0.700337	0	0	Moderate uncertainty
110	HF0203	Kibuli HOSPITAL	General hospital	PNFP	KAMPALA	0.079391	1.89421	-3.70154	3.741175	7.442711	0.516716	0	0	Low uncertainty
111	HF2791	Kida HOSPITAL	General hospital	PNFP	TORO	-0.22052	1.077492	-2.30596	1.938363	4.244322	0.41892	0	0	Low uncertainty
112	HF3330	North Kigezi HC IV	HC IV	PNFP	KIGEZI	2.518137	4.044693	-5.4344	10.4742	15.9086	0.733219	0	0	High uncertainty
113	HF1965	Arua REGIONAL REF HOSPITAL	Hospital	Government	WEST NILE	-0.76201	1.292275	-3.3016	1.787611	5.089209	0.277709	0	0	Low uncertainty
114	HF0588	Nkozi HOSPITAL	General hospital	PNFP	SOUTH BUGANDA	-0.18724	1.182641	-2.54155	2.120689	4.66224	0.437101	0	0	Low uncertainty
115	HF0777	Kalisizo HOSPITAL	General hospital	Government	SOUTH BUGANDA	-0.15712	1.151917	-2.44916	2.091248	4.540403	0.445752	0	0	Low uncertainty
116	HF2927	Kagadi HOSPITAL	General hospital	Government	BUNYORO	-0.61984	1.478681	-3.52588	2.286033	5.811914	0.33754	0	0	Low uncertainty
117	HF1475	Kiyunga HC IV	HC IV	Government	BUSOGA	2.08546	3.979325	-5.70615	9.931451	15.6376	0.699886	0	0	High uncertainty
118	HF1903	Atiak HC IV	HC IV	Government	ACHOLI	1.399446	3.273275	-5.03178	7.830942	12.86272	0.665506	0	0	Moderate uncertainty
119	HF0887	Kisubi HOSPITAL	General hospital	PNFP	KAMPALA	-0.19377	1.907312	-4.01565	3.474013	7.489666	0.459541	0	0	Low uncertainty
120	HF1688	ATIRIR HC IV	HC IV	Government	TESO	0.580365	2.141912	-3.61669	4.791322	8.408009	0.606788	0	0	Low uncertainty
121	HF3241	Rubaare HC IV	HC IV	Government	ANKOLE	0.710604	2.447985	-4.09702	5.520944	9.617969	0.614199	0	0	Moderate uncertainty
122	HF2695	Hamurwa HC IV	HC IV	Government	KIGEZI	1.911003	4.166617	-6.2681	10.12153	16.38963	0.676756	0	0	High uncertainty
123	HF2522	Comboni HOSPITAL	General hospital	PNFP	ANKOLE	-0.3857	1.245531	-2.83543	2.071867	4.907294	0.378406	0	0	Low uncertainty
124	HF3271	Buhunga HC IV	HC IV	Government	KIGEZI	1.021901	3.022213	-4.91428	6.972719	11.887	0.632367	0	0	Moderate uncertainty
125	HF1970	Oriajini HOSPITAL	General hospital	PNFP	WEST NILE	-0.40591	0.772351	-1.97242	1.187617	3.160035	0.2996	0	0	Low uncertainty
126	HF2749	Mparo HC IV	HC IV	Government	KIGEZI	1.816138	4.161053	-6.3593	10.00865	16.36796	0.668749	0	0	High uncertainty
127	HF2734	Muko HC IV	HC IV	Government	KIGEZI	2.050929	4.656751	-7.09058	11.22383	18.31441	0.670184	0	0	High uncertainty
128	HF1698	Soroti REGIONAL REF HOSPITAL	Hospital	Government	TESO	-1.51916	1.53237	-4.55814	1.467213	6.02535	0.160751	0	0	Low uncertainty
129	HF1934	Apac HOSPITAL	General hospital	Government	LANGO	-0.78169	1.353259	-3.47269	1.850724	5.323412	0.281756	0	0	Low uncertainty
130	HF2211	Amach HC IV	HC IV	Government	LANGO	0.32957	2.110979	-3.81981	4.466499	8.286314	0.562032	0	0	Low uncertainty
131	HF0052	Gombe HOSPITAL	General hospital	Government	SOUTH BUGANDA	-0.50234	1.159951	-2.78538	1.786598	4.571976	0.332481	0	0	Low uncertainty
132	HF1421	Kadami HOSPITAL	General hospital	PNFP	TESO	-1.51437	1.296373	-4.04856	1.050734	5.099289	0.121371	0	0	Low uncertainty

133	HF0883	Luwunga Barracks HC IV	HC IV	Other	KAMPALA	-0.49946	2.948588	-6.31417	5.284256	11.59842	0.432745	0	0	Moderate uncertainty
134	HF2457	Nyapea HOSPITAL	General hospital	PNFP	WEST NILE	-0.28805	0.590128	-1.51103	0.850808	2.361836	0.312735	0	0	Low uncertainty
135	HF0854	Bamu HOSPITAL	General hospital	PFP	SOUTH BUGANDA	-0.59332	1.281768	-3.12685	1.915107	5.04196	0.321721	0	0	Low uncertainty
136	HF1805	Mungula HC IV	HC IV	Government	WEST NILE	0.341255	2.6217	-4.81138	5.481792	10.29317	0.551791	0	0	Moderate uncertainty
137	HF3264	Bugangari HC IV	HC IV	Government	KIGEZI	1.725603	4.810585	-7.73505	11.18855	18.92361	0.640094	0	0	High uncertainty
138	HF1431	Kumi NGO HOSPITAL	General hospital	PNFP	TESO	-0.65834	1.294135	-3.2094	1.885498	5.094893	0.305477	0	0	Low uncertainty
139	HF3291	Kebisoni HC IV	HC IV	Government	KIGEZI	1.351676	4.558874	-7.61202	10.32097	17.93299	0.616573	0	0	High uncertainty
140	HF3226	Ntungamo HC IV	HC IV	Government	ANKOLE	0.094242	2.320419	-4.46545	4.651808	9.117259	0.516198	0	0	Moderate uncertainty
141	HF2501	Kikyo HC IV	HC IV	Government	TORO	0.548787	3.367834	-6.0712	7.173604	13.24481	0.564721	0	0	High uncertainty
142	HF2376	Aber Ngo HOSPITAL	General hospital	PNFP	LANGO	-0.97443	1.288047	-3.55114	1.520325	5.071467	0.224669	0	0	Low uncertainty
143	HF0948	Ndejje HC IV	HC IV	Government	KAMPALA	0.376641	3.301621	-6.15943	6.822939	12.98237	0.545412	0	0	Moderate uncertainty
144	HF0304	Kayunga Hospital	General hospital	Government	NORTH BUGANDA	-3.42886	0.970689	-5.03887	-1.25166	3.787213	2.06E-04	0	0	Low uncertainty
145	HF2830	Ntara HC IV	HC IV	Government	TORO	-0.02589	2.591784	-5.12036	5.066357	10.18672	0.496016	0	0	Moderate uncertainty
146	HF1207	Budondo HC IV	HC IV	Government	BUSOGA	0.028472	2.759845	-5.41503	5.438783	10.85381	0.504116	0	0	Moderate uncertainty
147	HF2761	Kataraka HC IV	HC IV	Government	TORO	0.323512	3.482708	-6.52573	7.172342	13.69807	0.537005	0	0	High uncertainty
148	HF0039	Butenga HC IV	HC IV	Government	SOUTH BUGANDA	-0.01093	2.736492	-5.39208	5.36262	10.7547	0.498407	0	0	Moderate uncertainty
149	HF3118	Masindi Military/Army Barracks HC IV	HC IV	Government	BUNYORO	0.101521	3.135552	-6.06182	6.261265	12.32309	0.512915	0	0	Moderate uncertainty
150	HF2494	Busaru HC IV	HC IV	PNFP	TORO	0.115276	3.264158	-6.29708	6.531375	12.82846	0.514086	0	0	Moderate uncertainty
151	HF2249	Obongi HC IV	HC IV	Government	WEST NILE	-0.15475	2.832306	-5.71883	5.405341	11.12417	0.478213	0	0	Moderate uncertainty
152	HF2448	Midigo HC IV	HC IV	Government	WEST NILE	-0.31114	2.41697	-5.05972	4.431673	9.491395	0.448786	0	0	Moderate uncertainty
153	HF0103	St. Joseph of the Good Shephard Kyamulibwa Ngo HC IV	HC IV	PNFP	SOUTH BUGANDA	-0.16714	2.989923	-6.04601	5.706901	11.75291	0.47771	0	0	Moderate uncertainty
154	HF1591	Ngora Ngo HOSPITAL	General hospital	PNFP	TESO	-1.14314	1.272829	-3.67095	1.339677	5.01063	0.184563	0	0	Low uncertainty
155	HF0092	Bukulula HC IV	HC IV	Government	SOUTH BUGANDA	-0.05199	3.508517	-6.95051	6.840698	13.79121	0.494089	0	0	High uncertainty
156	HF3171	Holy Innocents Children's HOSPITAL	General hospital	PNFP	ANKOLE	-1.17294	1.244829	-3.61282	1.291038	4.903862	0.173032	0	0	Low uncertainty
157	HF2753	Bukuku HC IV	HC IV	Government	TORO	-0.36765	2.462953	-5.21698	4.469942	9.686922	0.44067	0	0	Moderate uncertainty
158	HF2445	Yumbe Hospital	General hospital	Government	WEST NILE	-0.86757	0.781694	-2.54712	0.62494	3.172065	0.133529	0	0	Low uncertainty
159	HF3207	Mitooma HC IV	HC IV	Government	ANKOLE	-0.40903	2.48131	-5.28958	4.46348	9.753062	0.434534	0	0	Moderate uncertainty

160	HF2622	Ruhoko HC IV	HC IV	Government	ANKOLE	-0.41345	2.792691	-5.9033	5.068858	10.97216	0.441152	0	0	Moderate uncertainty
161	HF3214	Karugutu HC IV	HC IV	Government	TORO	-0.10816	4.351162	-8.67445	8.443495	17.11794	0.490084	0	0	High uncertainty
162	HF1390	Toroma HC IV	HC IV	Government	TESO	-0.62428	2.833211	-6.17984	4.953303	11.13314	0.412802	0	0	Moderate uncertainty
163	HF2186	Madi-Opei HC IV	HC IV	Government	ACHOLI	0.228052	5.154921	-9.9235	10.37159	20.29509	0.517643	0	0	High uncertainty
164	HF1994	Omugo HC IV	HC IV	Government	WEST NILE	-0.75607	2.313982	-5.30418	3.784591	9.088769	0.371933	0	0	Moderate uncertainty
165	HF3103	Bwijanga HC IV	HC IV	Government	BUNYORO	-0.68658	2.997974	-6.5806	5.203241	11.78384	0.409429	0	0	Moderate uncertainty
166	HF3083	Kyarusozi HC IV	HC IV	Government	TORO	-0.7205	2.95878	-6.5311	5.097267	11.62837	0.403804	0	0	Moderate uncertainty
167	HF2778	Kibiito HC IV	HC IV	Government	TORO	-0.78665	2.439585	-5.59259	4.002988	9.595577	0.373555	0	0	Moderate uncertainty
168	HF2019	Dokolo HC IV	HC IV	Government	LANGO	-0.78459	1.953551	-4.62732	3.041877	7.669197	0.343981	0	0	Low uncertainty
169	HF1643	Serere HC IV	HC IV	Government	TESO	-0.83299	1.723603	-4.21915	2.551022	6.770172	0.314447	0	0	Low uncertainty
170	HF3146	Mayanja Memorial HOSPITAL	General hospital	PNFP	ANKOLE	-1.32302	1.358998	-4.01608	1.331559	5.347641	0.165146	0	0	Low uncertainty
171	HF0921	Saidina Abubakar Islamic HOSPITAL	General hospital	PNFP	KAMPALA	-1.64212	1.894455	-5.42613	2.014237	7.440365	0.193025	0	0	Low uncertainty
172	HF2513	Ishaka Adventist HOSPITAL	General hospital	PNFP	ANKOLE	-1.40851	1.255099	-3.85706	1.087188	4.94425	0.130882	0	0	Low uncertainty
173	HF2916	Nyamirami HC IV	HC IV	Government	TORO	-0.73433	3.847337	-8.30603	6.827579	15.13361	0.424315	0	0	High uncertainty
174	HF0873	Buwambo HC IV	HC IV	Government	KAMPALA	-0.86897	3.365688	-7.52381	5.712604	13.23641	0.398132	0	0	Moderate uncertainty
175	HF0429	Kiwangala HC IV	HC IV	Government	SOUTH BUGANDA	-0.91494	3.215198	-7.24103	5.401204	12.64224	0.387988	0	0	Moderate uncertainty
176	HF0718	Nakaseke HOSPITAL	General hospital	Government	NORTH BUGANDA	-4.189	1.034481	-6.01564	-1.93148	4.084161	2.57E-05	0	0	Low uncertainty
177	HF2190	Padibe HC IV	HC IV	Government	ACHOLI	-0.22805	5.154921	-10.3716	9.9235	20.29509	0.482357	0	0	High uncertainty
178	HF0678	Mukono T.C. HC IV	HC IV	Government	KAMPALA	-1.00271	2.472679	-5.90729	3.816583	9.723876	0.342549	0	0	Moderate uncertainty
179	HF0583	Mpigi HC IV	HC IV	Government	SOUTH BUGANDA	-0.99861	2.218713	-5.41047	3.317987	8.728462	0.326325	0	0	Low uncertainty
180	HF2586	Kyangwali HC IV	HC IV	Government	BUNYORO	-0.91751	3.723616	-8.22989	6.397366	14.62726	0.402685	0	0	High uncertainty
181	HF3033	Rubuguri HC IV	HC IV	Government	KIGEZI	-0.69536	4.467883	-9.49017	8.078014	17.56818	0.438161	0	0	High uncertainty
182	HF1419	Kumi HC IV	HC IV	Government	TESO	-1.06913	2.203231	-5.39029	3.266455	8.656747	0.313748	0	0	Low uncertainty
183	HF2057	Lalogi HC IV	HC IV	Government	ACHOLI	-0.37679	5.268966	-10.7395	9.997352	20.73682	0.471496	0	0	High uncertainty
184	HF0202	International Hospital Kampala (IHK)	General hospital	PFP	KAMPALA	-1.86026	2.165495	-6.20013	2.305591	8.505721	0.195158	0	0	Low uncertainty
185	HF2140	Naam Okora HC IV	HC IV	Government	ACHOLI	-0.17334	5.683858	-11.3537	11.0134	22.36707	0.487835	0	0	High uncertainty
186	HF2886	St. Paul HC IV	HC IV	PNFP	TORO	-1.12303	2.9257	-6.88321	4.624599	11.50781	0.350545	0	0	Moderate uncertainty
187	HF3255	Rugazi HC IV	HC IV	Government	ANKOLE	-1.07947	3.366789	-7.70145	5.535303	13.23676	0.374248	0	0	Moderate uncertainty
188	HF1147	Kidera HC IV	HC IV	Government	BUSOGA	-0.0691	5.947477	-11.7679	11.63341	23.40132	0.495365	0	0	High uncertainty

189	HF2512	Kiu Teaching HOSPITAL	General hospital	PNFP	ANKOLE	-1.5571	1.249564	-4.02525	0.897646	4.922896	0.106361	0	0	Low uncertainty
190	HF3137	Bugamba HC IV	HC IV	Government	ANKOLE	-1.12272	3.286429	-7.59145	5.334438	12.92589	0.366318	0	0	Moderate uncertainty
191	HF0007	Kawolo HOSPITAL	General hospital	Government	NORTH BUGANDA	-2.83827	0.846858	-4.47997	-1.05028	3.429694	4.02E-04	0	0	Low uncertainty
192	HF0446	Lwengo HC IV	HC IV	Government	SOUTH BUGANDA	-1.15449	3.288562	-7.62597	5.305241	12.93121	0.362771	0	0	Moderate uncertainty
193	HF1927	Aduku HC IV	HC IV	Government	LANGO	-1.20735	2.513765	-6.16039	3.715403	9.875797	0.315509	0	0	Moderate uncertainty
194	HF2108	Karenga HC IV	HC IV	Government	Other	-0.74758	4.840577	-10.2663	8.773344	19.03969	0.438631	0	0	High uncertainty
195	HF0438	Kyazanga HC IV	HC IV	Government	SOUTH BUGANDA	-1.22978	2.896545	-6.93182	4.457804	11.38962	0.335575	0	0	Moderate uncertainty
196	HF0860	Ssembabule HC IV	HC IV	Government	SOUTH BUGANDA	-1.01743	4.183074	-9.24873	7.199494	16.44822	0.403915	0	0	High uncertainty
197	HF2280	Tokora HC IV	HC IV	Government	Other	-0.84373	4.750863	-10.1858	8.500263	18.68608	0.42952	0	0	High uncertainty
198	HF3153	Mbarara Municipal Council HC IV	HC IV	Government	ANKOLE	-1.27705	3.076966	-7.33487	4.766533	12.10141	0.339058	0	0	Moderate uncertainty
199	HF0715	Kiwoko HOSPITAL	General hospital	PNFP	NORTH BUGANDA	-3.18763	1.02674	-5.17212	-1.08952	4.082594	9.53E-04	0	0	Low uncertainty
200	HF1595	Ngora Gvt HC IV	HC IV	Government	TESO	-1.21141	3.747425	-8.56258	6.165415	14.728	0.373247	0	0	High uncertainty
201	HF3186	Kinoni HC IV	HC IV	Government	ANKOLE	-1.27958	3.398213	-7.97158	5.39418	13.36576	0.353256	0	0	High uncertainty
202	HF2983	Kazo HC IV	HC IV	Government	ANKOLE	-1.05812	4.393379	-9.69749	7.579046	17.27653	0.404838	0	0	High uncertainty
203	HF2292	Nabilatuk HC IV	HC IV	Government	Other	-0.97667	4.638744	-10.098	8.146283	18.2443	0.41662	0	0	High uncertainty
204	HF0002	Buikwe St. Charles Lwanga HOSPITAL	General hospital	PNFP	NORTH BUGANDA	-3.77431	1.011587	-5.73532	-1.71142	4.023895	9.53E-05	0	0	Low uncertainty
205	HF0926	Kasangati HC IV	HC IV	Government	KAMPALA	-1.32205	2.344374	-5.99026	3.227743	9.218007	0.286403	0	0	Moderate uncertainty
206	HF2570	Kigorobya HC IV	HC IV	Government	BUNYORO	-1.37292	3.637187	-8.52568	5.771646	14.29732	0.352913	0	0	High uncertainty
207	HF1751	Benedictine Eye HOSPITAL	General hospital	Government	BUKEDI	-1.36678	2.238591	-5.77394	3.023298	8.797238	0.270747	0	0	Low uncertainty
208	HF1209	Bugembe HC IV	HC IV	Government	BUSOGA	-1.45366	3.268339	-7.90974	4.943554	12.85329	0.328243	0	0	Moderate uncertainty
209	HF2984	Rushere Community HOSPITAL	General hospital	PNFP	ANKOLE	-2.06074	1.358394	-4.68242	0.667831	5.35025	0.064628	0	0	Low uncertainty
210	HF0362	Bombo General Military HOSPITAL	General hospital	Government	NORTH BUGANDA	-4.96247	0.959778	-6.57146	-2.80511	3.766355	1.17E-07	0	0	Low uncertainty
211	HF0940	Mildmay Uganda HOSPITAL	General hospital	PNFP	KAMPALA	-1.81926	1.919473	-5.67657	1.858151	7.53472	0.171618	0	0	Low uncertainty
212	HF2819	Rukunyu HC IV	HC IV	Government	TORO	-1.4458	4.244232	-9.79675	6.892713	16.68946	0.366683	0	0	High uncertainty
213	HF2009	Kuluva HOSPITAL	General hospital	PNFP	WEST NILE	-2.20066	0.819763	-3.66	-0.40329	3.256709	0.003632	0	0	Low uncertainty
214	HF0253	Murchison Bay HOSPITAL	General hospital	Government	KAMPALA	-1.9439	2.162392	-6.25195	2.24214	8.494089	0.184338	0	0	Low uncertainty
215	HF2565	Azur HC III	HC IV	PNFP	BUNYORO	-1.53452	4.087722	-9.57391	6.49493	16.06884	0.353683	0	0	High uncertainty

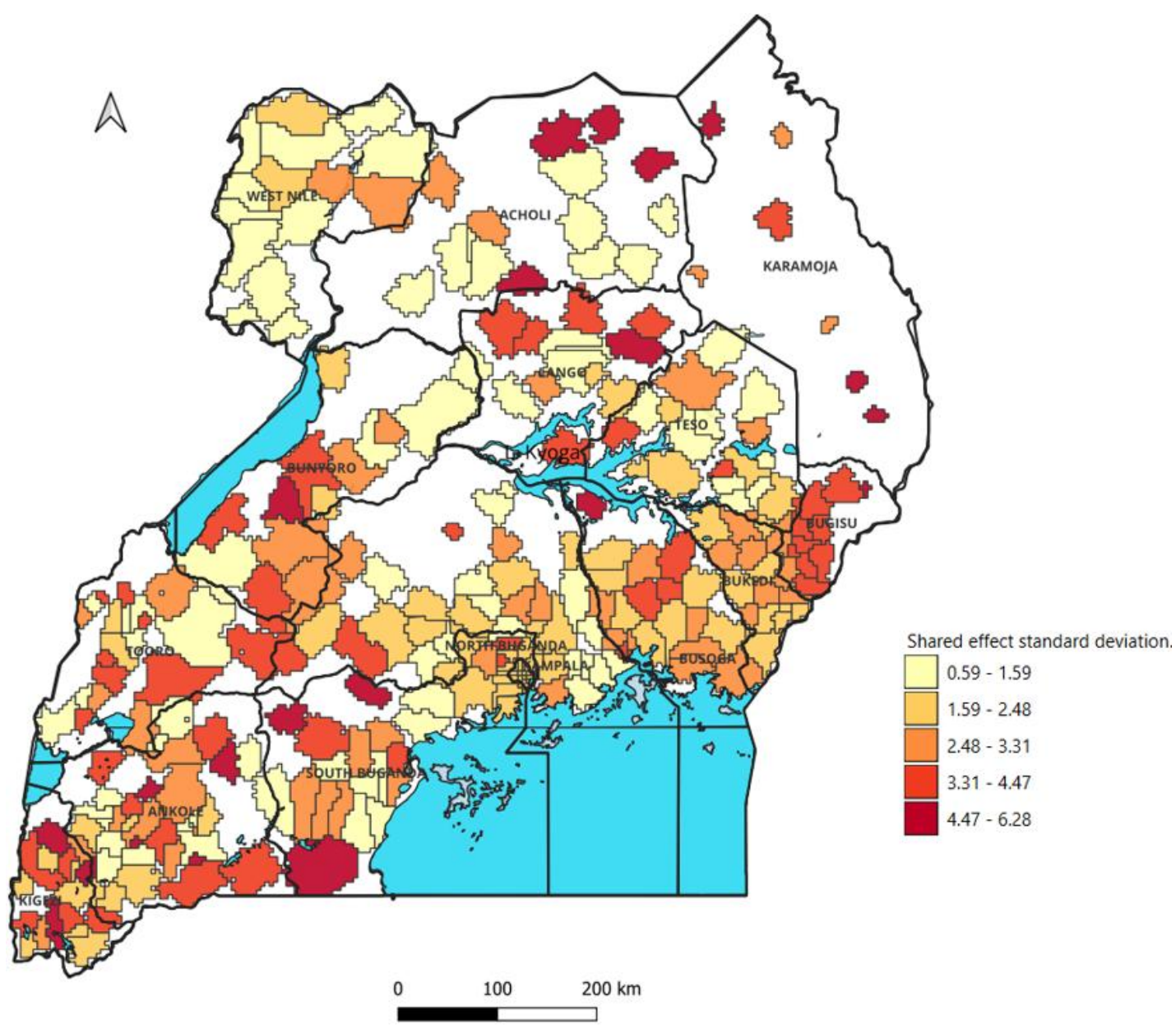
216	HF2900	Rwesande HC IV	HC IV	PNFP	TORO	-1.60385	3.912428	-9.31784	6.07822	15.39606	0.340926	0	0	High uncertainty
217	HF3188	Bwizibwera HC IV	HC IV	Government	ANKOLE	-1.68513	3.036259	-7.66237	4.276679	11.93905	0.289447	0	0	Moderate uncertainty
218	HF0720	Ngoma HC IV	HC IV	Government	NORTH BUGANDA	-1.6894	3.895285	-9.34889	5.950237	15.29913	0.332252	0	0	High uncertainty
219	HF2677	Rugaaga HC IV	HC IV	Government	ANKOLE	-1.6982	3.921858	-9.4103	6.008296	15.4186	0.332504	0	0	High uncertainty
220	HF2167	Kotido HC IV	HC IV	Government	Other	-1.64028	4.246337	-9.98933	6.708544	16.69787	0.349644	0	0	High uncertainty
221	HF0008	Lugazi Scoul HOSPITAL	General hospital	PFP	NORTH BUGANDA	-6.48856	1.406036	-9.16494	-3.62852	5.536417	1.97E-06	0	0	Low uncertainty
222	HF2879	Hiima Iaa (Uci) HC IV	HC IV	PFP	TORO	-1.7728	3.829909	-9.32339	5.748812	15.0722	0.321724	0	0	High uncertainty
223	HF0859	Ntuusi HC IV	HC IV	Government	SOUTH BUGANDA	-0.92786	6.276812	-13.2772	11.4067	24.68395	0.441241	0	0	High uncertainty
224	HF2375	Orum HC IV	HC IV	Government	LANGO	-1.77629	3.879295	-9.41054	5.833802	15.24434	0.323516	0	0	High uncertainty
225	HF3353	Kitagata HOSPITAL	General hospital	Government	ANKOLE	-2.78794	1.313155	-5.33058	-0.16194	5.168638	0.016873	0	0	Low uncertainty
226	HF0022	St. Francis Nyenga HOSPITAL	General hospital	PNFP	NORTH BUGANDA	-4.73304	1.111539	-6.73834	-2.35071	4.387627	1.03E-05	0	0	Low uncertainty
227	HF2673	Nyamuyanja HC IV	HC IV	Government	ANKOLE	-1.36137	5.516446	-12.2273	9.477224	21.7045	0.402538	0	0	High uncertainty
228	HF2652	Kabuyanda HC IV	HC IV	Government	ANKOLE	-1.8452	3.841138	-9.39862	5.70202	15.10064	0.315479	0	0	High uncertainty
229	HF0550	Mityana HOSPITAL	General hospital	Government	NORTH BUGANDA	-4.64023	0.909538	-6.18776	-2.60034	3.587418	1.68E-07	0	0	Low uncertainty
230	HF1269	Kaberamaido HC IV	HC IV	Government	TESO	-1.89673	3.841185	-9.44478	5.66007	15.10485	0.310728	0	0	High uncertainty
231	HF0773	Kakuuto HC IV	HC IV	Government	SOUTH BUGANDA	-1.82207	4.718132	-11.1051	7.445443	18.55057	0.34968	0	0	High uncertainty
232	HF1249	Mpumudde HC IV	HC IV	Government	BUSOGA	-2.02318	3.386935	-8.72691	4.591919	13.31883	0.275137	0	0	High uncertainty
233	HF0327	Kiboga HOSPITAL	General hospital	Government	NORTH BUGANDA	-5.63065	0.98096	-7.31983	-3.45404	3.865785	4.74E-09	0	0	Low uncertainty
234	HF2578	Kikuube HC IV	HC IV	Government	BUNYORO	-1.94815	5.058787	-11.883	7.997138	19.88012	0.350081	0	0	High uncertainty
235	HF1246	Walukuba HC IV	HC IV	Government	BUSOGA	-2.39066	3.450766	-9.22163	4.34757	13.5692	0.244221	0	0	High uncertainty
236	HF2991	Kiruhura HC IV	HC IV	Government	ANKOLE	-1.92788	5.961415	-13.6731	9.781429	23.45454	0.373199	0	0	High uncertainty
237	HF1882	Amolatar HC IV	HC IV	Government	LANGO	-2.46339	3.562978	-9.47762	4.521995	13.99962	0.244662	0	0	High uncertainty
238	HF3362	Shuuku HC IV	HC IV	Government	ANKOLE	-2.50676	4.734924	-11.845	6.782496	18.62745	0.298257	0	0	High uncertainty
239	HF2931	Kakindo HC IV	HC IV	Government	BUNYORO	-2.61567	3.31199	-9.13224	3.887565	13.0198	0.214834	0	0	Moderate uncertainty
240	HF2228	Ogur HC IV	HC IV	Government	LANGO	-2.6506	3.48588	-9.51402	4.182092	13.69611	0.223513	0	0	High uncertainty
241	HF0919	Namayumba HC IV	HC IV	Government	KAMPALA	-2.40051	2.270157	-6.89554	2.027383	8.922922	0.14516	0	0	Low uncertainty
242	HF2400	Anyeke HC IV	HC IV	Government	LANGO	-2.75092	3.450083	-9.54428	4.010929	13.55521	0.212624	0	0	High uncertainty
243	HF0698	Kojja HC IV	HC IV	Government	KAMPALA	-2.80804	2.780195	-8.33678	2.596387	10.93316	0.156244	0	0	Moderate uncertainty
244	HF0741	Nakasongola Military HOSPITAL	General	Government	NORTH BUGANDA	-5.94915	0.630798	-7.16985	-4.68008	2.489766	0	0	0	Low uncertainty

			hospital											
245	HF0019	Nkokonjeru HOSPITAL	General hospital	PNFP	NORTH BUGANDA	-6.8764	1.126084	-8.95177	-4.50438	4.447394	5.09E-10	0	0	Low uncertainty
246	HF0957	Wakiso HC IV	HC IV	Government	KAMPALA	-3.25961	4.042994	-11.3026	4.598772	15.90133	0.210053	0	0	High uncertainty
247	HF0290	Bbaale HC IV	HC IV	Government	NORTH BUGANDA	-2.96459	2.290117	-7.48937	1.50087	8.99024	0.097744	0	0	Low uncertainty
248	HF2428	Pajule HC IV	HC IV	Government	ACHOLI	-2.47645	1.134351	-4.75954	-0.28808	4.471462	0.014513	0	0	Low uncertainty
249	HF2508	Bushenyi HC IV	HC IV	Government	ANKOLE	-3.10628	2.549558	-8.14742	1.880954	10.02838	0.111543	0	0	Moderate uncertainty
250	HF0537	Mwera HC IV	HC IV	Government	NORTH BUGANDA	-3.15284	1.911615	-6.93218	0.572655	7.504831	0.049543	0	0	Low uncertainty
251	HF0376	Kalagala HC IV	HC IV	Government	NORTH BUGANDA	-3.23303	2.010201	-7.2272	0.669559	7.896756	0.053883	0	0	Low uncertainty
252	HF2152	Aboke HC IV	HC IV	Government	LANGO	-3.66848	4.216792	-11.9993	4.589405	16.58871	0.192158	0	0	High uncertainty
253	HF0732	St. Franciscan HC IV	HC IV	PNFP	NORTH BUGANDA	-3.7714	3.164273	-10.0112	2.413107	12.42435	0.116656	0	0	Moderate uncertainty
254	HF3339	Kabwohe HC IV	HC IV	Government	ANKOLE	-3.73466	2.772882	-9.22377	1.683822	10.90759	0.089014	0	0	Moderate uncertainty
255	HF2937	Kibaale HC IV (Kibaale)	HC IV	Government	BUNYORO	-3.88198	3.467206	-10.7266	2.910368	13.63693	0.131436	0	0	High uncertainty
256	HF0316	Bukomero HC IV	HC IV	Government	NORTH BUGANDA	-3.67072	2.029251	-7.68493	0.281343	7.966271	0.035233	0	0	Low uncertainty
257	HF0724	Semuto HC IV	HC IV	Government	NORTH BUGANDA	-3.78152	1.760844	-7.26503	-0.35187	6.91316	0.015874	0	0	Low uncertainty
258	HF2475	Nsiika HC IV	HC IV	Government	ANKOLE	-4.12559	4.949013	-13.9057	5.564851	19.47051	0.202248	0	0	High uncertainty
259	HF2410	5Th Military Division HOSPITAL	General hospital	Government	ACHOLI	-4.64502	1.256702	-7.12824	-2.19495	4.933287	1.09E-04	0	0	Low uncertainty
260	HF0264	Naguru Hospital - China Uganda Friendship	Hospital	Government	KAMPALA	-3.9697	2.300756	-8.59328	0.440727	9.034011	0.042229	0	0	Moderate uncertainty
261	HF0540	Kyantungo HC IV	HC IV	Government	NORTH BUGANDA	-3.99714	2.240429	-8.42996	0.366392	8.79635	0.037204	0	0	Low uncertainty
262	HF0889	Uganda Martyrs HOSPITAL	General hospital	PNFP	KAMPALA	-4.36845	2.077702	-8.55102	-0.40002	8.151	0.017753	0	0	Low uncertainty
263	HF2521	Kyabugimbi HC IV	HC IV	Government	ANKOLE	-4.54801	3.632741	-11.7375	2.551654	14.28913	0.105294	0	0	High uncertainty
264	HF0402	Bishop Asili Ceaser HC IV	HC IV	PNFP	NORTH BUGANDA	-4.74753	2.760304	-10.2167	0.626835	10.84358	0.042723	0	0	Moderate uncertainty
265	HF0411	Nyimbwa HC IV	HC IV	Government	NORTH BUGANDA	-4.90052	2.168354	-9.1995	-0.68297	8.516529	0.01191	0	0	Low uncertainty
266	HF0608	Kassanda HC IV	HC IV	Government	NORTH BUGANDA	-5.399	2.074941	-9.52196	-1.37218	8.149777	0.004634	0	0	Low uncertainty
267	HF1868	Alebtong HC IV	HC IV	Government	LANGO	-4.98405	4.818359	-14.5163	4.443505	18.95982	0.150478	0	0	High uncertainty
268	HF0401	Luwero HC IV	HC IV	Government	NORTH BUGANDA	-5.19636	2.697708	-10.5451	0.051762	10.59691	0.027039	0	0	Moderate uncertainty
269	HF3036	Mutolere (St. Francis) HOSPITAL	General hospital	PNFP	KIGEZI	-5.20321	3.309812	-11.7326	1.275805	13.00843	0.057969	0	0	Moderate uncertainty
270	HF0283	Namungoona Orthodox Hospital HC III	General hospital	PNFP	KAMPALA	-5.89394	2.226392	-10.3845	-1.64931	8.735149	0.004057	0	0	Low uncertainty
271	HF2932	Kakumiro HC IV	HC IV	Government	BUNYORO	-5.62431	2.897436	-11.3656	0.02687	11.39248	0.026121	0	0	Moderate uncertainty
272	HF0354	Ntwetwe HC IV	HC IV	Government	NORTH BUGANDA	-5.74974	3.065234	-11.8218	0.223675	12.04547	0.030342	0	0	Moderate uncertainty

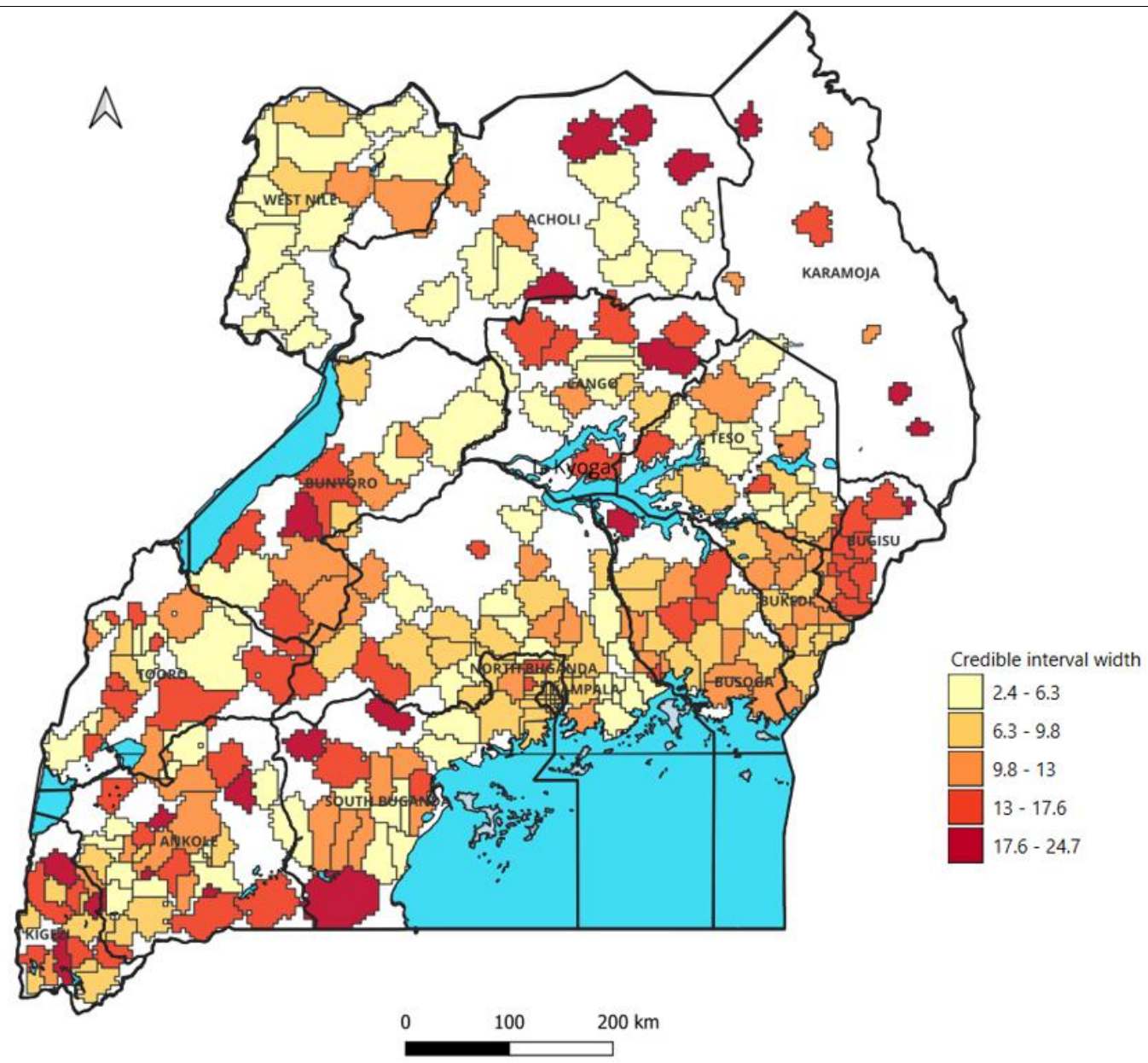
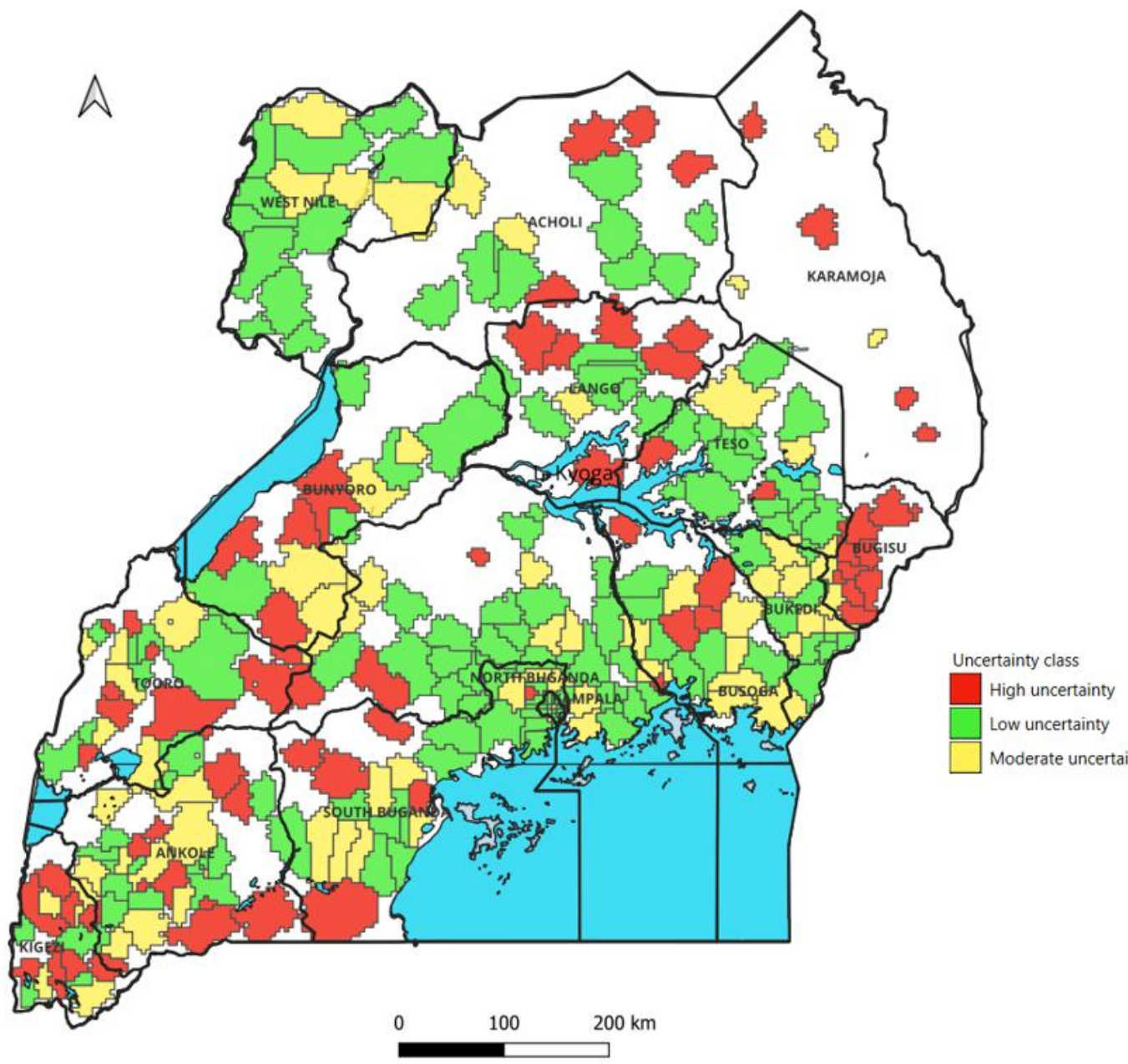
273	HF2528	Hunter Foundation HC	General hospital	PFP	ANKOLE	-7.26753	2.112145	-11.458	-3.16243	8.295597	2.90E-04	0	0	Low uncertainty
274	HF3145	UPDF 2nd Div.Hospital	General hospital	Government	ANKOLE	-7.72981	2.254593	-12.1943	-3.34113	8.853207	3.03E-04	0	0	Low uncertainty
275	HF3047	Kyegegwa HC IV	HC IV	Government	TORO	-6.34157	3.53801	-13.3507	0.562437	13.91313	0.036533	0	0	High uncertainty
276	HF2027	Awach HC IV	HC IV	Government	ACHOLI	-7.06255	3.03042	-13.0978	-1.18811	11.90964	0.009888	0	0	Moderate uncertainty
277	HF0083	Maddu HC IV	HC IV	Government	SOUTH BUGANDA	-6.09715	6.126845	-18.1968	5.901711	24.09853	0.159831	0	0	High uncertainty
278	HF0615	Kiganda HC IV	HC IV	Government	NORTH BUGANDA	-7.1461	3.470259	-14.0375	-0.39298	13.64455	0.019736	0	0	High uncertainty
279	HF0650	Mubende RR HOSPITAL	Hospital	Government	NORTH BUGANDA	-10.556	1.649906	-13.7592	-7.27702	6.482205	7.87E-11	0	0	Low uncertainty
280	HF0966	Amuria HC IV	HC IV	Government	TESO	-15.122	3.14637	-21.4474	-9.09346	12.35396	7.69E-07	0	0	Moderate uncertainty

Supplementary Figure S6. Posterior uncertainty in the adjusted shared malaria-SCD spatial effect.

(A) Posterior standard deviation map.



(B) Credible interval width map.



(C) ***Uncertainty classification of the adjusted shared malaria-SCD spatial effect.*** Map showing catchments classified as low, moderate, and high uncertainty according to posterior standard deviation and credible interval width of the adjusted shared spatial effect.